

**E-PATUNAY: A CRYPTOGRAPHIC-ASSURANCE
GOVERNANCE SYSTEM FOR THE BARANGAY
COMMUNITY RELATIONS DEPARTMENT ENABLING
STORAGE OPTIMIZATION AND INTER-AGENCY
VERIFICATION**

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LIST OF ABBREVIATIONS

BCRD - Barangay Community Relations Department

COA - Commission on Audit

CSC - Civil Service Commission

DICT - Department of Information and Communications Technology

LGU - Local Government Unit

PNP-ACG - Philippine National Police Anti-Cybercrime Group

MFA - Multi-Factor Authentication

RBAC - Role-Based Access Control

PKI - Public Key Infrastructure

API - Application Programming Interface

TLS - Transport Layer Security

AES - Advanced Encryption Standard

UAT - User Acceptance Testing

SUS - System Usability Testing

PDS - Personal Data Sheet

PDF - Portable Document File

SALN - Statement of Assets, Liabilities, and Net Worth

CIA - Confidentiality, Integrity, Availability

ABSTRACT

The Barangay and Community Relations Department (BCRD) currently relies on manual document tracking and unverified email communications. This traditional approach leads to operational inefficiencies, physical storage limitations, and critical vulnerabilities regarding data integrity, tamper-proofing, and unauthorized access. To address these challenges, this study developed *e-Patunay*, a secure, web-based governance system designed to provide cryptographic assurance, storage optimization, and seamless inter-agency record verification.

The system implements a Public Key Infrastructure (PKI) to apply server-side digital signatures to approved documents. To guarantee absolute data integrity, the system utilizes an append-only cryptographic ledger powered by transaction hash chaining and daily Merkle Tree consolidations. This architecture allows external oversight agencies, such as the Commission on Audit (COA), to independently verify the authenticity of official records through an isolated Verification API, providing mathematical proof of untampered data without exposing the underlying sensitive citizen information. Additionally, the system automates storage optimization by transitioning inactive records to Microsoft Azure Blob Storage, ensuring compliance with the retention mandates of the National Archives Act (RA 9470).

Developed using the Agile Kanban framework, the system's security architecture was rigorously evaluated against the STRIDE threat model. The platform's overall software quality and operational effectiveness were assessed using a mixed-methods approach, utilizing the ISO/IEC 25010 standard and the System Usability Scale (SUS). Ultimately, *e-Patunay* successfully modernizes BCRD operations by enforcing strict Role-Based Access Control (RBAC) and Multi-Factor Authentication (MFA), eliminating paper-based vulnerabilities, and establishing a zero-trust environment for government document management.

Keywords: *Cryptographic Assurance, Merkle Tree, Public Key Infrastructure (PKI), Data Integrity, Inter-Agency Verification, E-Governance, Storage Optimization.*

Chapter 1

INTRODUCTION

“Digitalization is the call of today, not the call of the future... We have fallen behind. We must fully embrace digitalization... We should not allow the people to continue to suffer from this antiquated, corrupt, and inefficient system.” - President Ferdinand “Bong Bong” Marcos Jr.

1.1 Background of the Study

Modern transactional systems rely on continuous user interaction and real-time data processing while handling sensitive information. These systems depend on integrated services and APIs, creating vulnerability points throughout the transaction lifecycle. Any weakness in these systems whether in data handling, authentication, or communication channels can jeopardize the confidentiality of information, the integrity of transactions, the availability of services, and ultimately, the trust that users place in the system.

In government institutions such as the BCRD, digitalization initiatives have primarily focused on basic digitization practices, including scanning physical documents and storing them as portable document format (PDF) files. While this approach addresses physical storage limitations and improves accessibility, it does not guarantee record authenticity, integrity, or accountability. Digitized records alone cannot verify whether documents are genuine, unaltered, or submitted by authorized individuals, which limits their reliability for auditing and legal compliance.

As national mandates such as the E-Governance Act (RA 12254) and the Philippine e-Government Master Plan of 2023-2028 require government agencies to

adopt secure and transparent digital systems, the limitations of traditional digitization methods become more evident. The BCRD requires a system that not only stores records electronically but also applies cryptographic mechanisms to ensure data integrity, authenticity, and non-repudiation. Without cybersecurity controls embedded in system design, digital records remain vulnerable to manipulation, unauthorized access, and loss of institutional credibility.

This study addresses the critical gap between basic digitization and secure cryptographic digital record management. By integrating cybersecurity principles into a governance-oriented transaction system, the proposed solution enables the BCRD to comply with national digitalization requirements while maintaining trust, legal validity, and operational reliability. Beyond secure storage, the system supports governance intelligence through structured data management, allowing compliance tracking, audit readiness, and verifiable inter-agency collaboration.

Cybersecurity is therefore fundamental to the development of government transaction systems, transforming digitalization from a procedural requirement into a strategic capability. Through the application of cryptographic controls, this research demonstrates how secure digital governance can strengthen institutional accountability, protect sensitive public records, and reinforce public trust in local government operations.

1.1 Purpose and Description

The Barangay Community Relations Department (BCRD) operates under the Quezon City Government as a specialized administrative body responsible for managing official records such as Personal Data Sheets (PDS), Statement of Assets Liabilities and Net Worth (SALN), and documentation from 142 barangays across the

city. The department functions under the Acting City Government Assistant Department Head, supported by the Legal Unit for compliance oversight.

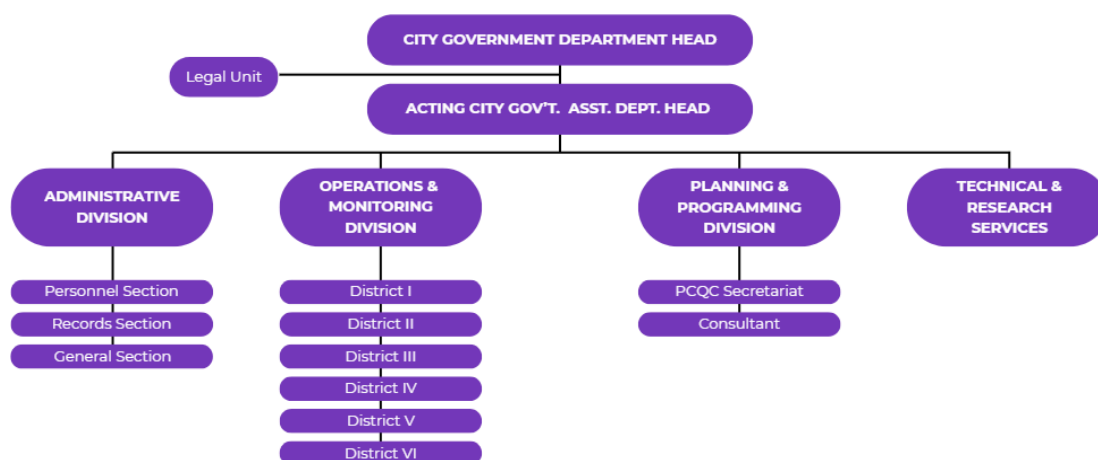


Figure 1. *BCRD Organizational Chart*

Organizationally, BCRD comprises four main divisions:

1. Administrative Division - Manages personnel, records, and general administrative functions through three sections (Personnel, Records, and General Section).
2. Operations & Monitoring Division - Oversees six district offices (District I through VI) responsible for direct barangay coordination and document processing.
3. Planning & Programming Division - Handles the PCQC Secretariat and consultant coordination for strategic initiatives.
4. Technical & Research Services - Provides technical support and research capabilities for department operations.

The current BCRD workflow follows a five-stage manual process that is characterized by security vulnerabilities and operation inefficiencies. During the Document Submission and Initial Receipt phases, the reliance on physical transit and insecure emails combined with their manual paper-based logging results in a lack of identity verification and security, making the system susceptible to data loss and human error.

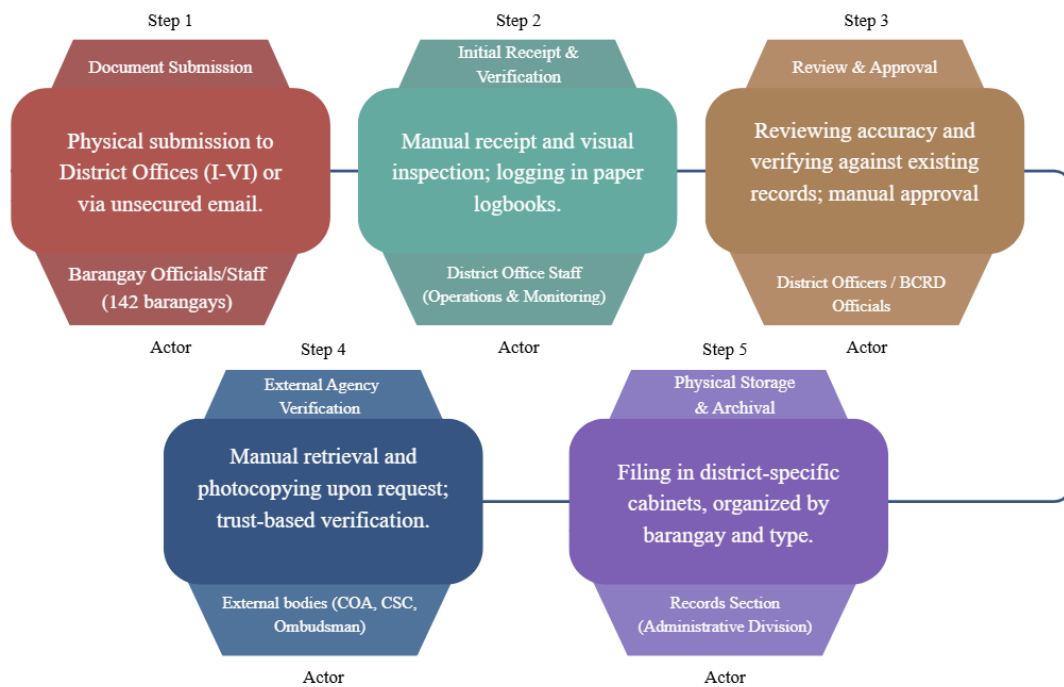


Figure 2. *BCRD Workflow*

The diagram illustrates the existing end-to-end workflow of the Barangay Community Relations Department (BCRD) for handling official documents, highlighting its predominantly manual and paper-based nature. The process begins with document submission, where barangay officials or staff from 142 barangays physically submit documents to district offices or send them through unsecured email channels. In Step 2, district office staff conduct an initial receipt and verification through manual visual inspection, recording entries in paper logbooks for tracking

purposes. This is followed by review and approval, where district officers or BCRD officials manually check the accuracy of documents by comparing them against existing records before granting approval. Once approved, documents may be subject to external agency verification,

The evolution of government operations necessitates a transition from manual, paper-based procedures to interconnected transactional systems. Effective digitalization would require the integration of web applications, cloud infrastructure, and APIs into a cohesive environment. While each integration improves efficiency and accessibility, it also introduces complex security challenges, as each additional service creates new authentication boundaries and data exchange points that may be exploited by attackers. As a result, traditional perimeter-based security approaches are insufficient for systems that process sensitive personal data across multiple stages, including submission, validation, storage, and utilization.

Local Government Unit implementations struggle to adopt digital processes that go beyond basic electronic services and fully modernize local government operations (Department of Finance, 2022). Studies on e-government adoption note that most public sector modernization efforts are limited to scanning physical documents and storing them as static digital files, a practice that improves storage efficiency but fails to ensure data integrity, authenticity, or legal accountability (United Nations, 2022).

Such systems function as passive “data graveyards”, incapable of tracking and verifying records trails, falling short of legal expectations. It also lacks intelligence, failing to actively support governance functions. Agencies such as the Barangay Community Relations Department (BCRD) remain dependent on manual verification

processes, limiting their ability to visualize workforce trends, monitor compliance status, or generate timely operational insights. This reliance on manual oversight reduces efficiency and increases exposure to human error and procedural delays.

To address these gaps, this study proposes the development of *e-Patunay*, an active compliance and oversight system designed to move beyond static digital repositories. The system proposes cryptographic verification mechanisms that bind each submitted document to a verifiable human identity, ensuring data integrity and preventing unauthorized modification. It further incorporates automated governance analytics that process submission data into real-time visualizations of barangay demographics and operational metrics. In addition, a compliance monitoring module actively tracks document validity and official tenures, issuing automated alerts for upcoming expirations. Finally, the system generates portable digital credentials, enabling external agencies to independently verify an official's status without requiring direct database access.

By this context, the study contributes to the field of cybersecurity by showcasing secure-by-design development for government operations. Practically, it strengthens the security posture of the BCRD by replacing trust-based manual procedures with verifiable and automated workflows. Academically, it contributes to system development literature by illustrating how advanced security controls can be embedded directly into administrative processes, demonstrating that government systems can be both resilient against modern cyber threats and operationally proactive in supporting effective governance.

1.2 Project Context

The problem environment of this study is the document lifecycle management of the BCRD. Currently, their workflow uses disjointed tools that do not work well together. Barangay Officials submit documents physically or via an unsecured email, which District Officers then verify by simply looking at them before saving them in separate file cabinets. This fragmented approach keeps data trapped in separate locations, making it difficult for internal staff to manage and impossible for external oversight bodies, like the Commission on Audit, to verify records independent of the BCRD.

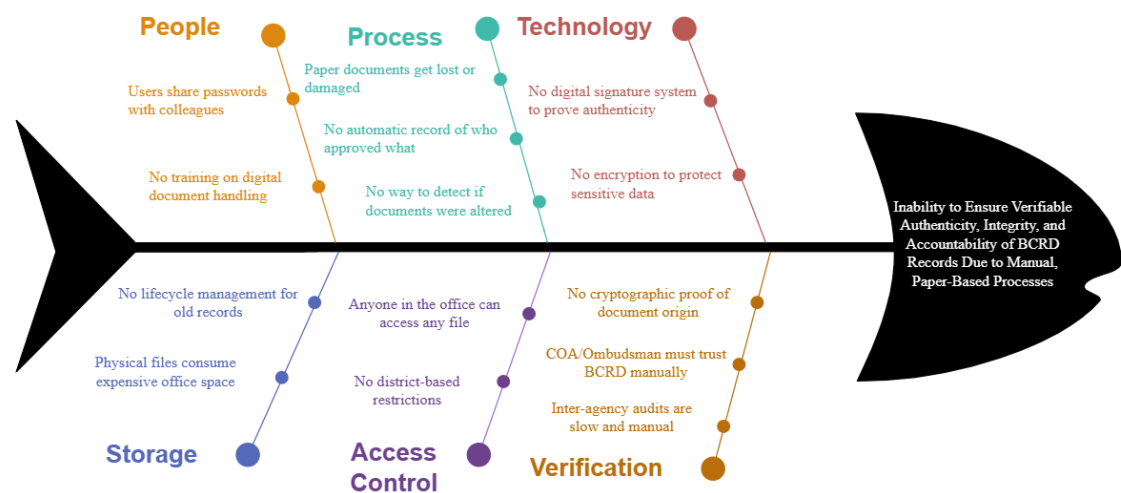


Figure 3. *Ishikawa Diagram: Showing the constraints of the existing manual filing workflow*

The **People** category highlights human factors that create security vulnerabilities in the current system. Users sharing passwords with colleagues undermines individual accountability, as multiple people may access the system using the same credentials, making it impossible to determine who performed specific actions. The absence of training on digital document handling means staff lack

awareness of security best practices, inadvertently creating risks through unsafe behaviors such as sending sensitive documents via unencrypted email or leaving workstations unlocked.

The **Process** category identifies operational failures inherent in manual, paper-based workflows. Paper documents getting lost or damaged during handling, transit, or storage creates gaps in the official record that cannot be recovered. The absence of automatic records showing who approved what means BCRD has no audit trail for accountability. If a questionable document is later discovered, there is no forensic evidence showing which official reviewed and approved it. Manual verification processes cannot detect if documents have been altered after submission, as sophisticated modifications to paper documents or digital files leave no visible trace.

The **Technology** category reveals critical infrastructure deficiencies that prevent BCRD from implementing modern security controls. The absence of a digital signature system means documents have no cryptographic proof of authenticity. Any claim that a document came from BCRD relies purely on trust rather than technical verification. Sensitive personal data stored and transmitted without encryption violates the Data Privacy Act of 2012 (RA 10173) and exposes information to interception or unauthorized access. Without cryptographic proof of document origin, BCRD cannot provide external agencies with verifiable evidence that documents are genuine and unaltered.

The **Storage** category demonstrates how physical document management creates unsustainable operational burdens. Managing records from 142 barangays consumes expensive office space that requires continuous expansion as the volume of

documents grows. The lack of lifecycle management for old records means BCRD must maintain the same level of accessibility and storage cost for 10-year-old documents as for current submissions, even though older records are rarely accessed. Physical files face existential risk from disasters such as fire, flood, or structural damage, with no backup or recovery mechanism ensuring records survive catastrophic events.

The **Access Control** category identifies the inability to enforce granular permissions in physical filing systems. The current situation where "anyone in the office can access any file" means junior staff can view sensitive personnel records meant only for senior administrators, and District I staff can access District VI files despite having no operational need. The absence of district-based restrictions prevents BCRD from implementing geographic separation that aligns access with job responsibilities. Physical access controls like locked cabinets are easily circumvented and provide no audit trail showing who accessed what and when.

The **Verification** category highlights a critical gap in inter-agency accountability. External oversight bodies such as the Commission on Audit, Civil Service Commission, and Office of the Ombudsman currently cannot verify documents independently. They must either trust BCRD's manual assertions or conduct time-consuming physical inspections. The requirement for COA/Ombudsman to "trust BCRD manually" creates a trust-based rather than evidence-based audit model that is vulnerable to fraud and inconsistent with modern accountability standards. Inter-agency audits being "slow and manual" creates delays that can impede investigations, compliance reviews, or legal proceedings requiring timely access to verified records.

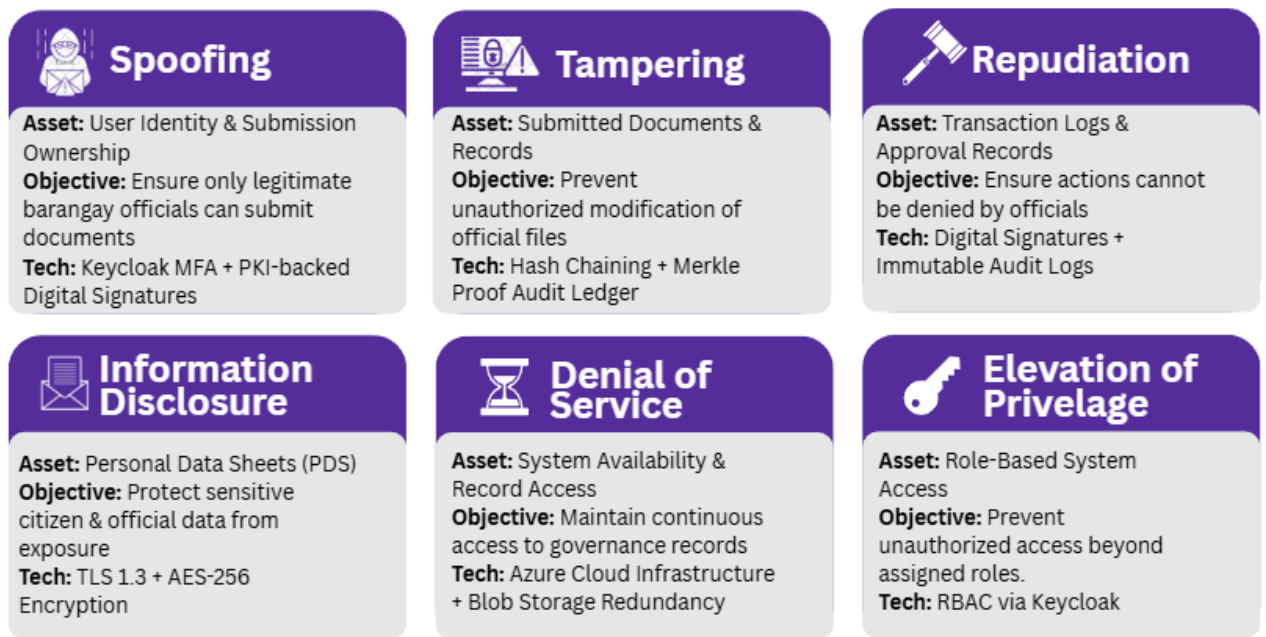


Figure 4. *STRIDE Threat Model: Vulnerabilities in the Current BCRD Workflow*

An analysis of the current BCRD workflow through the STRIDE threat model reveals critical vulnerabilities across all six security domains. In terms of Spoofing, the reliance on in-person confirmation and physical signatures allows malicious actors to submit fraudulent documents using forged credentials or unverified emails. Tampering poses a high risk because physical records can be altered or replaced without leaving a trace, making it impossible to verify the integrity of sensitive details like financial figures. This is exacerbated by a lack of Repudiation mechanisms; without forensic evidence to link specific actions to individuals, officials can deny involvement in submissions or approvals, creating significant accountability gaps during audits. Information disclosure is a constant threat as sensitive citizen data, including home addresses and government IDs, is transmitted via unencrypted emails or physical exchange, risking unauthorized exposure and violations of the Data Privacy Act. Furthermore, the system is susceptible to Denial of Service, where manual bottlenecks slow operations and the lack of digital backups means a single

physical disaster could result in unrecoverable data loss. Finally, Elevation of Privilege occurs due to a lack of strict access controls, as unlocked filing cabinets and shared office computers allow junior staff or unauthorized individuals to access or process documents meant only for senior directors.

If left unresolved, these vulnerabilities create legal and operational risks. With the enforcement of the E-Governance Act, continuing to use manual systems that cannot be verified exposes the BCRD to legal liability. Furthermore, the escalating storage requirements violate retention mandates under the National Archives Act. The inability to prove that public records are authentic weakens trust in the barangay governance system and separates the department from the government's modernization efforts.

1.3 Objectives of the Study

General Objectives

To develop a web-based governance system for the Barangay Community Relations Department that transforms manual record lifecycle processes into legally compliant digitized workflows, ensuring verifiable document authenticity, data integrity, and non-repudiation.

Specific Objectives

1. Develop a secure web-based document management platform that automates record lifecycle management and applies EJBICA-based PKI digital signatures for legal document authenticity.
2. Implement a tamper resistant audit ledger using hash chaining and merkle proofs.

3. Enforce multi-layered access control and encryption using Keycloak MFA/RBAC, TLS 1.3, and AES-256 in compliance with the Data Privacy Act.
4. Use Microsoft Azure utilizing its Blob archiving tiers to regulate storage overflow in compliance with the National Archives Act.
5. Implement an RBAC-enabled RESTful API for inter-agency verification to ensure authorized, role-specific data retrieval.
6. Use ISO/IEC 25010 in evaluating system quality and success utilizing User Accessibility Testing, System Usability Scale, and Jmeter load testing and conduct STRIDE-aligned security testing using industry standard tools to validate system defenses against specific threat categories:
 - Spoofing - Hydra for test authentication strength & brute-force resistance
 - Tampering - SQLMap to detect injection & modification vulnerabilities
 - Repudiation - OWASP ZAP to validate logging integrity & session tracking
 - Information disclosure - Nikto to identify data leakage & misconfigurations
 - Elevation of Privilege - Burp Suite for testing access control bypass & authorization flaws
 - Denial of Service - JMeter to Evaluate system performance under load

7. Automate governance monitoring by transforming submission data into live demographic dashboards to provide decision-makers with real-time visualizations.

1.4 Scope and Limitations

This capstone study focuses on the development of e-Patunay: a Cryptographic Assurance Governance System replacing BCRD Quezon City's manual paper-based certification processes with a secure web-based digital workflow that ensures document authenticity, chain-of-custody integrity, and BCRD official non-repudiation within defined academic project constraints.

e-Patunay handles the complete document lifecycle from submission to verification. It supports secure data entry by barangay officials, review and approval by BCRD staff, generation of signed official records, tamper-proof activity logging, controlled user access, encrypted data storage, and external agency verification capabilities.

The system serves three user groups with defined access levels: Barangay officials and staff submit documents and check their own status. BCRD Personnel process submissions, apply official approvals, manage users within their districts, and generate reports. External agencies like COA, CSC and Ombudsman access read-only verification functions.

Cybersecurity protections over data confidentiality through encryption, access control with multi-factor authentication over data confidentiality through encryption, access control with multi-factor authentication and role separation, audit

accountability with verifiable logs, document authenticity through digital signatures, and threat mitigations addressing spoofing, tampering, repudiation, disclosure, service disruption, and privilege escalation.

This study is conducted within defined academic, technical, and resource-based constraints. The design, development, implementation, and evaluation of e-Patunay are planned to be completed within the limited capstone project timeframe. As a result, the study does not cover city-wide development across all 142 barangays, continuous system operation beyond the pilot period, or extended performance observation past the academic timeline.

The web-based system is developed and tested using available team resources and academic infrastructure. Consequently, enterprise-level deployment scenarios serving the full Quezon City scope and integration with production municipal systems fall outside project boundaries. System evaluation will primarily be conducted within BCRD, representing typical operational conditions but not replicating all production-scale variables, with further evaluation in a controlled district 2 environment if BCRD's timeline permits.

From a cybersecurity perspective, security evaluation focuses on common and well-documented threat scenarios through structured testing and vulnerability assessment. The study covers simulated attack conditions within the pilot environment rather than continuous real-time security monitoring or operational security operations center functions. Advanced threat actors such as nation state attacks, zero-day exploits, or persistent threats remain beyond research boundaries.

The system employs established open standards and accessible security approaches rather than proprietary or highly specialized commercial solutions

requiring licensing. Findings apply specifically within the defined controlled testing environment, with applicability to broader organizational deployments subject to production-scale validation.

1.5 Significance of the Study

This study addresses the growing need to integrate cybersecurity principles into government system development, particularly in the context of digital record management and transactional governance systems. The research emphasizes the importance of embedding cybersecurity concepts throughout the system development lifecycle rather than treating security as a secondary or post-deployment concern. It demonstrates risk-aware development within Philippine e-governance contexts, focusing on cryptographic assurance rather than basic digitization.

Academically, this study contributes to the field of cybersecurity by demonstrating the application of PKI-Based trust models within a real-world governance context. The research illustrates how digital certificates and server side digital signatures can be used to establish custodial authenticity and non-repudiation for officially issued electronic documents, rather than relying solely on basic digitization methods.

Furthermore, the study offers insights into a hybrid trust model for government document attestation. It combines centralized server-side digital signing by authorized officials with the practical need to avoid large-scale client certificate distribution, contributing to discussions on suitable trust models for government-managed record systems.

Finally, the study may serve as a reference for future academic research related to cybersecurity, secure system development, and digital governance. It provides an example of how legal, technical and security considerations can be aligned within a project scope contributing to academic discourse on responsible and security-oriented system implementation.

From a practical perspective, the proposed e-Patunay system is expected to support the BCRD in achieving safer, more efficient digital operations by incorporating security mechanisms that protect user data, ensure document authenticity, and provide verifiable proof of system operations. The project directly addresses operational inefficiencies inherent in manual, paper-based processes including susceptibility to physical data loss, lack of auditability, and excessive storage requirements.

The system may assist the BCRD and similar local government agencies in understanding the importance of integrating cybersecurity principles into system development from inception, thereby improving system reliability, operational transparency, and user trust in digital platforms. If successfully implemented, the system may serve as a replicable proof-of-concept and reference model for other local government units and barangays facing similar challenges in transitioning from manual to secure digital records management.

1.6 Conceptual Framework

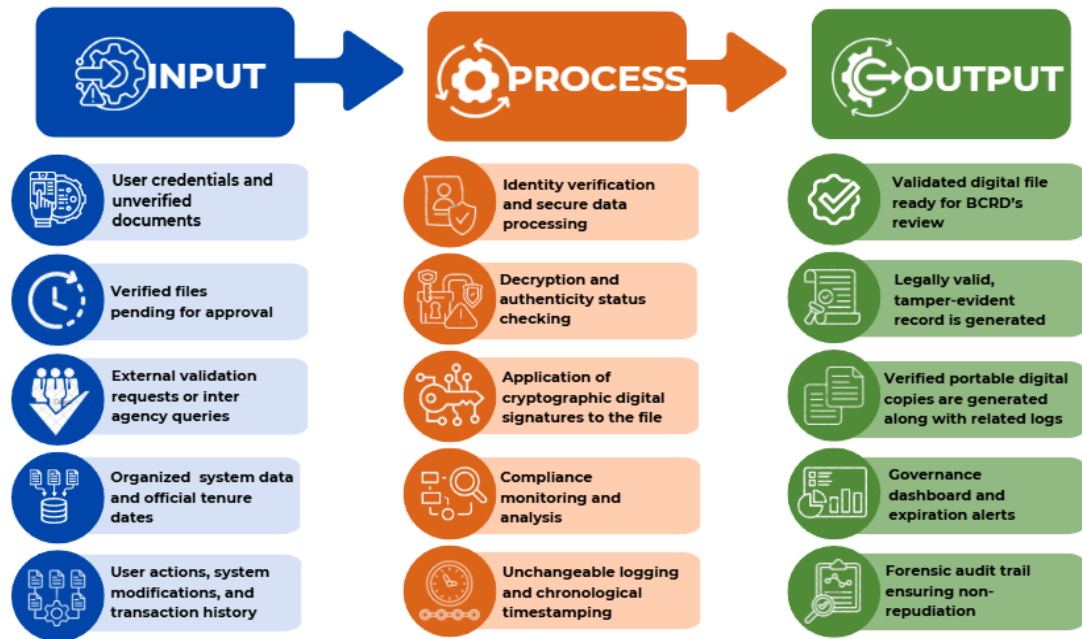


Figure 5. *Conceptual Framework of e-Patunay*

The conceptual framework illustrates how the proposed system, e-Patunay, operates by presenting the flow of information from when it is inputted, how the system processes it, and how it is outputted to the users. Cybersecurity is prioritized all throughout the system's flow.

The input component illustrates the raw data entering the system. These inputs range from unverified user credentials and documents submitted by barangay officials, and external validation queries from agencies like the COA. Additionally, the system ingests official tenure dates and real-time user actions, which serve as the raw data for compliance tracking and security auditing.

The process component represents how the system conceptually handles these inputs. The system validates data through identity verification and compliance monitoring. Importantly, this stage is when cryptographic digital signatures are bound

to documents and their authors to create an unchangeable log of every transaction history. This ensures that data is secured against tampering during its lifecycle.

The output component defines what is delivered to the stakeholders. The primary output is a legally valid, tamper-evident record that can be trusted by auditors. For external users, the system generates verified and portable digital copies that prove authenticity without requiring access to the database. Finally, the system reflects a governance dashboard for decision-makers and a forensic audit trail that facilitates non-repudiation, ensuring that every action can be traced back to a specific individual.

1.7 Cybersecurity Principles



Figure 6. *CIA Triad: e-Patunay's Theoretical Foundation*

The proposed study is grounded in established cybersecurity theories and principles that guide the secure design, development, and operation of government record management systems. These theories provide the conceptual foundation for

ensuring data authenticity, protecting sensitive information, and establishing verifiable accountability within local governance contexts. The CIA Triad serves as the primary theoretical foundation for this study, directly informing the system's security architecture and operational objectives:

Confidentiality asserts that sensitive information must be accessible only to authorized people. In the context of the Barangay Community Relations Department (BCRD), which processes personally identifiable information such as personal data sheets, financial disclosures, residency details, and employment records, confidentiality is necessary to prevent identity theft and privacy violations (NIST, 2024). For the proposed e-Patunay system, confidentiality conceptually guides the implementation of encryption mechanisms, authentication controls, and role-based access policies to protect data during storage, transmission, and processing.

Integrity emphasizes that data should remain accurate and protected from unauthorized modification. International information security standards highlight that integrity violations undermine trust in official records and may compromise their admissibility in legal proceedings. In the context of this study, integrity guides the need for the use of cryptographic hashing structures and immutable audit trails. This shifts verification from human trust to cryptographic certainty, where any modification to a record mathematically invalidates its integrity. In the context of this study, integrity supports the use of cryptographic techniques such as hashing, digital signatures, and tamper-evident audit records to detect and prevent unauthorized changes to submitted documents and system logs. (NIST, 2024)

Availability supports the system's operational requirement of maintaining continuous access to governance records for authorized users while protecting against

service disruptions. Conceptually, availability informs the e-Patunay system's need for a redundant and resilient cloud infrastructure capable of sustaining service continuity despite local physical disasters that would typically destroy paper-based archives.

Non-repudiation is the principle that ensures parties cannot deny their actions within the system. In the context of e-Patunay, this principle directly addresses the repudiation threats identified in the STRIDE model, where barangay officials might deny submitting, approving, or modifying documents. This theoretical foundation necessitates that to digitalize BCRD's accountability mechanisms, processes and actions should not be trust dependent, but cryptographically verifiable. This principle follows both internal governance requirements and external audit compliance.

The Zero Trust Model informs the system's approach to authentication and access control by challenging traditional security assumptions. Rather than granting implicit trust to users based on network location or initial authentication, Zero Trust requires continuous verification of identity and strict enforcement of least-privilege access. Each transaction requires fresh authentication verification, and access permissions are strictly limited to the minimum necessary for each user role whether barangay official, BCRD district officer, or external agency auditor. Defense-in-depth is a layered security approach that implements multiple security controls at different levels, ensuring that if one layer fails, others continue to provide protection. This principle guides the system's multi-layered security architecture.

The Department of Information and Communications Technology's National Cybersecurity Plan highlights that adopting a zero-trust architecture as a security policy for government information assets is a minimum standard. In the context of the

study, this theoretical framework necessitates that BCRD officials can only process documents within their assigned jurisdiction.

Secure-by-design emphasizes integrating security considerations throughout the system development lifecycle rather than applying security controls as an afterthought. This principle is particularly relevant for government systems where security failures can undermine public trust and violate legal obligations.

1.8 Definition of Terms

General Cybersecurity Terms

- **Access Control** – A security mechanism that restricts system or data access to authorized users or processes. In *e-Patunay*, access control is enforced with login authentication and multi-factor methods to ensure only legitimate users can submit or view records.
- **Audit Log** – A verifiable record of system or user activities, used for accountability. The proposed system uses audit logging so that every action (e.g., document approval) is logged in an unalterable way, providing “audit accountability with verifiable logs”.
- **Confidentiality** – The property that data is kept secret from unauthorized parties. In this study, data confidentiality is preserved through encryption (e.g., submitted records are encrypted) so that sensitive information (personal or financial) remains private.
- **Data Integrity** – Assurance that information is accurate and has not been altered. The system enforces data integrity by “digitally locking” each

submitted document to a specific user via cryptographic verification, so any unauthorized change to a record can be detected.

- **Denial-of-Service** – A threat where service availability is disrupted, e.g. flooding a system to slow operations. In the chapter's STRIDE analysis, Denial of Service is described as the risk that manual workflows can be overwhelmed during high-volume submissions.
- **Encryption** – The process of encoding data so only authorized parties can read it. *E-Patunay* uses encryption to protect the confidentiality of stored documents and credentials, ensuring that transmitted or stored records remain unreadable without proper keys.
- **Multi-Factor Authentication (MFA)** – A security control requiring users to provide two or more credentials (e.g. password + code) before granting access. The chapter notes the system uses multi-factor authentication as part of its access control strategy to strengthen user login security.
- **Non-Repudiation** – Assurance that a user cannot deny having performed an action. In *e-Patunay*, non-repudiation is achieved by digitally signing records; the system's logs and signatures provide proof of who submitted or approved a document.
- **Repudiation** - The risk or act of a user denying an action without proof. In the STRIDE Model it is identified as a threat (e.g. “an official denies submitting a document with no evidence to counter it).
- **Role Separation** – A design principle that different users have distinct permissions (e.g. submitter vs. approver). The system implements role

separation so that, for example, barangay officials can submit documents while only BCRD staff can approve them, reducing insider misuse.

- **Spoofing** – An attack where an adversary pretends to be another user. The chapter describes Spoofing as a threat: e.g. a malicious actor could “submit a document posing as a Barangay Official” if no strong identity check is in place.
- **Tampering** – Unauthorized alteration of data. The study notes existing records “are stored as physical files that can be replaced without leaving a trace,” making tampering possible. The new system uses cryptographic locks to prevent such undetected changes.
- **Vulnerability** – A weakness in a system that can be exploited by threats. The introduction explains that integration of multiple services creates many “vulnerability points” in the transaction lifecycle. *E-Patunay* addresses these by adding security controls to eliminate such vulnerabilities.

Technical Terms

- **Automated Governance Analytics** – Data-processing tools that generate insights on operations and compliance. In *e-Patunay*, automated analytics “process submission data to provide decision-makers with real-time visualizations” (e.g. barangay demographics charts), helping officials spot trends.
- **Chain-of-Custody** – The documented history of ownership or handling of a record. The system maintains chain-of-custody integrity for documents (from submission to archive) so that every handoff is tracked and tamper-evident.

- **Compliance Monitoring module** – Software that checks and reports on regulatory or policy compliance. *E-Patunay* includes a compliance module “that actively tracks document validity and official tenures,” alerting staff when certifications expire or rules change.
- **Cryptographic Verification** – Using cryptographic methods (e.g. hashes, signatures) to prove authenticity. The study implements cryptographic verification to bind each document to a specific user, thereby preventing unauthorized alterations and ensuring integrity.
- **Digital Certificate** – An electronic credential issued by a trusted authority to bind a public key with an identity. The chapter notes that digital certificates (as part of PKI) are used to establish the authenticity of electronically issued documents.
- **Digital Credential** – An electronic form of identification. The system generates portable digital credentials for officials (e.g. a verifiable ID card) that external agencies can use to independently verify status without accessing the main database.
- **Digital Signature** – A cryptographic signature attached to a document to prove origin and integrity. *E-Patunay* uses server-side digital signatures on official records, as noted in the significance section, so that any issued document can be verified as coming from an authorized official.
- **Document Management Platform** – Software for storing, organizing, and tracking documents. The project’s core is a “web-based document

management platform” that digitizes the Barangay records workflow, replacing manual file cabinets with a secure online system.

- **Tamper-proof Logging** – Immutable logging that cannot be altered once written. *E-Patunay* records all actions (e.g. approvals) in tamper-proof logs (such as blockchain or append-only databases) so that the history of each record is guaranteed unchangeable.
- **Web-based Governance System** – A browser-accessible application for managing and overseeing processes. The proposed solution is a web-based system for the BCRD (“web-based governance system”) that transforms paper workflows into an online platform with built-in security.

Project-Specific Terms

- **Barangay Official** – An elected leader of a barangay (e.g. barangay captain or staff). In the workflow, barangay officials prepare and submit their community’s certification documents to the BCRD.
- **Data Graveyard:** In this study, a data graveyard refers to a digital repository where documents are merely scanned and stored (e.g., as PDFs) without mechanisms for verification, integrity checking, or active use. Such repositories accumulate large volumes of data but cannot confirm authenticity, detect unauthorized changes, or support analytics, effectively rendering the stored records operationally and legally inert.

Compliance and Standards

- **Data Privacy Act (Philippines)** – Republic Act No. 10173, the Philippine law governing personal data protection. The chapter notes that improper handling of resident information (e.g. unencrypted emails) would violate the Data Privacy Act.
- **E-Governance Act (RA 12254)** – Philippine legislation promoting digital government. The study warns that using unverifiable manual systems violates this Act; with its enforcement, government offices (like the BCRD) must adopt systems (such as e-Patunay) that provide accountability.
- **Legal Standards** – Requirements set by law or regulation. The BCRD needs a digital system that “meets legal standards” (e.g. retention and audit requirements) and gives auditors “verifiable proof of record integrity” to comply with regulations.
- **National Archives Act (RA 9470)** – Philippine law governing public records retention. The chapter notes that failing to properly store records in the BCRD risks violating archival retention rules under the National Archives Act.
- **Open Standards** – Technical specifications (for data formats, protocols, etc.) that are publicly available and free to use. *E-Patunay* is built on open standards so that it avoids proprietary lock-in; the chapter specifies using “established open standards” rather than licensed commercial solutions.
- **Philippine e-Government Master Plan (2023–2028)** – A government blueprint for digital transformation. The background cites this Plan as a mandate requiring Local Government Units to “transition from paper-based workflows to digital operations,” aligning with the e-Patunay project’s goals.

- **Secure-by-Design** – A development principle where security is integrated from the start. The study emphasizes a “Secure-by-Design” approach (rather than adding security later), demonstrating that *e-Patunay* embeds strong controls (like encryption and digital signatures) throughout the system’s architecture.

Chapter 2

REVIEW OF RELATED LITERATURE

This section establishes the study's scholarly and professional foundation by examining existing theories, literature, industry standards, and prior studies related to the research topic. It consolidates relevant academic discourse, legal frameworks, and technical specifications that the study is grounded in existing knowledge while identifying the critical gaps that necessitate the system's development.

2.1 Related Literature

Mandates and Initiatives to Digitalize Governance

In the past years, the Philippine government is evidently working towards the modernization of public administration found in legislative and executive mandates designed to eliminate antique and inefficient manual systems. This motive grew stronger as it turned into a national focus on September 5, 2025 after the Philippine President Ferdinand “BongBong” Romualdez Marcos Jr. signed the E-Governance Act (RA 12254), pushing the nation's digital transformation. This institutionalizes the digital transformation of all branches in the government to a digitally empowered framework. This entails the transformation from isolated paper-based workflows into secure back-end coordination, efficient service delivery, and seamless inter-agency collaboration.

Furthermore, the DICT's E-Government Master Plan (EGMP) of 2023-2028 serves as the blueprint for executing this massive digital transformation. The EGMP necessitates that decentralized data production across municipalities are harmonized through a consolidated process architecture. In conjunction with the Data Privacy Act

of 2012 (RA 10173), which demands compliance to the principles of privacy-by-design, data proportionality, and data minimization, these mandates create a regulated legal environment. LGUs must balance the transparent sharing of open data required for inter-agency audits with the protection of PII contained within personal data sheets.

The mandate for secure digital transformation extends into the management of government personnel records, which is a core function of departments like the BCRD. The Philippine Civil Service Commission (CSC) has set policies to begin the modernization of human resource processes, moving agencies away from traditional record-keeping methods. Physical archives require repetitive manual tasks which lead to processing limitations, delayed inter-agency audits, and data loss due to physical degradation.

To streamline these administrative operations, the CSC wrote the 2025 Omnibus Rules on Appointments and Other Human Resource Actions (ORAOHRA). Of note, this regulatory framework is the legal acceptance of certified electronic and digital signatures on personnel documents. However, it notes that these digital signatures are only valid if they comply with the DICT's Philippine National Public Key Infrastructure (PNPKI) standards. Furthermore, the CSC's Integrated Records Management Office (IRMO) emphasizes that records management and retention tracking systems should be deployed and monitored. This policy shifts the standard of physical paper documents having inked signatures to cryptographically verifiable digital files, showing the need for local departments to deploy cryptographically-assured systems rather than basic digital filing cabinets.

Emerging Threat Landscape from Digitalization

However, from a digital shift emerges new threats. The centralization of LGU data without controls increases an organization's risk profile. According to the IBM X-Force Threat Intelligence Index (2025), identity-based attacks accounted for 30% of global network intrusions. The report also highlights an 84% yearly increase in infostealer malware delivered via phishing, emphasizing that attackers increasingly exploit valid user credentials. Furthermore, IBM (2025) also identifies AI-assisted threats, making traditional phishing more dangerous.

In the Philippines, local threat data aligns with these identity-focused trends. The Philippine National Police Anti-Cybercrime Group (PNP-ACG) reported that while overall cybercrime decreased in 2025, financially motivated scams remain prevalent. Illegal system access was reported for a total of 3368 times in 2023, being the second most reported online crime (PNP-ACG, 2023; DICT, 2023). PNP-ACG and DICT's threat landscape report and cyber-related crimes from 2023-2025 shows that without cryptographic binding of identity to the transaction, government portals can be flooded with fraudulent submissions (Spoofing) that overwhelm verification teams and dilute the integrity of public records.

BCRD's collection of sensitive personal data such as financial disclosures and residency certificates into centralized but unsecured data graveyards creates a high-value target for malicious actors. Recent reports on the Philippine threat landscape indicate a surge in data leak incidents involving government agencies, driven by collection of Personally Identifiable Information (PII) in systems that lack "defense-in-depth" architectures (Villanueva, 2025).

Compliance Compromise and Static System Limitations

Internal vulnerabilities further arm these external threats. BCRD states that their manual and semi-digital workflows often leveraged shared passwords, violating the principle of least privilege (Jibin, 2025). Currently, operational convenience supersedes security; staff may share credentials or access during peak volume periods to speed-up document processing (Conchas & Salapa, 2025). This practice violates non-repudiation, as administrative actions cannot be reliably traced to a specific individual during an audit. Systems lacking cryptographic binding and auditing are also susceptible to internal tampering, where privileged users can alter records and delete logs to hide modifications (Hicks, 2023).

Evidently, LGUs' traditional manual verification methods or weak digitalization efforts fail against sophisticated social engineering and AI generated fabrications (Jibin, 2025). The lack of data usability in conjunction with the centralization of data without a corresponding centralization of trust and security controls not only slows down LGU operations, but also increases the risk profile of the organization. In this study's context, a governance system must possess the capability to cryptographically attest to the origin and integrity of every submission, rendering AI-generated fakes detectable through signature verification failures.

Public Key Infrastructure as a Trust Architecture

To mitigate identity and integrity threats, public sector systems must implement cryptographic architectures of trust. The DICT and CSC mandates the user of the PNPKEI to establish secure online transactions. PNPKEI digital certificates utilize a 2048-bit key for identity verification and a 4096-bit for encryption, mathematically ensuring that any alteration invalidates the signature

Efforts of the DICT document that LGUs nationwide are being trained to utilize PKI. Some LGUs such as the Department of the Interior and Local Government in Central Luzon received training in 2022, and the Bangsamoro Autonomous Region in Muslim Mindanao in 2025. This shows the future of nationwide usage of PKI in establishing trust in digital governance.

PKI Models: Client-Side vs Server-Side Consideration

PKI implementation generally follows client-side or server-side architectures. Client-side signing requires users to manage private keys on physical hardware tokens, which introduces logistical barriers and usability issues (Ege et al., 2025; Tullis & Black., 2024). Server-side signing on the other hand centralizes cryptographic keys within a Hardware Security Module (HSM), shifting the technical burden away from the end-user while maintaining non-repudiation (Nutrient). The latter is a capability important to the general age group that transact with BCRD.

The implementation of cryptography in government systems must align with international standards. NIST's Special Publication 800-63-4 (2025) establishes phishing-resistant MFA as the baseline requirement for identity assurance. Additionally, NIST SP 800-57 instructs the functional separation of cryptographic keys to limit the impact of potential key compromises. In the context of PKI implementation in the system, the rules of implementation matters in ensuring the storage separation of the BCRD's private key and the public key.

Audit Integrity Mechanisms

While digital signatures protect individual documents, the integrity of system history and chronological user actions requires a different set of cryptographic

controls. Tamper-evident structures must be built to secure audit logging over a modifiable model.

The ISO/IEC 27001:2022 framework informs the standard practices for information security management. The use of cryptography requires organizations to define and apply cryptographic policies to protect data confidentiality and authenticity. Information deletion mandates the secure and verifiable deletion of data when it is no longer required, preventing the formation of data graveyards.

On the other hand, chronological integrity of system history requires a different set of controls. Standard relational databases are vulnerable to log manipulation by malicious administrators (Sadu, 2025). Hash chaining offers sequential binding of log entries using cryptographic hashes wherein altering one record results in invalidating all subsequent entries. Comparatively, Blockchain is frequently discussed for government purposes, offering the highest security of cryptographic trust. However, the application of Blockchain necessitates high performance requirements and complex mechanisms, as well as the decentralization of records (Rajuroy, 2025).

2.2 Related Studies

This section provides a comparative analysis of specific studies that have attempted to analyze and address similar problems in government record management. This analysis highlights the methodologies used, their strengths, and their limitations, further defining the specific gap that e-Patunay aims to fill.

Limitation of Static System Implementation

Romero (2021) finds that many databases are created primarily for basic case management and tallying rather than for analysis and utilization. These systems frequently rely on pre-written queries that facilitate gathering routing reports but offer limited utility for analysis or strategic planning.

This is further evident in the research of Agapito et al. (2025), where it is found that in Barangay Sto. Domingo, computers are merely used for encoding (typing up certifications and clearances) rather than managing a dynamic database. Despite the presence of digital tools, the data remains functionally static and difficult to leverage for broader governance goals. The lack of capabilities means that agencies like the BCRD cannot leverage their vast archives to visualize workforce trends or predict compliance lapses, effectively rendering the digitized data dormant.

Ironically, this static form of digitization often replicates the inefficiencies of manual filing, moving the problem from physical cabinets to digital files. Agapito identifies poor records retrieval as a persistent challenge in local governance, ranking as one of the highest operational difficulties encountered by officials.

The same is evident in Calansan's LGUs. Balangat (2025) reports that retrieval processes were rated as 'Somewhat Not Agree' in terms of effectiveness. The lack of proper indexing and searchability in digital systems means that retrieved records are not tracked or returned properly, contributing to data loss. Furthermore, existing digital storage options are often underutilized due to a lack of technical competence among staff, leaving valuable data stranded in inaccessible formats.

The Compliance Gap in Record Management

Due to the absence of a standardized records management policy, record-keeping tends to be inconsistent across different LGU departments. This lack of uniformity prevents the utilization of data for comprehensive analysis, creating isolated pockets of information that cannot serve the wider organization (Balangat, 2025). In the context of the BCRD, this prevents the seamless verification of records by external agencies (such as the Commission on Audit), making them rely on manual, trust-based validation methods that are prone to service delay.

This is consistent with the findings of Tabia (2025) about records management and data security in local government units. The study highlights that while LGUs have the capacity to manage large volumes of information, the mere conversion of physical files to digital formats does not equate to “digital governance”. Documents are often stored in file systems or basic databases without metadata integrity protection. Such systems are particularly susceptible to tampering and information disclosures, two core components of the STRIDE threat model. Unlike physical records, where alteration requires physical access and often leaves visible forensic traces, digital records in basic database systems can be modified via SQL injections or compromised administrative credentials without leaving an immediately obvious trail.

Tabia (2025) identifies that significant compliance gaps regarding the National Archives Act (RA 9470) and the Data Privacy Act of 2012 (RA 10173) are present. While LGUs are required to adhere to strict retention schedules and disposal protocols, manual or static-digital systems often lack the automated lifecycle management features necessary to enforce these rules.

Collectively, the assessment of these studies show that passive digitization of records is inadequate and legally insecure. To address the operational and security gaps identified by Romero, Agapito, Balangat, and Tabia, subsequent research has shifted focus toward developing concrete technological interventions. The succeeding studies evaluate various architectural implementations.

Implementation Attempts at Government System Modernization

Abubo et al. (2024) developed a blockchain based system designed to address the inefficiencies of manual entry and the security risks of unauthorized data modification. The system strengthened trust through decentralization. It also increased efficiency with the implementation of Optical Character Recognition (OCR) to digitize records automatically. However, the system introduced performance challenges due to the complexity and computational overhead that blockchain brings. Furthermore, the OCR component performed poorly in low-light conditions indicating hardware and environment dependency. In contrast, the proposed e-Patunay system focuses on a lighter web-based government system, foregoing the need for blockchain and utilizing only Merkle trees, encryption, and hashing algorithms to satisfy non-repudiation. This keeps the simplicity of centralization and the strength of immutability. Moreover, rather than physical document scans, e-Patunay proposes actual digital document entries, circumventing the weaknesses of OCR.

Aliling et al. (2024) focused on the transition of administrative functions from manual methods to a streamlined digital system to improve efficiency, accuracy, and accessibility of services and record management. The system demonstrated a successful transition to digital governance by speeding up transaction and service delivery. While the study achieved efficiency, the system operates as a standard

database application, entirely trusting on the database administrator and the application logic. e-Patunay improves upon this by adding the layer of cryptographic assurance that the study lacks.

Melendres & Aranda (2024) developed a web-based system to replace manual logbooks simplifying the storage and access processes of residents. The system features a centralized database, QR code authentication, and data visualization through heatmaps to aid decision-making. The system's analytic features moved data from simple storage to active utility. e-Patunay differs by going beyond user-security by also strengthening system and data security through cryptography.

Yongson (2024) explored the integration of the Philippine National Public Key Infrastructure (PNPKI) into digital services. The study aimed to create a security model that fortifies digital transactions with the CIA triad and non-repudiation. The system provided strong legal validity and high security to digital transactions. However, challenges arise from adoption as success depends heavily on user awareness and training; the complexity of PKI can be a barrier to most officials. e-Patunay addresses this adoption barrier by implementing Custodial-Attestation. In this framework, the submission official utilizes a standard electronic signature for identity declaration, while the cryptographic burden is shifted to the BCRD who acts as the custodian applying a digital signature to the record.

Overall, existing studies contribute valuable techniques for government record management, ranging from blockchain-based immutability and data analytics to national PKI integration. However, these implementations often address operational efficiency, forensic security, or user accessibility as isolated components. Few systems successfully integrate cryptographic non-repudiation, active decision support, and

usable custodial attestation within a single platform tailored for local government units. This gap supports the development of e-Patunay as a unified, legally compliant solution that balances the rigorous security of PKI with the operational agility required by the Barangay Community Relations Department.

2.3 Synthesis

On one side, there are functional digitization initiatives (Aliling et al., 2024) that successfully address the physical limitations of manual records, solving storage issues and speeding up retrieval, but fail to address forensic limitations. These systems cannot cryptographically prove who created a record or if it has been altered, leaving them vulnerable to the STRIDE threats of Spoofing, Tampering, and Repudiation.

On the other side, there are decentralized security initiatives (Abubo et al., 2024) that prioritize immutability above all else. While secure, they introduced complexity, cost, and privacy compliance issues that make them impractical for the resource-constrained and regulatory-heavy environment of a Local Government Unit.

There is a lack of systems that bridge these two implementations. There is no existing local study that implements a centralized cryptographic assurance system tailored for LGUs which uses the efficiency of a centralized web architecture but embeds the forensic security of PKI-based digital signatures and Merkle Tree Audit Logging to ensure legal admissibility and non-repudiation.

This study fills that gap. It rejects the singular choice between efficient but insecure and secure but impractical. By adopting a server-side signing architecture and implementing hash-chained audit logs, e-Patunay proposes a secure-by-design

framework. It mobilizes the mandates of the E-Governance Act by providing a system that is not just a digital filing cabinet, but a forensic utility for governance. The development e-Patunay is a necessary evolution toward cryptographic governance, aligning local operations with national cybersecurity standards and legal requirements. This provides a clear rationale for the methodology and system design presented in the succeeding chapter.

Chapter 3

METHODOLOGY

The methodological foundation and technical roadmap for the design, development, and assessment of the e-Patunay system are established in this section. It outlines the Agile Kanban model and developmental research approach used to systematically move the BCRD's manual workflows into a safe, cryptographically secure environment. This chapter offers a methodical approach to guarantee that the suggested solution is technically sound, complies with the law, and is feasible to implement by outlining the system's architectural designs, multi-layered security controls, and evaluation criteria, all of which are based on ISO/IEC 25010 and NIST standards.

3.1 Research Design

The study adopts a Developmental Research Design utilizing a Mixed-Methods approach. This selection is centered on the systematic creation and evaluation of a functional intervention to address a specific organizational challenge.

The study addresses the Barangay Community Relations Department's reliance on vulnerable manual workflows by developing a secure digital alternative. By moving beyond insecure digitization, the research employs an iterative design, implementation, and evaluation, using both quantitative and qualitative data gathering techniques to build a system that ensures data integrity, legal compliance, and accountability.

As such, a developmental research approach is appropriate for this study as it focuses on the systematic creation of a technical solution to solve a specific organizational challenge. This approach aligns with the study's objectives to transform manual record lifecycle processes into legally compliant digitized workflows while addressing identified cybersecurity risks. By utilizing a mixed-methods strategy, the research can objectively measure technical performance through quantitative metrics while simultaneously capturing human-centered insights through qualitative assessment, ensuring the system is both operationally proactive and resilient against modern cyber threats.

The system will be designed as a secure, web-based platform aimed at replacing manual paper-based document lifecycles with a cryptographically verifiable workflow. The system encompasses seven integrated security and operational components: a PKI-based digital signature mechanism using EJBCA to ensure document authenticity and non-repudiation; a tamper-resistant audit ledger employing hash chaining and Merkle proofs to create immutable transaction records; Zero Trust access controls through Keycloak enforcing multi-factor authentication and role-based authorization aligned with organizational hierarchies; compliance monitoring capabilities that actively track validity periods and generate automated expiration alerts; automated analytics that transform submission data into real-time governance intelligence through demographic and operational visualizations; storage management utilizing Microsoft Azure Blob archiving tiers to optimize costs while maintaining retention compliance; and an RBAC-enabled RESTful API enabling secure inter-agency verification without direct database access. With an emphasis on practical utility as the main metric of success, developmental research offers the

required roadmap to guarantee that the production of this system is based on accepted theories, such as the CIA Triad and Zero Trust Architecture.

Iterative cycles of design, testing, and refinement are used in this method of study to make sure that every architectural choice is supported by theory and validated against threat models from the real world. This strategy offers an organized and substantiated way to develop a field-tested tool for safe digital governance.

3.2 Requirements Analysis

This section describes the rules and needs used to build and protect the e-Patunay system. It explains the functional, non-functional, and security requirements that guide the project, showing how the system is designed to be safe, reliable, and legally compliant. Overall, this section focuses on what the system must be able to achieve, rather than the exact coding or technical steps used to build it.

3.2.1. Functional Requirements for the e-Patunay System

This section defines the functional requirements for e-Patunay, transforming the Barangay Community Relations Department (BCRD) workflow into a secure, verifiable ecosystem compliant with the E-Governance Act (RA 12254). The system requirements are mapped to mitigate STRIDE threats and achieve the Specific Objectives defined in Chapter 1

3.2.1.1. Identity and Access Management

The system's security architecture shall be implemented by an Identity and Access Management (IAM) framework designed to guarantee accountability and data integrity. Central to this architecture, the system shall implement centralized authentication, requiring a unique identity for every user session to maintain a clear

audit trail. To protect capability functions, Multi-Factor Authentication (MFA) shall be used as a critical secondary layer of verification for all administrative and district-level accounts. Furthermore, in alignment with Zero Trust principles, the system shall enforce role-based access controls (RBAC) to strictly limit user access to data within specifically assigned geographic or administrative districts, effectively minimizing the risk of unauthorized data exposure.

3.2.1.2. Cryptographic Assurance

The system's Cryptographic Assurance framework shall be designed to establish a high level of trust and non-repudiation for all processed documents. To achieve this, the system shall implement Automated Credentialing to ensure that every user is assigned a unique digital identity at the time of account creation, enabling a smooth transition to Digital Signing capabilities. The system shall allow authorized officials to apply server-side digital signatures to documents to guarantee authenticity, thereby eliminating the requirement for complex client-side hardware. To preserve the integrity of these records over time, the system shall utilize Long-Term Validation (LTV) protocols. These protocols shall embed necessary timestamps and revocation data, ensuring that all documents remain verifiable and legally valid even after the original signing certificates have expired.

3.2.1.3. Storage and Data Management

To maintain data integrity and protection, the system's Storage and Data Management framework shall prioritize both security and operational efficiency. A core security requirement shall be Data Encryption, which necessitates that all information stored at rest is encrypted to prevent unauthorized access and mitigate risks associated with physical or cloud-based data breaches. Parallel to these security

measures, the system shall implement automated Lifecycle Management to optimize resource allocation. This process shall ensure that records are dynamically transitioned between active, referenced, and archive storage tiers based on their age and access frequency, maintaining system performance while ensuring long-term data availability.

3.2.1.4. Audit and Integrity

To ensure high-assurance tracking and accountability, the system's Audit and Integrity framework shall be structured to guarantee total data reliability and permanence. The system shall implement Immutable Logging by recording all operations in an audit log where each entry shall be mathematically linked to its previous entry. This cryptographic chaining structure shall prevent retroactive tampering, as any unauthorized modification shall render all subsequent connections invalid. Furthermore, the Integrity Verification system shall allow daily transaction logs to be periodically validated against a read-only transparency anchor. This procedure shall guarantee that the system's operational history remains consistent and verifiable, providing a provable audit trail for both internal management and community-level transparency.

3.2.1.5. Interoperability & Security Quality

To facilitate secure interoperability and continuous validation, the system shall provide a Secure External Interface designed for standardized, programmatic verification by external agencies. This interface shall allow these entities to confirm the authenticity and status of documents while strictly prohibiting direct access to the internal database to maintain core system security and data isolation. Additionally, the system shall ensure Security Testing Compatibility by including a configurable testing

environment. This dedicated environment shall be specifically designed to facilitate automated security assessments and formal security testing conducted by authorized audit entities, ensuring the system's defenses remain verified against potential vulnerabilities.

3.2.1.6. Compliance Monitoring Module

The system shall incorporate a Compliance Monitoring Module designed to ensure compliance on established administrative protocols and regulatory requirements. Central to this module, the Real-Time Procedural Audit feature shall perform a real-time comparison of every document issuance transaction against the procedural steps defined in the BCRD Citizen's Charter to identify and log any deviations. To prevent unauthorized processing, the system shall implement Prerequisite Validation, which shall automatically flag and log any attempts to issue a certification that fails to meet mandatory requirements. To maintain flexibility, an administrative Compliance Rule Library shall provide a mechanism to update these rules in response to policy changes without requiring core code modifications. Furthermore, the system shall enforce security maintenance through Automated Term Revocation, which shall automatically remove administrative access credentials once a user's formal term is officially concluded. Finally, the system shall issue Proactive Expiration Alerts, sending automated notifications to authorized staff 30 days prior to the expiration of critical certifications, credentials, or official Fidelity Bonds.

3.2.1.7. Automated Governance Analytics

Serving as the system's "intelligence layer," the Automated Governance Analytics (AGA) module shall be designed to convert raw submission data into actionable, real-time insights for decision-makers. The system shall implement

Real-Time Service Metrics to automatically aggregate certification data, providing a clear view of service volume categorized by type and geographic district. To optimize operations, the module shall perform statistical analysis for Bottleneck Identification, highlighting districts experiencing significant administrative delays. Furthermore, the system shall utilize text mining for Qualitative Trend Analysis to identify recurring themes in citizen complaints, while simultaneously providing Demographic Visualizations via a real-time dashboard based on submission metadata. To ensure regulatory alignment, the system shall calculate and display a Compliance Maturity Score for each barangay, visualizing their adherence to the E-Governance Act and BCRD policies. Finally, the AGA module shall support long-term sustainability through Predictive Resource Allocation, forecasting future demand for certifications, and Data Lifecycle Forecasting, which shall visualize storage growth to identify records ready for transition to the archive tier.

Table 1. *Summary of the Functional Requirements of e-Patunay*

Functional Area	Requirement Description	Supported Objective
Identity and Access Management	The system shall implement a centralized authentication mechanism for every session , requiring multi-factor authentication (MFA) for privileged accounts , and enforce Role-Based Access Control (RBAC) with strict district-level data isolation.	Mitigate Spoofing and Elevation of Privilege while enforcing Zero Trust (Specific Obj 5).
Cryptographic Assurance	The system shall facilitate a secure server-side digital signing process , ensure signatures remain verifiable long-term after certificate expiration , and automatically issue unique digital credentials to users upon account creation.	Address tampering and ensure non-repudiation (Specific Obj 2).
Storage Optimization	The system shall encrypt all data at rest and implement an automated lifecycle policy to transition records between storage tiers (active, referenced, and archive) based on document age and access patterns.	Resolve "Data Graveyard" issues and ensure data-at-rest security (Specific Obj 4 & 5).

Immutable Ledger	Audit	The system will keep an audit log with linked entries to guard against manipulation and a way to check integrity against a read-only transparency foundation.	Mitigate Repudiation and ensure mathematically detectable integrity (Specific Obj 2).
Interoperability & Security Quality		The system shall provide a secure, standardized interface for external document status verification and include a configurable testing mode to facilitate automated security and security assessments.	Facilitate external audits and ensure system robustness (Specific Obj 6).
Compliance Monitoring module		The system shall perform real-time procedural audits against the BCRD Citizen's Charter, automatically revoke access upon tenure expiration, and flag unauthorized or incomplete transactions	Ensure regulatory adherence, automate accountability, and mitigate administrative negligence (Specific Obj 2, 3, 5, 6).
Automated Governance Analytics		The system will use classify complaints, anticipate resource requirements using historical trend analysis, and offer real-time dashboards for service metrics and demographics.	Foster data-driven decision-making, identify bottlenecks, and enable predictive governance (Specific Obj 7).

3.2.2 Non-Functional Requirements

The e-Patunay system is engineered to provide data access to ensure the BCRD workflow remains efficient. To support continuous document processing, the platform is built for availability and reliability, minimizing any technical downtime. Furthermore, the system prioritizes user-centric design, allowing BCRD staff and barangay officials to manage complex cryptographic security features with user-friendly and minimal technical training.

Furthermore, the e-Patunay system is designed with conceptual scalability, ensuring it can accommodate additional barangays or larger datasets in the future without compromising processing speeds. These non-functional requirements serve as a foundation to ensure the platform remains stable and responsive. By prioritizing

these attributes, the study guarantees that the system is not only effective for current users but also resilient under standard operational conditions within the project's scope.

Table 2. *Non-Functional Requirements of e-Patunay*

Quality Attribute	Requirement Description	Requirement Description
Performance	The system shall maintain low latency and high response times under concurrent loads.	Prevents operational bottlenecks during peak PDS submission periods across 142 barangays.
Reliability	The system shall implement automated validation and data integrity checks to ensure record accuracy.	Mitigates human-factor risks and "data graveyard" tendencies among non-technical staff.
Usability	The system shall provide a standardized, web-based interface for external auditing access.	Ensures bodies like COA and CSC can verify records independent of manual BCRD retrieval.
Scalability	The architecture shall utilize elastic cloud resources to accommodate increasing data volumes.	Complies with RA 9470 by handling cumulative document growth without performance degradation.

3.2.3. Security Requirements

Security is necessary because simple document digitization (like saving files as PDFs) does not ensure authenticity, integrity, or accountability, creating a “data graveyard” of vulnerable records. Without proper cybersecurity controls, records can be manipulated and trust is lost. Therefore, the system shall implement strict security requirements and cryptographic protections to ensure data integrity, authenticity, and non-repudiation, in line with the Philippine e-Government Master Plan 2023–2028.

The security requirements of the system shall be guided by Philippine laws on data privacy, e-governance, and records management. The Data Privacy Act of 2012

mandates the implementation of appropriate technical and organizational safeguards, including the NPC-recommended AES-256 encryption for data protection. The E-Governance Act further requires government systems to ensure secure data storage and transmission, while the National Archives Act emphasizes the protection and preservation of vital public records to support continuity of operations.

Table 3. *Security Requirements of e-Patunay*

Regulatory Framework	Focus Area	Impact on Security Requirements
RA 12254 (E-Governance Act)	Digital Transformation	Mandates secure interoperability and centralized security oversight via DICT.
RA 10173 (Data Privacy Act)	Data Protection	Requires mandatory Privacy Impact Assessments and AES-256 encryption.
RA 9470 (National Archives Act)	Record Lifecycle	Implements automated record retention and protects archived data stored offsite.
RA 8792 (E-Commerce Act)	Legal Validity	Provides the legal basis for digital signatures to replace wet-ink signatures.

This table highlights the major laws that will shape the system’s security requirements, covering data protection, secure system integration, records management, and the legal use of digital signatures to ensure compliance and system reliability.

The system is grounded in the Confidentiality, Integrity, and Availability (CIA) Triad, with the strategic elevation of Non-Repudiation to a primary design objective. Confidentiality ensures that sensitive records like PDS are accessible only to authorized personnel, preventing privacy violations and identity theft. Integrity ensures that data remains accurate and protected from unauthorized modification,

shifting the burden of verification from human trust to cryptographic certainty. Availability guarantees that the system and its records remain accessible despite potential local disasters, supported by redundant cloud infrastructure.

Non-repudiation is critical for legal defensibility in government operations. It ensures that an official who approves a document or a barangay staff member who submits a record cannot later deny the occurrence of that action. This is conceptually framed by the Zero Trust Architecture, which rejects the assumption that internal network users are inherently safe. By adhering to the principles of "Verify Explicitly," "Least Privilege," and "Assume Breach," the E-Patunay system minimizes the risk posed by insider threats and compromised credentials.

The first and most critical security requirement for the E-Patunay system is the implementation of a robust authentication framework that ensures every action can be attributed to a verified identity. The system must enforce strict authentication requirements prior to granting access to any departmental functionality or record repository. This is achieved by integrating Keycloak, an open-source identity and access management solution that supports modern authentication protocols such as OAuth 2.0 and OpenID Connect.

The system shall implement a centralized identity management framework to ensure every action is attributed to a verified identity. To eliminate the risks associated with shared credentials, the system shall require Multi-Factor Authentication (MFA) for all users with administrative or approval authority, requiring at least two independent credential categories.

Furthermore, the system shall enforce a granular Role-Based Access Control (RBAC) model mapped to the BCRD organizational hierarchy. Access to records

must be restricted based on the "need-to-know" principle, ensuring that staff can only view or manage data within their specific geographic or functional jurisdiction

3.2.3.1 Mapping Roles to the BCRD Organizational Hierarchy

The authorization module must be configured to ensure that District I staff, for instance, are restricted to viewing and managing records from the specific barangays under their jurisdiction, while staff in the Administrative Division's Personnel Section are restricted to HR-related records. Access to sensitive PDS documents should be limited to senior directors and the Legal Unit for compliance oversight.

Table 4. *Mapping Roles to the BCRD Organizational Hierarchy*

User Role	Permissions and Access Scope	Security Objective Supported
Barangay Official	Submission of records for their specific barangay only.	Least Privilege, Accountability.
District Staff	Review and receipt of records for barangays in their district.	Need-to-Know, Confidentiality.
Legal Unit	Read-only access to all records for compliance audits.	Oversight, Integrity.
System Admin	Infrastructure management; no access to document content.	Separation of Duties, Confidentiality.

Keycloak's authorization services will be used to define resource-based and policy-based permissions that are embedded within the access tokens. This ensures that every request to the system's backend API is verified not just for validity but for the specific authorization claims required to perform the requested action.

To ensure the legal validity and non-repudiation of digital records, the system shall utilize a custodial-attestation model where a digital signature is applied to documents upon receipt using a protected, server-side private key. All issued

documents must support long-term validation to remain verifiable even after the initial signing certificates have expired.

All sensitive data, including personal identifiers and government IDs, must be encrypted at rest to prevent unauthorized disclosure in the event of a storage breach. Additionally, all data transmission between users and the platform must be secured via industry-standard encryption protocols to mitigate interception and "man-in-the-middle" attacks.

The system shall maintain a tamper-resistant audit ledger to ensure accountability and detect unauthorized modifications. Every security-relevant event, including authentication attempts, administrative actions, and API requests, must be recorded with a timestamp and actor identity. To prevent the deletion of logs by privileged insiders, the system shall link audit entries using cryptographic hash chaining, making any retroactive tampering mathematically detectable.

Accountability is a cornerstone of digital governance. The system shall record all security-relevant events to ensure that administrative actions are traceable and that anomalous behavior can be detected in real time. The system shall maintain comprehensive audit logs that include:

- Authentication Events: All successful and failed login attempts, multi-factor authentication challenges, and identity token issuances.
- Administrative Actions: Every instance of record creation, modification, or deletion, including the specific identity of the actor and a precise timestamp.
- System Access: All instances where internal staff or external agencies access sensitive document repositories.

- Interface Usage: All requests made to the inter-agency verification interface, including the unique service identifier of the requesting agency.

Because the BCRD handles documents with enduring archival value, the availability and long-term preservation of records are high-priority requirements. The system shall be designed for high availability, utilizing redundant infrastructure to ensure that local physical disasters do not disrupt government services

To manage a high volume of records efficiently, the system shall implement automated lifecycle management policies. These policies shall automatically transition records between different storage tiers such as active, referenced, and archival, based on the age of the document and its frequency of access to optimize storage costs while remaining compliant with national retention schedules.

The system shall satisfy requirements for vital records protection by performing regular backups to offsite locations and maintaining a disaster recovery plan that enables the resumption of operations after a catastrophic event. Redundant storage shall be configured to ensure that copies of the document repository and the audit ledger are maintained in separate geographic regions, protecting the data from wide-scale regional disasters.

Furthermore, to enable verifiable inter-agency collaboration, the system shall provide a secure, standardized interface that allows authorized external agencies to verify document status without requiring direct access to internal databases. To ensure these rigorous security requirements are met, the system shall undergo extensive security testing prior to deployment.

3.2.4. Feasibility of the study

This section examines whether the e-Patunay system and the related research activities are realistic and achievable within the scope of this study and the defined academic timeline. In line with the purpose of the requirements analysis presented in the previous sections, this feasibility assessment focuses on the viability of conducting and completing the study rather than on enterprise-level deployment or continuous city-wide operation. After defining the functional, non-functional, and security requirements in Sections 3.2.1 to 3.2.3, it is necessary to determine whether these requirements can be implemented, tested, and evaluated using the available time, resources, and academic setting. Therefore, the feasibility of the study is discussed in terms of technical, operational, economic, schedule, and legal and ethical aspects

3.2.4.1. Technical Feasibility

The e-Patunay system is technically feasible because the required technologies are available, compatible, and sufficient to meet the system's functional, non-functional, and security requirements.

Identity and Access Management: The system utilizes Keycloak as the foundation for Single Sign-On (SSO) and Multi-Factor Authentication (MFA). Compatibility with various client applications is ensured by its reliance on OpenID Connect and OAuth 2.0 protocols. The availability of thorough technical documentation and active community support for configuring security realms and roles further reduces implementation uncertainty.

Cryptographic Infrastructure: The use of EJBCA for PKI implementation is technically viable due to its platform-independent and scalable nature. The system employs a server-side signing model (custodial attestation), which centralizes

cryptographic operations. This approach simplifies logistical requirements for the 142 barangays by removing the need for physical hardware token distribution, thereby streamlining the user experience without compromising the integrity of the digital identity framework.

Application Development Stack: The system is built using the Laravel (PHP) framework for backend application development and API handling, paired with Vue.js for the user interface. These tools are commonly used in contemporary system development and are suitable for implementing the defined system features. Specialized libraries such as TCPDF and laravel-dompdf are integrated to support the required document generation and signing functionalities.

Cloud Infrastructure and Scalability: Integration with Microsoft Azure provides a managed environment for hosting and storage. Azure Blob Storage offers a tiered data management strategy, facilitating compliance with the National Archives Act (RA 9470) by allowing the migration of inactive records to cost-optimized storage tiers. The availability of initial cloud resources through the Azure for Students program ensures that the infrastructure can be provisioned and tested within current research constraints.

Data Integrity and Security: The system utilizes PostgreSQL for data storage and retrieval. Security is maintained through the application of AES-256 encryption for data at rest and TLS 1.3 for data in transit. Furthermore, the implementation of hash-chained audit logs and Merkle proofs provides the necessary technical capability to ensure verifiable document authenticity and non-repudiation.

Table 5. Hardware Requirements for e-Patunay

Environment	Hardware Minimum
Development	i5 / Ryzen 5, 16GB RAM, 256GB SSD
End-User	i3 / Dual-core, 4GB RAM

The hardware specifications demonstrate that it can be used in both academic and real-world contexts. In order to simultaneously manage the EJBCA PKI, Keycloak IAM, and Laravel backend during construction and security evaluation, a mid-range processor, 16GB of RAM, and a 256GB SSD are needed for the development phase.

Table 6: Software Requirements for e-Patunay

Environment	Software Minimum
Development	Windows 10/Linux, PHP, Node.js, PostgreSQL, EJBCA, Keycloak, Microsoft Azure
End-User	Updated Modern Web Browser with TLS 1.3 support

The software requirements detail the different technologies required for both the development of the system and its secure operation. The platform is built using the Laravel (PHP) framework, Vue.js, PostgreSQL, and Microsoft Azure cloud services. To support the identity management and cryptographic functions identified in the technical feasibility, the system integrates Keycloak and EJBCA.

For end-users, the web-based and cloud-hosted system requires a stable internet connection and a modern web browser that supports TLS 1.3. These requirements verify that the system is capable of functioning efficiently within both the development environment and the BCRD infrastructure.

3.2.4.2. Operational Feasibility

The study of e-Patunay is operationally feasible because it directly addresses the Barangay Community Relations Department's (BCRD) problems in manual document processing and record verification while fitting into their established organizational hierarchy. The system supports the same tasks currently done using paper logbooks and physical file cabinets, but in a more organized and secure way, making it practical for regular use during the study.

Table 7. Operational Feasibility of e-Patunay

Operational Aspect	Consideration	Assessment
Departmental Roles	Role-Based Access Control (RBAC) mapped to BCRD's actual organizational structure	Feasible
Daily Administrative Tasks	Covers document intake, review, approval, and automated archiving in one system	Feasible
Current Practice Fit	Replaces manual paper logbooks and unsecured emails with a secure web platform	Feasible
User Effort	Uses custodial attestation to shift cryptographic burdens away from non-technical staff	Feasible
Problem Solving Value	Reduces "data graveyards," unauthorized access, and physical record loss	Feasible
Use During Study	Can be used in pilot district operations (District II) for real-world evaluation	Feasible

The table further supports the discussion by showing how e-Patunay fits into the actual operations of the BCRD in several important ways. It first shows that the system is designed around the real roles found in the department, which means each

user only interacts with data relevant to their specific roles through the "principle of least privilege". Building on this, the table also shows that the system covers the most important daily activities, including document lifecycle management and compliance monitoring, which makes it useful in regular operations.

Additionally, the table demonstrates that the process is a good fit for the department's current procedures because, rather than significantly changing the way administrative services are delivered, it replaces manual, distributed, and vulnerable methods with a single secure platform. Regular use is supported by the system's use of simple custodial attestation, in which the system handles the cryptographic task rather than the user or staff, allowing them to complete their tasks without requiring extensive technical training. Lastly, the table demonstrates that e-Patunay is useful for evaluation and verifies that the system is operationally feasible within the conditions of this study since it can be utilized in actual BCRD operations during the pilot phase.

3.2.4.3. Economic Feasibility

The economic aspects of the study are feasible because the resources required to build, test, and evaluate the system are available within the academic setting and do not require large-scale enterprise funding. The study is designed to utilize open-source frameworks, student-specific cloud access, and existing school systems, making the project realistic to complete within the financial limits of a university research environment.

Table 8. *Economic Feasibility of e-Patunay*

Resource Category	Description	Use in the Study	Availability
System Platform	Laravel (PHP) and Vue.js	Main system development and UI	Free and open-source
Database	PostgreSQL	Storing records, appointments, and logs	Free and open-source
Cloud Service	Microsoft Azure	Hosting, storage, and demo deployment	Free tier / Azure for Students
Security & PKI	EJBCA and Keycloak	Identity management and digital signatures	Free and open-source
Development Tools	Code editors and documentation tools	Writing and testing the system	Free or school-provided
Testing Tools	OWASP ZAP, SQLMap, Nikto, etc.	Security validation and penetration testing	Free or academic-use tools
Computers	School and BCRD computers	Development and user testing	Already available
References	Legal mandates (RA 10173, RA 12254)	System design and documentation	Free or school access

The table demonstrates that the tools and resources needed to develop and test the system are already accessible or can be utilized for free within the academic ecosystem. By prioritizing open-source tools over proprietary alternatives, the study ensures the proposed digital solution remains economically feasible without requiring external funding for a research-led prototype.

Large-scale physical setup and ongoing paid services are not required because the system's goal is to develop and assess a functional system for academic use. The long-term financial burden of maintaining conventional filing systems is also addressed by the use of cold-tiering and scalable Azure Blob storage. All things considered, this demonstrates that the study can be finished within the financial

limitations while continuing to be useful and assessable throughout the research lifecycle.

3.2.4.4. Schedule Feasibility

The project is organized into three main stages that follow the school’s academic calendar. The first stage is dedicated to planning and research, where the focus is on understanding the requirements and designing a solid foundation for the system. The second stage moves into the building phase, where the actual digital platform is assembled and the core features are developed. The final stage focuses on testing the system to ensure it is secure and easy to use, while also gathering feedback from stakeholders to verify its effectiveness. This structured approach ensures the project moves smoothly from an initial concept to a fully tested and completed solution.

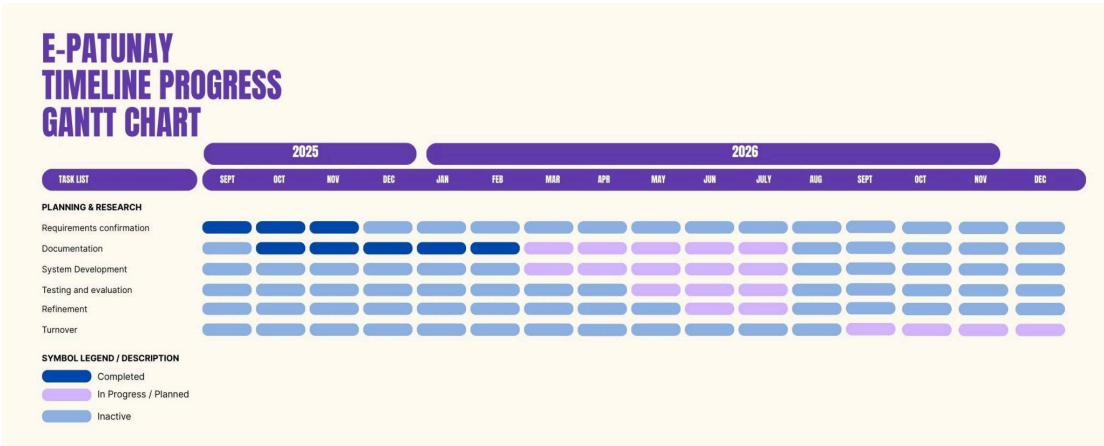


Figure 7. *e-Patunay System Gantt Chart Project Timeline*

The system is schedule-feasible because its development cycle is strategically aligned with the academic calendar, ensuring that all security-critical objectives are achievable within the set timeframe. The study reduces technical risk and avoids delays by breaking down complex features into manageable tasks. With this planned

approach, system quality will be continuously validated without interfering with the main implementation timeline. By the end of the research lifecycle, a fully tested, legally compliant governance solution shall be demonstrated and given to the BCRD because of its ability to prioritize essential functionality.

3.2.4.5 Security Feasibility

Table 9. *Security Feasibility Matrix of e-Patunay*

Feasibility Dimension	Challenge / Requirements	Technical Justification and Strategy	Feasibility Verdict
Access Control Implementation	Enforce identity verification and Role-Based Access Control (RBAC) for 142 barangays.	Implement Keycloak to manage SSO and MFA, enforcing the principle of least privilege across all administrative modules.	High
Document Integrity & Non-Repudiation	Prevent tampering of records and verify the origin of digital approvals.	Utilize EJBCA for PKI-based digital signatures and Merkle proofs to provide evidence of document authenticity and state.	High
Data Protection in Transit and at Rest	Protect sensitive data from interception or unauthorized disclosure.	Apply AES-256 encryption for data at rest within Azure SQL and TLS 1.3 for all data in transit between the client and cloud.	High
Auditability & Accountability	To maintain a traceable and tamper-resistant record of the system.	Implement hash-chained audit logs that link events chronologically, ensuring that logs themselves are protected.	High
Security Validation & Testing	Test the system against common web vulnerabilities and threats.	Perform authorized security testing using OWASP ZAP, sqlmap, Nikto, and Hydra to detect and fix injection, XSS, and authentication vulnerabilities .	Moderate-High

The security feasibility of the system shall be implemented through the use of hash-chaining for verifiable audit trails, Keycloak for centralized identity management, and EJBCA for cryptographic assurance. To make sure the platform is resilient to threats, security validation is carried out through security testing using tools like OWASP ZAP and sqlmap. In compliance with the Data Privacy Act (RA 10173) and the E-Governance Act (RA 12254), these controls directly support the safe handling of sensitive information. The system offers a practical framework for pilot testing in the BCRD environment by using these viable security modules.

3.2.4.6. Legal and Ethical Feasibility

The system's architecture integrates PKI-based digital signatures in accordance with the E-Governance Act (RA 12254), which promotes secure digital approvals and paperless transactions. By adhering to these technical standards, the e-Patunay system ensures that digital records maintain their integrity and meet the criteria for legal validity. Furthermore, the system is designed to facilitate compliance with the Data Privacy Act (RA 10173) through the implementation of encryption and Role-Based Access Control (RBAC), ensuring that Personally Identifiable Information (PII) is handled according to the principles of privacy-by-design.

Operational adherence to the National Archives Act (RA 9470) is managed through a lifecycle engine that incorporates official retention schedules. This prevents the indefinite storage of records and ensures data is disposed of in a manner consistent with privacy regulations. Ethically, the study follows established institutional research protocols by utilizing anonymized datasets for development and ensuring that any human-centered usability evaluations are conducted only after obtaining informed consent from participants.

Table 9. *Legal and Ethical Feasibility of e-Patunay*

Legal / Ethical Aspect		Assessment
Data Privacy (RA 10173)	Protects PII through lawful processing, encryption, and strict access controls.	Compliant
E-Governance Act (RA 12254)	Employs PKI-based digital signatures to ensure the legal validity of digital records.	Compliant
National Archives Act (RA 9470)	Automates archival and disposal per official schedules via a lifecycle management engine.	Compliant
Research Ethics	Uses anonymized data for testing and requires informed consent for all participants.	Compliant
System Design	Enforces data minimization and security-by-default through MFA and custodial attestation.	Enforcing security-by-default through mandatory Multi-Factor Authentication (MFA) and custodial attestation models.

The legal and ethical feasibility of e-Patunay is established through its alignment with national mandates and institutional research standards. By utilizing PKI-based signatures and lifecycle management, the system facilitates compliance with RA 12254 and RA 9470, ensuring digital records are legally valid and properly archived. Furthermore, the integration of encryption and RBAC enforces Data Privacy Act (RA 10173) principles. Ethically, the study protects sensitive information by using anonymized data and obtaining informed consent for all testing phases.

3.3 Project Development Model

This section describes the structured development process used to design, build, and evaluate the e-Patunay system. It explains the development model used to guide the project, showing how the study's system development is structured, systematic, and security-aware. This section establishes methodological discipline, not implementation detail.

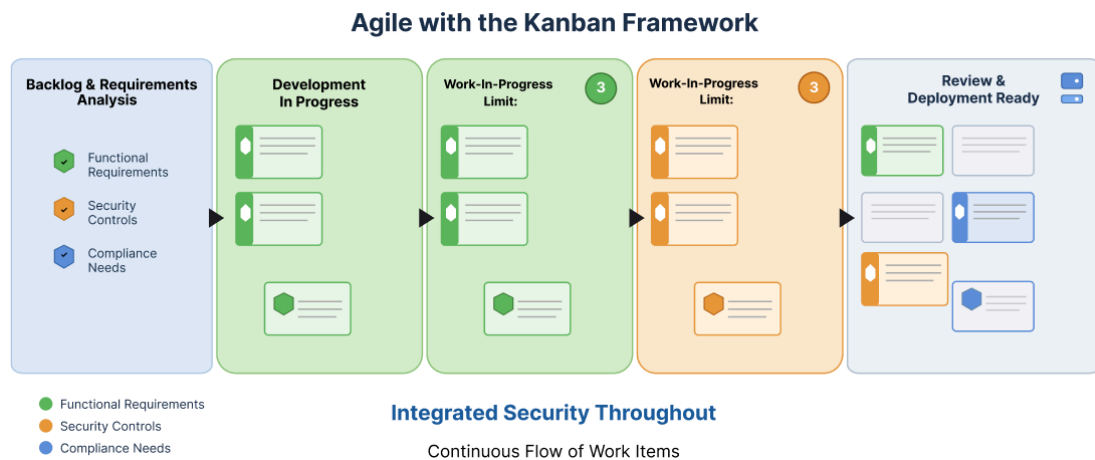


Figure 8. *Kanban Development Model for e-Patunay System*

This study uses an Agile Kanban flow to support continuous development and checking of both document workflows and strict security controls based on input from the BCRD and results from security testing. The BCRD currently relies on paper-based tracking and unverified emails, leaving them with limited control over data security. Because of this, the study requires a flexible process to continuously prioritize, build, test, and deploy both the system features and security tasks together, without rigid deadlines. The project backlog tracks all operational features and security requirements in one central list. Prioritization rules select a set of high-priority items to be pulled into the workflow, controlled by Work-In-Progress (WIP) limits that ensures both system features or security tasks are completed first

before new tasks are started. During design and development, continuous progress tracking on the Kanban board supports control of workflow bottlenecks and allows immediate handling of vulnerabilities found during the implementation of Two-Step Login (MFA), encryption handling, and hash chaining. Testing and evaluation cover both functionality and security (aligned with the STRIDE threat model) so that bugs, access issues, and policy gaps are captured as results that can be immediately corrected before moving to the next stage. Completed features are regularly presented to BCRD officials, and their feedback on the system's workflow and security is immediately added to the project backlog. Deployment and documentation support the study outputs by producing a secure, working build for validation and creating records needed for verification, such as test results, audit ledger logs, and security compliance evidence, with any deployment findings continuously returned to the backlog for ongoing refinement.

3.4 System Architecture and Design

This section explains the structural blueprint and visual models used to design the e-Patunay system. It looks at the project through various diagrams including the overall architecture, user interactions, data flow, database relationships, step-by-step activities, and network deployment to show how the system's different parts connect and work together.

3.4.1 Architecture Overview

The system is organized into five logically distinct layers that together enforce separation of concerns, support modularity, and embed cybersecurity controls at every stage of the document lifecycle. As illustrated in Figure 8, the architecture flows from the user-facing Presentation Layer through an Authentication and Access Control

Gate, an Application Logic Layer, a Cryptographic Services Layer, and finally a Data and Storage Layer. Each layer has a defined responsibility; no layer exposes its internal operations directly to another without passing through the controls of the intervening gate.

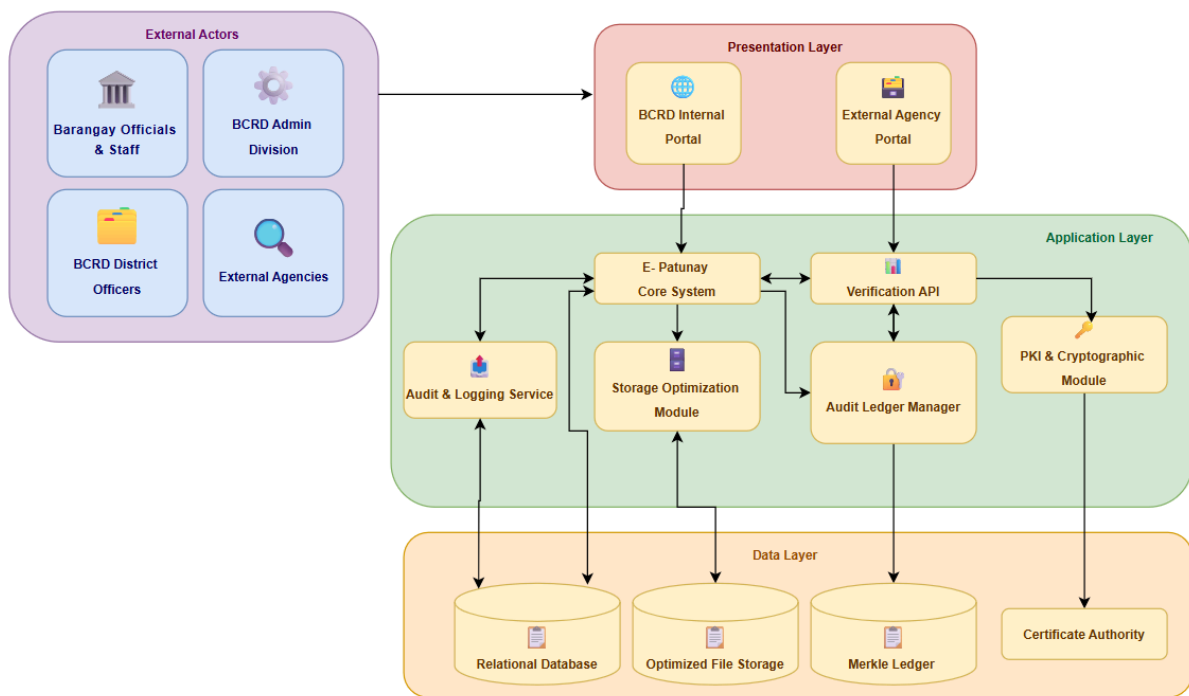


Figure 8. *e-Patunay System Architecture Diagram*

The Presentation Layer serves as the interface through which all three user groups – Barangay Officials, BCRD Personnel (including District Officers), and External Agency Verifiers interact with the system. This layer renders secure web pages, forms, and dashboards. It transmits all data over Transport Layer Security (TLS) 1.3, ensuring that no credentials or document payloads traverse the network in plaintext.

The Application Logic Layer coordinates all business workflows: document intake and validation, routing for review and approval, generation of governance

analytics, compliance monitoring and alert generation, and the production of verifiable portable credentials for external agency use. This layer does not store or sign documents directly; instead, it delegates those responsibilities to dedicated lower layers. This design principle ensures that application logic remains testable and auditable independently of cryptographic operations.

The Data and Storage Layer is responsible for persistent storage across three tiers: an active relational database for current records and compliance tracking, a warm archive for recently closed records, and a cold archive on Microsoft Azure Blob Storage tiered according to the retention schedules mandated by the National Archives Act (RA 9470). The separation of storage tiers directly addresses the Storage vulnerabilities identified in the Ishikawa analysis: it eliminates physical document loss risk, enforces automated lifecycle management, and reduces the long-term operational cost of maintaining records from all 142 barangays under the BCRD's jurisdiction.

Table 10 summarizes the responsibilities and primary security functions of each architectural layer.

Table 10. *e-Patunay Architectural Layers Responsibilities and Security Functions*

Layer	Primary Responsibility	Security Function
Presentation	Web interface for Barangay Officials, BCRD Personnel, and External Verifiers	TLS 1.3 encryption of all data in transit; secure session management
Auth & Access Control	Identity verification and role-based permission enforcement via Keycloak + MFA	Mitigates Spoofing and Elevation of Privilege (STRIDE)
Application Logic	Document workflow, compliance monitoring, analytics, credential generation	Enforces business rules; supports audit accountability via structured processing

Cryptographic Services	PKI/EJBCA signing, AES-256 encryption, hash-chaining, Merkle audit ledger	Mitigates Tampering, Repudiation, and Information Disclosure (STRIDE)
Data & Storage	Active DB, warm archive, Azure Blob cold archive with lifecycle tiering	Mitigates Denial of Service via redundancy; enforces RA 9470 retention compliance

3.4.2 Context Diagram

The context diagram defines the boundary of the system and identifies the external entities that interact with it. As illustrated in Figure 9, the system sits at the center of a network of actors and data flows.

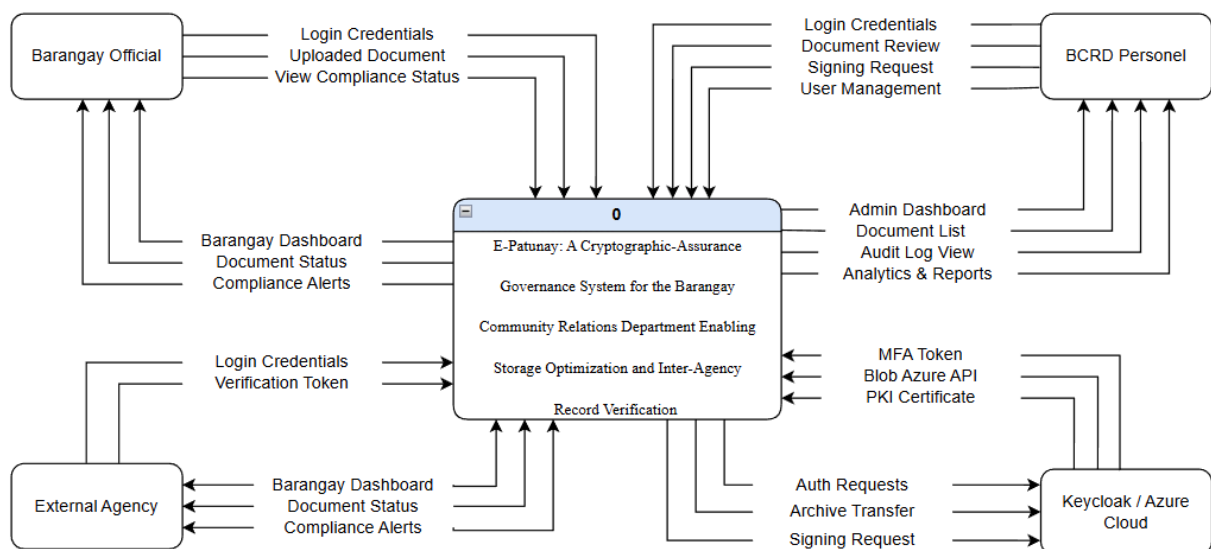


Figure 9. *e-Patunay Context Diagram*

Figure 9 illustrates the overall data interactions between the system and its external entities during routine document governance operations. The diagram presents the system as a single process (Process 0) and shows how Barangay Officials and Staff, BCRD District Officers, the BCRD Administrative Division, External Agencies, and Microsoft Azure exchange information with the system. This

high-level view defines the system boundary and identifies the primary sources and destinations of data without detailing internal processing logic.

The flow begins when Barangay Officials and Staff representing personnel from the 142 barangays across Quezon City's six districts submit inputs such as login credentials, Personal Data Sheets (PDS), Statements of Assets, Liabilities, and Net Worth (SALN), and document submission requests. In response, the system generates outputs including the Barangay Dashboard, document submission status, and compliance alerts. These exchanges represent the routine document submission and status-monitoring transactions initiated by barangay personnel.

BCRD District Officers interact with the system by providing document review actions, approval or rejection decisions, and signing requests scoped to their assigned district. The system supplies District Officers with document queues, signed document records, and audit log views that support review workflows and geographic accountability. This interaction reflects the role of the District Officer in evaluating and authenticating submitted records within their designated territorial boundary.

The BCRD Administrative Division provides system-wide operational inputs such as user account creation, role assignments, and configuration updates. In return, the system delivers the Admin Dashboard, governance analytics, visualization graphs, and system-wide audit logs. These data flows represent the administrative oversight required to manage personnel accounts, enforce access controls, and monitor compliance activity across all six districts.

External Agencies, representing oversight bodies interact with the system through a dedicated and isolated channel. These agencies submit verification tokens and document query requests through the Verification API. In response, the system

returns Merkle proof results and document validity statuses, providing cryptographic evidence of record authenticity without exposing underlying sensitive citizen data.

Microsoft Azure serves as the system's external cloud infrastructure provider and does not initiate interaction with the system. The system transmits outbound archive transfer requests and inactive document payloads to Azure Blob Storage through a secure Managed Identity connection, in compliance with the retention mandates of the National Archives Act (RA 9470). Azure receives these transfers and maintains cold-tier storage, ensuring long-term record availability beyond the capacity of on-premise infrastructure.

Overall, the context diagram presents a consolidated view of how e-Patunay serves as the central processing system that connects barangay personnel, district-level reviewers, administrative oversight, external verification bodies, and cloud archiving infrastructure. By mapping these external data exchanges, the diagram establishes the scope of system interactions and provides a foundation for the more detailed process, security, and deployment discussions presented in the succeeding sections of the study.

3.4.3 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) illustrates how data moves through the system from input to output, identifying the processes that transform data, the data stores that persist it, and the trust boundaries that define where cryptographic controls are applied. Figure 10 presents a Level-1 DFD annotated with trust boundaries to support security analysis.

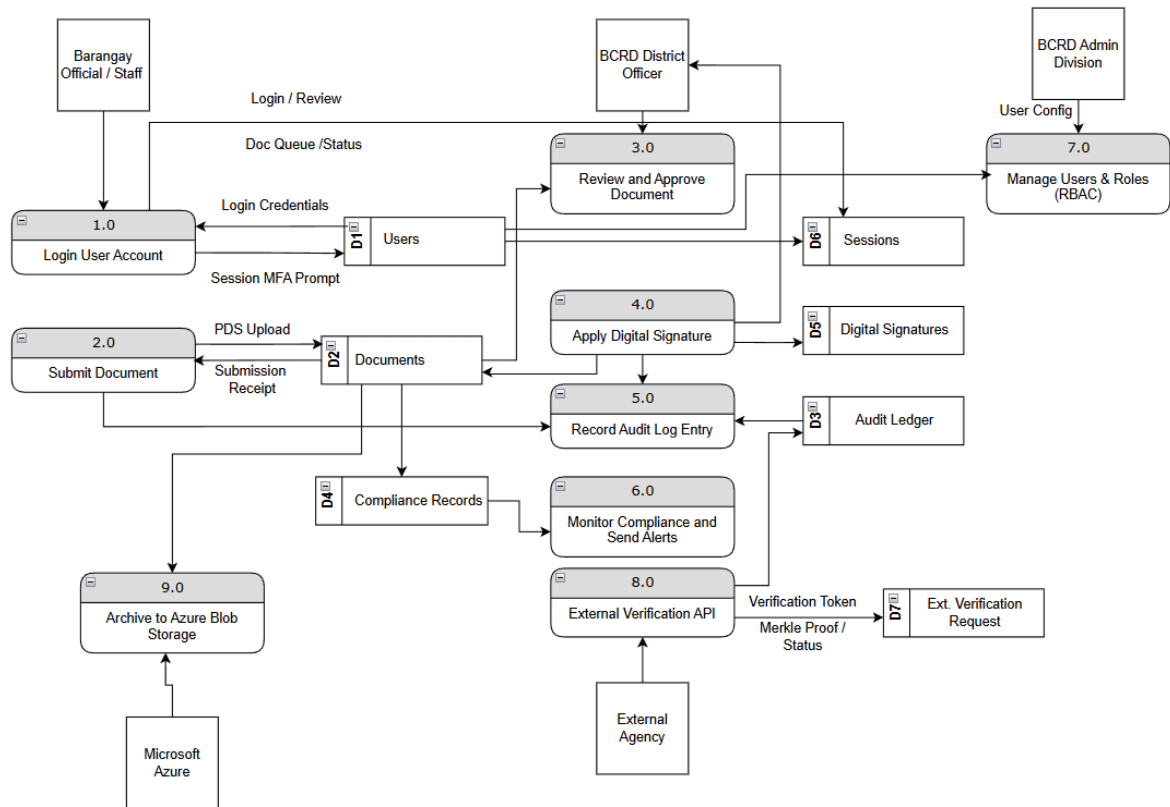


Figure 10. *e-Patunay Data Flow Diagram*

Figure 10 illustrates the high-level overview of how data flows from the external entities namely Barangay Officials and Staff, BCRD District Officers, the BCRD Administrative Division, External Agencies, and Microsoft Azure to the core system processes such as authentication, document submission, review and signing, audit logging, compliance monitoring, and external verification, and to the major data stores including Users, Documents, Audit Ledger, Digital Signatures, and Compliance Records.

3.4.4 Use Case Diagram

The Use Case Diagram defines the system interactions available to each actor and maps them to their required authorization level. Figure 11 illustrates the complete set of use cases organized by actors. This diagram supports the access control and role separation objectives of the system by making explicit which actions are permissible for each user type.



Figure 11. *e-Patunay Use Case Diagram*

The system recognizes three primary actor types, each associated with a distinct set of use cases. Barangay Officials and Staff are authorized to submit documents (PDS and SALN), monitor the processing status of their own submissions, and download their digitally signed approval records for their own records. Their access is strictly scoped to the records of their own barangay; district-level isolation prevents cross-district data exposure.

BCRD Personnel including District Officers and Administrative Division staff exercise a broader set of capabilities. District Officers receive, review, and approve or reject submitted documents within their assigned district. They can generate governance analytics reports and view compliance dashboards. The Administrative Division additionally manages user accounts, configures district-level access boundaries, and triggers archiving workflows. Importantly, the BCRD Personnel role is itself subdivided by district assignment, so a District II officer cannot access District IV records even after authentication. The enforcement of RBAC resolves the Access Control gap identified in the Ishikawa diagram under the category "Anyone in the office can access any file."

External Agency Verifiers access a single use case, verify document authenticity. This use case exposes only a cryptographic proof of authenticity and a confirmation of the signing officer's current employment status. No raw document content, personal data beyond what is necessary for verification, or system metadata is disclosed. The use case is intentionally narrow to implement the principle of least privilege for external actors and to comply with the Data Privacy Act of 2012 (RA 10173), which prohibits disclosure of personally identifiable information beyond the minimum necessary for the stated purpose.

Table 11. *e-Patunay Use Case Summary by Actor and RBAC Scope*

Actor	Available Use Cases	RBAC Scope
Barangay Official / Staff	Submit Document, Check Submission Status, Download Signed Record	Own barangay records only
BCRD District Officer	Review Document, Approve / Reject, Generate District Report, View Compliance Dashboard, Manage District Users	Assigned district only
BCRD Admin Division	All District Officer capabilities + Manage User Accounts, Configure RBAC, Trigger Archive Workflow	System-wide
External Agency Verifier (COA / CSC / Ombudsman)	Verify Document Authenticity (read-only via Verification API)	Cryptographic proof only; no raw data access

3.4.5 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) represents the logical structure of the system database, defining the entities managed by the system, their attributes, and the relationships between them. Figure 12 presents the complete ERD. The design prioritizes data integrity, controlled ownership, and auditability the three data-layer requirements that emerge most directly from the study's objectives.



Figure 12. e-Patunay Entity Relationship Diagram (ERD)

The core entities are: User, Barangay, District, Document, DocumentVersion, AuditLogEntry, DigitalSignature, ComplianceRecord, and ExternalVerificationRequest.

The User entity captures the system account of each registered actor. It stores a hashed credential reference (the plaintext password is never stored), the user's role assignment, their associated Barangay or District, and their PKI Certificate ID linking to the Cryptographic Services Layer. The relationship between User and Barangay is many-to-one: multiple staff members may represent a single barangay, but each user is associated with exactly one barangay.

The Document entity is the central entity of the data model. Each document record holds a reference to its submitting user, its current approval status, its encrypted file path in the Data Layer, a SHA-256 hash of the original file computed at submission time, and a foreign key reference to its associated DigitalSignature record (populated upon approval). The Document entity supports versioning through the DocumentVersion child entity, ensuring that the history of amendments to a document is preserved without overwriting the original submission.

The AuditLogEntry entity implements the tamper-resistant audit ledger. Each entry records the actor who performed an action, the action type, a timestamp, the hash of the affected document state, and a reference to the previous log entry's hash forming the hash chain that supports Merkle-proof verification. Because each entry includes the hash of its predecessor, any retrospective modification of a log entry invalidates all subsequent entries, making tampering immediately detectable. This structure directly implements the non-repudiation requirement identified in the study's Specific Objective 2.

The ComplianceRecord entity supports the compliance monitoring module described in the conceptual framework. It tracks the validity period of each submitted document type, the official tenure dates of the submitting barangay officer, and the automated alert schedule. External verification requests are captured in the ExternalVerificationRequest entity, which logs the requesting agency, the document identifier queried, the result returned, and the timestamp providing a complete audit trail of all external access events.

3.4.6 Network / Deployment Diagram

The Network and Deployment Diagram illustrates the physical and logical placement of system components across the network infrastructure, showing how the system's security architecture is realized in the deployment environment.

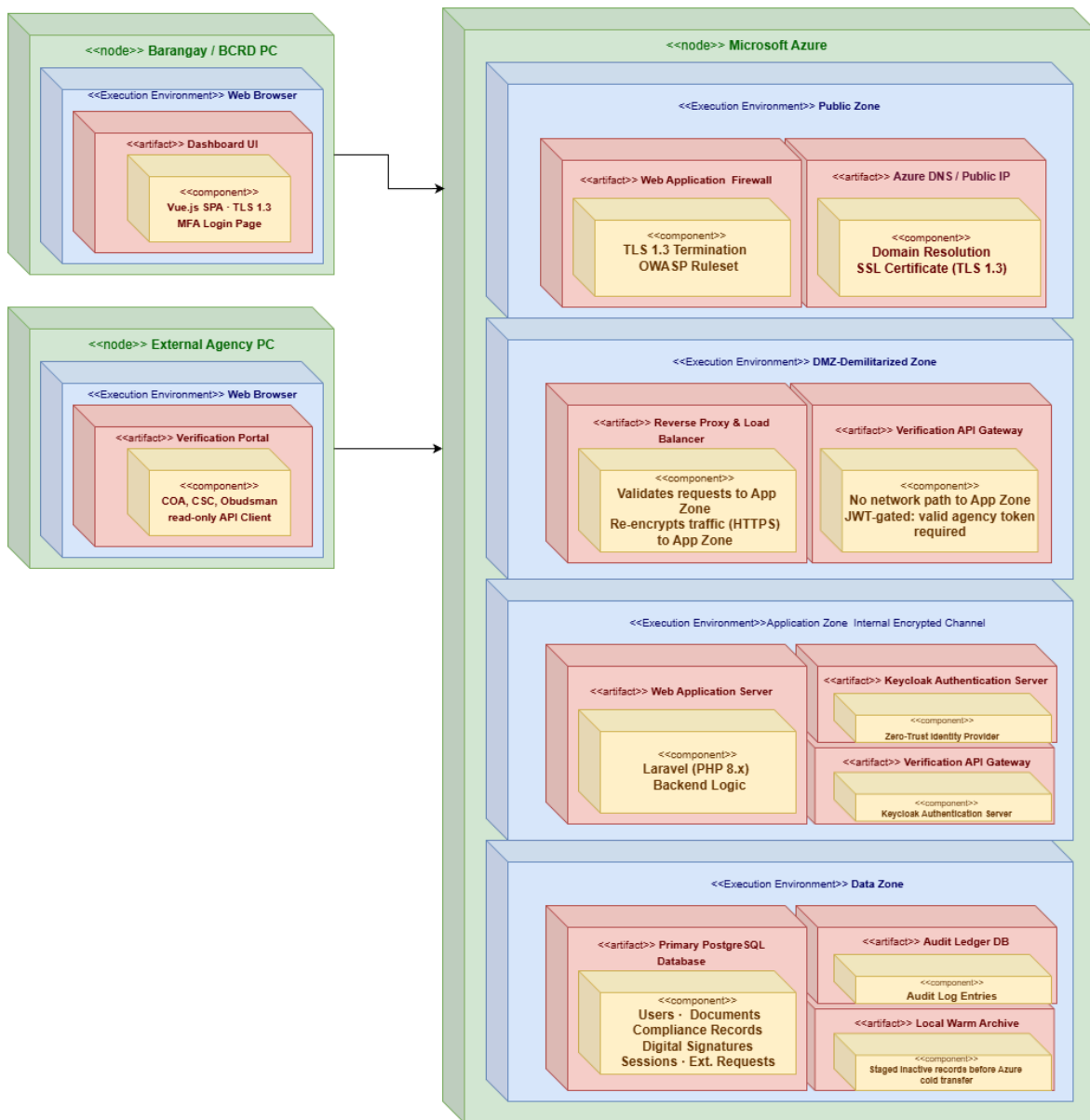


Figure 13. *e-Patunay Network and Deployment Diagram*

The deployment is organized across four network zones separated by firewall rules and access controls: the Public Zone, the DMZ (Demilitarized Zone), the Application Zone, and the Data Zone. This four-zone architecture implements the

defense-in-depth principle introduced in Section 1.7 and directly reduces the attack surface identified in the Ishikawa analysis's Technology category.

In the Public Zone, all three user groups access the system through a standard HTTPS connection over the public internet. Barangay Officials connect from district offices or barangay halls, BCRD Personnel connect from within the BCRD administrative network, and External Agency Verifiers connect from their respective institutions. All connections terminate at a Web Application Firewall (WAF) at the network perimeter, which filters malicious traffic and enforces TLS 1.3.

The DMZ hosts the Reverse Proxy and Load Balancer, which forwards validated requests to the Application Zone, and the Verification API Gateway, which handles all external agency verification requests in isolation from the main application. Placing the Verification API in the DMZ ensures that external actors never have a network path to the Application Zone or Data Zone, even if a misconfiguration were to occur.

The Application Zone contains the Web Application Server (hosting the system frontend and application logic), the Keycloak Authentication Server, and the Cryptographic Services Server (hosting the EJBCA CA and signing service). These three components communicate over an internal encrypted channel. The Application Zone is isolated from the Public Zone by a second firewall that permits only traffic forwarded by the Reverse Proxy.

The Data Zone is the most restricted segment of the network. It contains the Primary Relational Database, the Audit Ledger Database (a separate, append-only database instance that stores AuditLogEntry records), and the Local Warm Archive. A third firewall permits database connections only from the Application Zone servers,

and the Audit Ledger Database accepts insert operations only no update or delete commands are permitted at the database configuration level, providing a hardware-level enforcement of the tamper-resistant audit requirement.

The Microsoft Azure Blob Storage service sits outside all on-premise zones and communicates with the Data Zone through a secure API connection authenticated by an Azure Managed Identity. Cold-archive transfers are scheduled and logged; no interactive access path from the Application Zone to Azure bypasses this managed identity boundary.

Table 12. *e-Patunay Network Deployment Zones Components, Access Policy, and STRIDE Mitigations*

Zone	Components	Access Policy	STRIDE Threats Mitigated
Public Zone	User clients; WAF perimeter	HTTPS / TLS 1.3 enforced; WAF blocks malformed requests	Spoofing, Information Disclosure
DMZ	Reverse Proxy; Verification API Gateway	Forwards only authenticated traffic; external agency API isolated	Elevation of Privilege, Spoofing
Application Zone	Web App Server; Keycloak; EJBCA Cryptographic Server	Internal encrypted comms; no direct public access	Tampering, Repudiation
Data Zone	Primary DB; Append-only Audit Ledger DB; Local Warm Archive	DB connections from App Zone only; Ledger DB: insert-only	Tampering, Denial of Service
Cloud (Azure)	Azure Blob Storage (Cold Archive Tiers)	Managed Identity authentication; scheduled transfers only	Information Disclosure, Denial of Service

The network segmentation illustrated in Figure 15 ensures that a breach in any single zone does not automatically compromise adjacent zones. An attacker who

succeeds in injecting malicious traffic past the WAF in the Public Zone still faces the Reverse Proxy, the Keycloak authentication barrier, and the inner firewall before reaching any application logic. An attacker who compromises the Application Zone cannot access the Data Zone directly without traversing the database firewall. And the append-only policy on the Audit Ledger Database means that even an attacker with database write access in the Data Zone cannot retroactively alter the audit trail. This multi-layer containment strategy embodies the secure-by-design principle and provides the system with resilience against the range of threats identified in the STRIDE model.

Design Summary and Alignment with Objectives

The 5 diagrams presented in Section 3.4 collectively define the structural and behavioral design of e-Patunay. Each diagram has been developed in direct response to the root causes and STRIDE threats identified in Chapter 1. The architecture overview and deployment diagram implement defense-in-depth and zero-trust principles. The context and DFD diagrams make explicit the trust boundaries that determine where cryptographic controls are applied. The use case diagram implements least-privilege access by defining strict, role-differentiated capability sets. The ERD implements immutable audit accountability through the hash-chained AuditLogEntry entity.

Table 13 below summarizes the alignment between each design diagram, the Ishikawa root causes it addresses, the STRIDE threats it mitigates, and the specific objectives it supports.

Table 13. *Alignment Matrix Design Diagrams, Ishikawa Causes, STRIDE Threats, and Specific Objectives*

Diagram	Ishikawa Causes Addressed	STRIDE Threats Mitigated	Specific Objectives Supported
Architecture Overview	Technology (all), Access Control, Storage	All STRIDE categories	SO 1 Secure document management platform; SO 3 TLS & AES-256 encryption
Context Diagram	Verification, Access Control	Spoofing, Tampering	SO 5 RBAC-enabled RESTful verification API
DFD	Technology, Verification	Information Disclosure, Tampering	SO 2 Tamper-resistant audit ledger; SO 3 Encryption compliance
Use Case Diagram	People, Access Control	Spoofing, Elevation of Privilege	SO 3 Keycloak MFA and RBAC; SO 5 Role-specific API access
ERD	Technology, Storage	Tampering, Information Disclosure	SO 2 Hash-chaining audit ledger; SO 1 PKI-bound signatures
Network / Deployment Diagram	Technology, Storage, Access Control	DoS, Elevation of Privilege, Information Disclosure	SO 3 Encryption in transit; SO 4 Azure Blob archiving; SO 6 Pen test scope definition

3.5 Cybersecurity Measure

This section explains the defensive strategies and safety controls used to protect the system from digital threats. It looks at the project through two main areas. The specific security practices chosen and how these measures are actively applied within the system to show how the platform defends against unauthorized access and data tampering.

3.5.1 Security Practices

Multi-Factor Authentication (MFA) is enforced for all system users to prevent unauthorized access through compromised credentials. The system implements time-based one-time password (TOTP) authentication through Keycloak, requiring users to provide a knowledge factor (password) and a possession factor (an authenticator application code). This dual-factor architecture directly aligns with the identity assurance baselines established by NIST SP 800-63-4, addressing the Ishikawa root cause of "weak passwords are easy to guess or steal" and mitigating STRIDE spoofing threats where attackers attempt to impersonate legitimate users. MFA enforcement occurs at the authentication layer before any system functionality becomes accessible. Even if an attacker obtains a user's password through phishing, credential stuffing, or a database breach, they cannot authenticate without access to the user's physical device generating time-based codes. The system does not provide fallback mechanisms that would allow MFA bypass, ensuring consistent enforcement across all user sessions.

Role-Based Access Control (RBAC) implements the principle of least privilege by restricting user permissions based on their organizational role and assigned district. The system defines distinct roles including Barangay Official (document submitter), District Officer (document reviewer/approver), BCRD Administrator (system manager), and External Agency Verifier (read-only verification access). Each role receives only the permissions necessary to perform designated functions, preventing unauthorized actions even by authenticated users.

District-level access restrictions ensure geographic separation of responsibilities. District I officers can process only District I submissions, preventing

cross-district access that could enable unauthorized data viewing or manipulation. This access control addresses the Ishikawa root cause "anyone in the office can access any file" and mitigates STRIDE elevation of privilege threats where users attempt to exceed their authorized permissions.

Access control decisions are enforced server-side through Keycloak authorization policies integrated with the Laravel backend. Frontend UI elements may hide unauthorized functions for usability, but security enforcement occurs at the API and database layers to prevent bypass through direct HTTP requests or API manipulation.

3.5.1.1 Data Protection Measures

Encryption in Transit protects data confidentiality during transmission between users and the system. All HTTP traffic is enforced to use TLS 1.3, the latest and most secure version of the Transport Layer Security protocol. The Azure App Service hosting the e-Patunay application is configured to reject connections using older protocols (TLS 1.0, TLS 1.1, TLS 1.2) and weak cipher suites, preventing downgrade attacks that could expose transmitted data to interception. TLS 1.3 provides perfect forward secrecy, ensuring that even if long-term server keys are compromised in the future, previously captured encrypted traffic cannot be decrypted retroactively. Guided by NIST SP 800-57 for cryptographic key management and the OWASP Top 10 mitigations for cryptographic failures, this protection is critical for government records containing personally identifiable information subject to Data Privacy Act (RA 10173) requirements. The enforcement of HTTPS for all communications addresses the Ishikawa root cause "no encryption to protect sensitive data" and mitigates STRIDE information disclosure threats.

Encryption at Rest ensures that stored data remains confidential even if physical storage media or database files are compromised. The system utilizes Azure's native encryption services to protect data at multiple layers. Azure Database for PostgreSQL implements Transparent Data Encryption (TDE), automatically encrypting the entire database, log files, and backups using AES-256 encryption without requiring application-level changes. Azure Blob Storage, which stores digitally signed PDF documents, implements Storage Service Encryption (SSE) that encrypts all files at rest using AES-256 before persisting to disk.

These encryption mechanisms are managed by Azure's infrastructure, with encryption keys stored in Azure Key Vault under role-based access controls. This approach provides enterprise-grade key management without requiring the research team to implement custom cryptographic key storage, which could introduce vulnerabilities if implemented incorrectly. Encryption at rest compliance addresses Data Privacy Act requirements for protecting personal information and mitigates STRIDE information disclosure threats involving unauthorized database or file system access.

3.5.1.2 Network and System Protection

Secure Configuration Management ensures that all system components follow stringent security hardening best practices. To directly mitigate vulnerabilities highlighted in the OWASP Top 10, the Azure App Service hosting the Laravel application and Vue.js frontend is configured with strict security headers. These include HTTP Strict Transport Security (HSTS) to prevent protocol downgrade attacks, X-Frame-Options to prevent clickjacking, X-Content-Type-Options to prevent MIME-type sniffing attacks, and Content Security Policy (CSP) to mitigate

cross-site scripting (XSS) risks. PostgreSQL database access is restricted to the application server through Azure Virtual Network integration and firewall rules, preventing direct internet-facing database connections. Administrative access to database management interfaces requires VPN or Azure Bastion secure access, never direct public exposure. These configuration controls address the Ishikawa root cause "no cryptographic proof of document origin" by ensuring the infrastructure supporting cryptographic operations remains highly secure and impenetrable.

Input Validation and Output Encoding protect against injection attacks and cross-site scripting vulnerabilities. All user inputs received through web forms or API endpoints undergo server-side validation before processing. Laravel's built-in validation framework enforces data type constraints, length limits, format requirements, and whitelist-based acceptable value ranges. Inputs are validated against expected patterns using regular expressions where appropriate, and unexpected or malicious patterns are rejected before reaching business logic or database queries.

Database interactions exclusively use Laravel's Eloquent ORM or prepared statements with parameterized queries, preventing SQL injection by ensuring user inputs are never concatenated directly into SQL commands. Output encoding is applied when displaying user-generated content, converting potentially dangerous characters into safe HTML entities to prevent stored cross-site scripting attacks. These protections address STRIDE tampering and information disclosure threats involving malicious input designed to compromise system integrity or extract unauthorized data.

3.5.1.3 Monitoring and Logging

Comprehensive Audit Logging creates accountability and supports forensic

investigation by recording all security-relevant events in the cryptographic audit ledger. The system logs authentication events (successful and failed login attempts), authorization decisions (access granted or denied to specific resources), document lifecycle state transitions (submission, review, approval, rejection, signing), administrative actions (user creation, role assignment, configuration changes), and system errors or security exceptions.

Each audit log entry includes a timestamp, authenticated user identity, action performed, affected resource, and outcome (success or failure). Log entries are written to the append-only cryptographic audit ledger, which uses hash chaining to link each entry cryptographically to the previous entry, creating a tamper-evident sequence. Any attempt to modify or delete historical log entries would break the hash chain, making tampering immediately detectable through hash verification.

The cryptographic audit ledger addresses the Ishikawa root cause "no automatic record of who approved what" and mitigates STRIDE repudiation threats where users deny performing actions. By cryptographically linking authenticated user sessions to specific submissions through the audit log, the system establishes non-repudiation users cannot credibly deny submitting documents because the audit trail provides mathematical proof of their actions.

Merkle Tree Proofs enable efficient verification of audit log integrity at scale. Daily, the system constructs a Merkle tree from all audit log entries created during that period, computing a single Merkle root hash that represents the cryptographic summary of the entire day's logs. This Merkle root is stored and can be published or provided to external verifiers. To prove a specific audit entry exists in the log without revealing all entries, the system generates a Merkle inclusion proof, a small set of

hash values that, when combined with the target entry, reconstruct the published Merkle root.

External agencies accessing the Inter-Agency Verification Portal can request Merkle proofs to verify that specific actions (e.g., a BCRD official signing a particular document) are recorded in the tamper-evident audit log without requiring access to the entire database. This capability supports independent accountability and addresses the Ishikawa root cause "COA/Ombudsman must trust BCRD manually" by providing cryptographic verification rather than trust-based assertions.

3.5.2 Application of Cybersecurity Measures

Spoofing Mitigation protects against unauthorized users impersonating legitimate officials or system components. The combination of Keycloak-based Multi-Factor Authentication and electronic attestation workflows creates multiple verification points that attackers must bypass to successfully spoof identity. Barangay officials submitting documents must authenticate with both password and time-based OTP, then complete an electronic attestation ceremony that cryptographically binds their authenticated session to the submission through audit log entries. BCRD officers applying digital signatures must re-authenticate with TOTP immediately before signing, ensuring that even if a session is hijacked, the attacker cannot execute signing operations without physical access to the authenticator device.

Digital certificates issued through the EJBCA PKI are bound to specific BCRD officials through identity verification during certificate issuance. The certificate's embedded identity information and cryptographic validation prevent attackers from forging signatures or claiming documents were signed by officials who did not actually perform the action. This multi-layered spoofing defense addresses the

root cause "users share passwords with colleagues" by making password sharing insufficient for accessing protected functions.

Tampering Prevention ensures that unauthorized modifications to documents, database records, or audit logs are detectable and often preventable. Digital signatures applied to final PDF documents create cryptographic binding between document content and the signer's private key. Any modification to the signed document, even changing a single character, breaks the signature and causes verification failures, making tampering immediately evident to anyone validating the document.

Hash chaining in the audit ledger provides tamper-evidence for system logs. If an attacker attempts to modify or delete historical log entries, the hash chain breaks because the modified entry's hash no longer matches the subsequent entry's "previous hash" field. Periodic integrity validation tasks check the entire chain and alert administrators if tampering is detected. This addresses the root cause "no way to detect if documents were altered" and mitigates both document tampering and log tampering threats.

Azure's Transparent Data Encryption and Storage Service Encryption protect data at rest from tampering at the storage layer. Even if an attacker gains access to physical storage media or database files, the encryption prevents reading or modifying data without access to the encryption keys managed by Azure Key Vault. This defense-in-depth approach ensures that multiple barriers protect against tampering at different system layers.

Repudiation Prevention establishes cryptographic proof of user actions, eliminating credible deniability. The electronic attestation workflow requires barangay officials to explicitly affirm their identity and intent before document

submission. The system captures the authenticated user's session ID, user ID, and timestamp, then cryptographically logs this information in the tamper-evident audit ledger. This creates non-repudiable proof that a specific user submitted specific data at a specific time.

Digital signatures provide even stronger non-repudiation for BCRD officials. When an officer applies a digital signature, the cryptographic operation mathematically binds the signature to both the document content and the signer's private key. Because only the legitimate key holder can create a valid signature, and because the signature verification process is deterministic, the officer cannot later claim they did not sign the document. The combination of MFA-protected signature operations and audit logging of signing events creates comprehensive non-repudiation addressing the root cause "no way to prove who submitted a document."

Information Disclosure Prevention protects sensitive personally identifiable information from unauthorized access or leakage. TLS 1.3 encryption ensures that data transmitted between users and the system cannot be intercepted or read by network attackers. Even if an attacker captures network traffic, the strong encryption prevents extracting meaningful information.

Role-Based Access Control combined with district-level restrictions implements the principle of least privilege, ensuring users can access only data necessary for their job functions. District I officers cannot view District II submissions, preventing lateral information disclosure where authenticated users access data outside their authorization scope. This granular access control addresses the root cause "anyone in the office can access any file."

Azure's encryption at rest protects stored data from information disclosure through physical media theft or database file compromise. Even if an attacker extracts database files or blob storage contents, the AES-256 encryption renders the data unreadable without access to encryption keys secured in Azure Key Vault. This layered protection ensures that confidentiality is maintained even if individual security controls fail.

Denial of Service Mitigation ensures system availability for legitimate users despite potential attacks or resource exhaustion. API rate limiting prevents abusive clients from overwhelming the system with excessive requests. The system blocks IP addresses exceeding 100 requests per minute, protecting backend resources from exhaustion while allowing legitimate usage patterns.

Cloud-based Azure infrastructure provides elastic scalability, allowing the system to handle varying loads without manual intervention. During high-volume submission periods (e.g., before deadlines), Azure App Service and PostgreSQL can scale resources to maintain acceptable response times. Azure Blob Storage lifecycle management prevents storage exhaustion by automatically moving infrequently accessed documents to archive tiers, freeing hot storage for active use.

Performance testing using Apache JMeter validates that the system can handle expected operational loads without degradation. Load test results inform capacity planning and identify bottlenecks before they impact production availability. This proactive approach to availability addresses the root cause related to "142 barangays generating overwhelming document volume."

Elevation of Privilege Prevention ensures that users cannot exceed their authorized permissions even through technical exploitation. Server-side enforcement

of Role-Based Access Control prevents privilege escalation by validating every API request against the user's assigned role and district. Even if an attacker manipulates frontend code or crafts custom API requests, the backend authorization checks reject unauthorized actions.

JWT-based session management with cryptographic signatures prevents token tampering. If an attacker attempts to modify JWT claims (e.g., changing their role from "Barangay Official" to "Admin"), the signature verification fails and the request is rejected. Session tokens expire after 15 minutes, limiting the window of opportunity if a token is stolen.

Security testing specifically targets privilege escalation scenarios, attempting horizontal privilege escalation (accessing other users' data within the same role) and vertical privilege escalation (gaining administrative privileges from a regular user account). Testing validates that authorization controls function correctly under attack conditions. This systematic testing addresses the root cause "anyone in the office can access any file" by verifying technical enforcement of access restrictions.

3.5.2.1 Integration with System Objectives

Every cybersecurity measure implemented in e-Patunay directly supports specific research objectives defined in Chapter 1. Objective 1 (implement two-tier PKI using EJBCA) is fulfilled through the digital signature infrastructure described above, enabling legally recognized digital signatures for custodial document attestation. Objective 3 (implement tamper-proof audit ledger) is achieved through hash chaining and Merkle tree proofs providing verifiable data integrity and submitter non-repudiation. Objective 4 (establish the IAM framework with MFA and RBAC) is realized through Keycloak integration, enforcing least privilege and district-level

access restrictions. Objective 5 (implement data confidentiality through encryption) is accomplished via TLS 1.3 and Azure AES-256 encryption protecting data in transit and at rest per RA 10173. Objective 7 (deliver inter-agency verification portal) is enabled by digital signatures and Merkle proofs allowing cryptographic verification without BCRD manual intervention.

This explicit mapping from threat to control to objective demonstrates that the system's security architecture is purpose-driven rather than arbitrarily selected. Each control addresses validated risks identified through systematic analysis, ensuring that security investments provide measurable risk reduction aligned with research goals.

3.6 Tools and Technologies

This section documents the tools, technologies, and platforms used in the development and security evaluation of the e-Patunay system. It focuses on identifying the technical components utilized during system implementation and testing to support transparency and reproducibility. This section lists the tools used and does not provide justification or detailed explanations of design decisions, which are discussed in earlier sections of this chapter.

3.6.1 Development Tools

The development tools used for the system support the creation of a web-based application and its associated data management functions within the scope of this study. These tools are commonly used in contemporary system development and are suitable for implementing the system features defined in earlier sections.

Table 14. *Development Tools for e-Patunay*

Category	Tool/Technology	Purpose
Backend Framework	Laravel (PHP)	Backend application development and API handling
Frontend Framework	Vue.js	User interface development
Database Management System	PostgreSQL	Data storage and retrieval
PDF Generation	laravel-dompdf	Generation of PDF documents
PDF Signing	TCPDF	Application of digital signatures to PDF files
Cloud Platform	Microsoft Azure	System hosting environment
Cloud Storage	Azure Blob Storage	Storage of uploaded and generated files.

Laravel (PHP) is used to support server-side application logic and request handling. Vue.js is utilized for the development of the client-side user interface, enabling user interaction with the system during testing and evaluation. PostgreSQL serves as the database management system for storing system data generated throughout the study. Additional libraries are used to support document generation and signing functionalities required for system operation. Microsoft Azure provides the cloud-based environment used to host the application. Azure Blob Storage is utilized for storing system-generated records and uploaded files.

Collectively, these development tools provide a suitable technical foundation for implementing and evaluating the e-Patunay system in alignment with the scope and objectives of the study.

3.6.2 Security and Cryptographic Tools

The security and cryptographic tools listed in this section support identity management, encryption, and secure communication within the system. These tools are used to enable controlled security-related functions during system development and evaluation.

Table 15. *Security and Cryptographic Tools for e-Patunay*

Category	Tool/Technologies	Purpose
PKI Infrastructure	Enterprise Java Beans Certificate Authority (EJBCA)	Certificate issuance and management
Identity Management	Keycloak	Authentication and access control
Cryptographic Functions	PHP Native Crypto	Cryptographic hash generation
Transport Security	TLS 1.3	Encryption of data in transit
Storage Encryption	AES-256	Encryption of data at rest

EJBCA is used to support public key infrastructure functions related to digital certificates. Keycloak is utilized for identity and access management during system use and evaluation. Native cryptographic functions provided by PHP are used for hashing operations within the system. Standard encryption protocols are applied to protect data during transmission and storage throughout system testing. These tools collectively support secure handling of system data in accordance with the requirements identified earlier in the study.

3.6.3 Security Testing and Penetration Tools

This section details the software used to evaluate the defenses of the system. To ensure the platform can withstand real world attacks and meet its security

objectives, the testing process directly aligns with the STRIDE threat model. Each tool is specifically chosen to validate a corresponding vulnerability, making sure the system is fully protected before it is officially deployed

Table 16. *Security Testing and Penetration Tools for e-Patunay*

Testing Category	Tool	Purpose
Authentication Testing	Hydra	Network logon cracker used to perform brute-force and dictionary attacks to validate MFA and mitigate Spoofing
SQL Injection Testing	SQLMap	Automated testing of database interaction points to validate defenses against database tampering .
Vulnerability Scanning	OWASP ZAP	Scans for web weaknesses and verifies that system activity logs cannot be deleted, mitigating Repudiation .
Server Security Scanning	Nikto	Checks the web server for misconfigurations and hidden files to prevent Information Disclosure .
Load and Stress Testing	Apache JMeter	Simulates heavy user traffic to ensure the platform remains stable and resists Denial of Service attacks.
Penetration Testing	Burp Suite Community Edition	Intercepts web traffic to test access controls and prevent standard users from gaining administrative permissions, mitigating Elevation of Privilege .

These tools are utilized during the Testing and Validation stage of the Kanban workflow. For example, THC Hydra and Burp Suite ensure that user identities and access limits work properly. Meanwhile, SQLMap and Nikto verify that the database and server environments protect sensitive government records from unauthorized

viewing or modification. By using this targeted testing approach, the research team can systematically prove that the e Patunay platform successfully defends against the core threats identified in the study.

3.7 Testing and Evaluation

This section explains the methods used to check if the *e-Patunay* system works correctly and meets its goals. It looks at the project through three main areas. The step-by-step testing methods, the standards used to measure its success, and how the final scores and feedback will be calculated to show that the system is secure, user-friendly, and ready for real-world use.

3.7.1 Testing Methodologies

Testing methodologies describe the types of testing performed to verify system functionality, performance, security, and usability. The e-Patunay system undergoes multiple testing phases to ensure that each component operates correctly in isolation, that integrated components interact properly, that the complete system meets end-to-end requirements, and that security controls effectively mitigate identified threats.

3.7.1.1 Unit Testing

Unit testing verifies the correctness of individual system components and functions in isolation. Each module is tested independently to ensure it performs its intended task before integration with other components. For the e-Patunay system, unit testing focuses on core functional units including:

Document Hash Computation: Tests verify that SHA-256 hash functions produce consistent, deterministic outputs for identical inputs and that hash values change completely when even a single byte of input data is modified.

AES-256 Encryption/Decryption: Tests confirm that encryption produces ciphertext that cannot be read without the proper key, that decryption correctly restores original plaintext, and that encrypted data differs completely between encryption operations even with identical plaintext.

Digital Signature Generation: Tests validate that signatures are mathematically bound to both the signed data and the signer's private key, that signature verification correctly identifies valid signatures, and that modified data produces verification failures.

Role-Based Access Control Logic: Tests ensure that permission checks correctly grant or deny access based on user roles and resource ownership, that geographic district restrictions are properly enforced, and that unauthorized access attempts are consistently blocked.

Audit Log Hash Chain: Tests verify that each audit entry correctly links to the previous entry's hash, that hash chain computation produces consistent results, and that tampering with historical entries breaks the chain in detectable ways.

3.7.1.2 Integration Testing

Integration testing evaluates interactions between system components to confirm that data and control flows operate correctly when modules are combined. This testing phase verifies that independently tested units work together properly and

that interfaces between components function as designed. Integration testing for e-Patunay examines:

Application-Keycloak Authentication Flow: Tests verify that login requests are correctly forwarded to Keycloak, that JWT tokens are properly validated before granting access, that token expiration is enforced, and that authentication failures produce appropriate error responses.

Application-Database-Blob Storage Workflow: Tests confirm that document metadata is correctly stored in PostgreSQL with references to encrypted files in Azure Blob Storage, that retrieval operations successfully fetch both metadata and file content, and that referential integrity is maintained.

Application-EJBCA Certificate Issuance: Tests validate that certificate requests are properly authenticated, that issued certificates contain correct subject information, that signing operations use valid certificates, and that certificate validation correctly identifies expired or revoked certificates.

Frontend-Backend API Communication: Tests ensure that API requests include proper authentication headers, that responses conform to expected formats, that error conditions are handled gracefully, and that secure HTTPS communication is enforced.

3.7.1.3 System Testing

System testing validates the overall behavior of the e-Patunay system as a complete unit, ensuring that functional and non-functional requirements are met under normal operating conditions. This testing phase executes complete workflows from

end to end, simulating real-world usage scenarios by actual users. System testing scenarios include:

Complete Document Submission Workflow: Tests execute the entire process from barangay official login through document upload, encryption, storage, and submission confirmation, verifying that each step completes successfully and that proper notifications are sent.

Complete Approval Workflow: Tests perform the full approval process from district officer login through document review, integrity verification, approval decision, digital signature generation, and certificate creation, ensuring the complete chain of custody is maintained.

External Verification API Workflow: Tests simulate external agency requests to verify document authenticity, confirming that API authentication works correctly, that verification responses include proper cryptographic proofs, and that no unauthorized data is exposed.

Governance Dashboard Generation: Tests validate that analytical queries correctly aggregate submission data, that visualizations accurately reflect database state, that compliance alerts trigger at appropriate thresholds, and that dashboard access is restricted to authorized personnel.

3.7.1.4 Performance Testing

Performance testing assesses the system's response time, throughput, and efficiency under normal operational load. This testing ensures that cryptographic operations, database queries, and file storage operations complete within acceptable

time limits to provide a responsive user experience. Performance metrics measured include:

Document Upload Time: Measures time from file selection to upload confirmation, including hash computation, encryption, and blob storage transfer for documents of varying sizes.

Approval Processing Time: Measures time from approval button click to confirmation display, including database updates, certificate retrieval from EJBCA, digital signature generation, and audit log creation.

Verification API Response Time: Measures time from external API request to response delivery, including token validation, database queries, and cryptographic proof construction.

Dashboard Query Performance: Measures time required to execute analytical queries and render visualization components, ensuring governance dashboards remain responsive even with large datasets.

3.7.1.5 Load and Stress Testing

Load and stress testing evaluates system behavior under increased activity levels and peak usage scenarios. Testing simulates multiple concurrent users submitting documents, reviewing submissions, and accessing the system simultaneously to identify performance degradation, resource bottlenecks, and stability issues. Load testing scenarios include:

Concurrent Document Submissions: Simulates multiple barangay officials uploading documents simultaneously, measuring system throughput, response time degradation, and error rates under load.

Simultaneous Approval Operations: Tests system behavior when multiple district officers process approvals concurrently, verifying that database transaction isolation prevents conflicts and that EJBCA certificate issuance scales appropriately.

Peak Verification API Load: Simulates high-volume verification requests from external agencies, testing API rate limiting, connection pooling, and response consistency under stress.

3.7.1.6 Vulnerability Scanning

Vulnerability scanning identifies known security weaknesses in system components, dependencies, and configurations. Automated scanning tools detect common vulnerabilities including outdated libraries, misconfigurations, missing security headers, and exposure of sensitive information. Vulnerability scanning covers:

Application Code Scanning: Static analysis tools examine source code for common security flaws such as SQL injection vulnerabilities, cross-site scripting opportunities, insecure cryptographic implementations, and improper input validation.

Dependency Vulnerability Assessment: Tools identify known vulnerabilities in third-party libraries and frameworks, comparing installed versions against CVE databases to detect components with published security issues.

Infrastructure Configuration Review: Scanners check Azure service configurations for security weaknesses including overly permissive network rules, weak encryption settings, missing access controls, and deviation from security baselines.

3.7.1.7 Security Testing

Security testing simulates real-world cyberattacks in a controlled environment to evaluate the system's resistance to identified threats. Testing focuses on validating the effectiveness of implemented security controls by attempting to bypass authentication mechanisms, escalate privileges, tamper with data, and exploit system vulnerabilities. Security testing activities target threats identified in the STRIDE threat model presented in Chapter 1:

Spoofing Attacks: Tests attempt to impersonate legitimate users through credential guessing, token manipulation, session hijacking, and bypassing multi-factor authentication to verify that identity verification mechanisms properly prevent unauthorized access.

Tampering Attempts: Tests attempt to modify stored documents, alter approval records, manipulate audit logs, and break hash chains to confirm that integrity verification mechanisms detect unauthorized modifications.

Repudiation Testing: Tests verify that digital signatures provide non-repudiable proof of actions by attempting to deny approval decisions, forge signatures, or create ambiguity about document origins.

Information Disclosure: Tests attempt to access sensitive data through SQL injection, path traversal, API parameter manipulation, and unauthorized database access to verify that confidentiality controls prevent data leakage.

Denial of Service: Tests attempt to disrupt system availability through resource exhaustion, application-layer flooding, and algorithmic complexity attacks to evaluate service resilience and rate limiting effectiveness.

Elevation of Privilege: Tests attempt to escalate user privileges through RBAC bypass, administrative interface exploitation, and unauthorized function access to confirm that authorization controls properly restrict actions based on user roles.

3.7.1.8 Access Control Testing

Access control testing verifies that role-based permissions are correctly enforced throughout the system and that geographic access restrictions properly limit data visibility based on district assignments. Testing confirms that users can only access resources appropriate to their role and that unauthorized access attempts are consistently blocked and logged. Access control test scenarios include:

Role-Based Permission Enforcement: Tests verify that barangay officials can only submit documents and view their own submissions, that district officers can only review documents from their assigned districts, that administrative staff have appropriate management capabilities, and that external agencies have read-only verification access.

Geographic Access Restrictions: Tests confirm that District I officers cannot access District II documents, that approval attempts on out-of-district submissions are blocked, and that cross-district data leakage does not occur through API endpoints or database queries.

Authorization Failure Logging: Tests verify that unauthorized access attempts generate appropriate audit log entries capturing user identity, attempted action, timestamp, and denial reason for security monitoring and forensic investigation.

3.7.1.9 Logging and Audit Testing

Logging and audit testing validates that accountability mechanisms comprehensively record system activities and that tamper-evident audit ledgers maintain integrity over time. Testing confirms that all security-relevant events are logged with sufficient detail for forensic analysis and that hash-chained audit entries detect any attempts to modify historical records. Audit testing activities include:

Audit Log Completeness: Tests verify that authentication attempts, document submissions, approval decisions, access control failures, configuration changes, and administrative actions all generate corresponding audit entries with complete contextual information.

Hash Chain Integrity: Tests confirm that each audit entry correctly links to the previous entry through cryptographic hashing, that sequential hash validation succeeds for unmodified logs, and that tampering with historical entries breaks the chain in detectable ways.

Audit Data Immutability: Tests attempt to modify, delete, or reorder audit log entries through direct database manipulation and application-layer operations to verify that append-only enforcement prevents historical record alteration.

3.7.2 Evaluation Criteria

Evaluation criteria define how the system's effectiveness is measured based on the study's objectives. The e-Patunay evaluation framework incorporates technical measurements, security assessments, and user-based evaluations to provide a comprehensive assessment of system quality, functionality, and security posture. To ensure evaluation is structured, objective, and grounded in recognized standards, the

framework applies four complementary models: the FURPS Model, the ISO/IEC 25010 System and Software Quality standard, the NIST Cybersecurity Framework (CSF), and the OWASP Application Security Verification Standard (ASVS). Each model addresses a distinct dimension of system assessment and is selected specifically for its applicability to the e-Patunay system's technical architecture, security requirements, and academic context.

- FURPS Model: The FURPS model provides a foundational framework for evaluating general system quality across five key dimensions: Functionality, Usability, Reliability, Performance, and Security. This model ensures that e-Patunay is assessed not only for its cryptographic and security features, but also as a complete, usable software product that must operate efficiently for barangay officials and district officers in a real governance environment. Functionality assesses whether document submission, approval, and verification workflows perform correctly; Usability evaluates learnability and ease of task completion; Reliability measures system stability and uptime; Performance examines the responsiveness of encryption and signing operations; and Security assesses the effectiveness of access controls, data protection, and audit accountability mechanisms.
- ISO/IEC 25010 (System and Software Quality Models): This international standard extends foundational quality models by defining specific, measurable characteristics for software product quality. In this study, ISO/IEC 25010 is referenced in Specific Objective 6, which requires the use of User Accessibility Testing, System Usability Scale (SUS), and JMeter load testing as evaluation instruments. The standard provides a rigorous basis for assessing Functional Suitability (correctness of cryptographic and document

management functions), Performance Efficiency (response times under load), Compatibility (interoperability of Keycloak, EJBCA, and Azure Blob Storage integrations), Reliability (fault tolerance and recoverability), and Security (non-repudiation, confidentiality, and audit accountability). Grounding evaluation in ISO/IEC 25010 ensures that technical measurements are assessed against globally recognized engineering criteria rather than arbitrary benchmarks.

- NIST Cybersecurity Framework (CSF): Recognizing the cybersecurity-specialized nature of the study, the NIST CSF is employed to evaluate the system's ability to manage cybersecurity risks across the complete security lifecycle. While the STRIDE model identifies specific threat categories addressed through penetration testing, the NIST CSF structures evaluation around five continuous functions: Identify (asset inventory and risk assessment of barangay document governance), Protect (RBAC, MFA, AES-256, and TLS 1.3 controls), Detect (hash-chain tamper detection, OWASP ZAP log monitoring, and audit trail completeness), Respond (documented incident response and STRIDE-aligned remediation procedures), and Recover (Azure Blob archiving and business continuity provisions under the National Archives Act). This lifecycle perspective ensures that the system's automated security mechanisms and governance controls are systematically aligned with industry-standard cyber defense practices.

Evaluation Dimensions

The e-Patunay system is evaluated across five primary dimensions that comprehensively assess system quality and security effectiveness:

Table 17. *Evaluations Dimensions for e-Patunay*

FURPS Dimension	Description	Aligned Objective
Functionality	Ability of the system to perform intended cryptographic, document management, and governance functions correctly	To ensure core system functions operate as intended with cryptographic assurance (SO 1)
Usability	Ease of system use by barangay officials, district officers, and administrative staff for document submission, review, and governance tasks	To support effective document management and administrative operations (SO 6)
Reliability	Stability and availability of the system during continuous operation, including resilience to failures and data loss prevention	To maintain continuous access to governance records and ensure business continuity (SO 1, 6)
Performance	Efficiency and responsiveness of cryptographic operations, document processing, database queries, and API responses	To provide timely document processing and verification without operational delays (SO 1)
Security	Protection against STRIDE threats through cryptographic controls, access restrictions, integrity verification, and comprehensive audit logging	To improve overall system security posture and establish verifiable accountability (SO 2, 5, 6)

Key Performance Indicators (KPIs)

Key Performance Indicators translate evaluation dimensions into measurable metrics that allow objective assessment of system quality and security. Each KPI is clearly defined, measurable, and aligned with specific study objectives. KPIs for the

e-Patunay system reflect realistic performance expectations without claiming absolute security or perfection. The following table presents the KPI framework organized by FURPS dimensions:

Table 18. FURPS-Based KPI Matrix for e-Patunay System

Dimension	Evaluation Aspect	KPI	Target	Objective
Functionality	Document submission accuracy	% documents validated successfully	$\geq 95\%$	SO 1
	Digital signature success	% documents signed using EJBCA PKI	100%	SO 1
	Audit ledger integrity	% entries hash-chained correctly	100%	SO 2
Usability	Task completion rate	% users completing tasks without help	$\geq 85\%$	SO 6
	System Usability Scale	Mean SUS score from post-survey	≥ 70	SO 6
Reliability	System uptime	% availability during pilot period	$\geq 99\%$	SO 1,6
Performance	Upload response time	Avg time upload to validation confirm	≤ 3 sec	SO 1
	Signing latency	Avg time to sign and log	≤ 2 sec	SO 2,3
Security	Encryption compliance	% data using TLS 1.3 and AES-256	100%	SO 3
	Vulnerability mitigation	% vulnerabilities mitigated in pentest	$\geq 95\%$	SO 6

Technical Evaluation Metrics

Technical evaluation metrics provide quantitative measurements of system behavior, performance, and security based on controlled testing activities. These metrics offer objective evidence that the system meets its functional and cybersecurity requirements grounded in ISO/IEC 25010 quality characteristics. Technical metrics describe what will be measured and how measurement will be performed, without presenting actual results or interpretations.

The following technical metrics guide system evaluation:

Functional Accuracy: Measures whether core cryptographic functions (hashing, encryption, digital signatures), document management workflows (submission, approval, storage), and governance features (analytics, compliance monitoring) operate correctly during testing. Measured as percentage of successful function executions during system testing.

Performance Efficiency: Assesses response time for critical operations including document upload with encryption, approval processing with digital signing, verification API responses, and dashboard query execution. Measured in seconds or milliseconds depending on operation type.

System Reliability: Evaluates system stability and availability during continuous operation, measuring uptime percentage, successful transaction completion rate, and frequency of unhandled failures during testing period.

Security Effectiveness: Assesses the system's ability to detect, prevent, and mitigate identified STRIDE threats, measured as percentage of simulated attacks successfully blocked or mitigated during security testing activities.

Access Control Enforcement: Measures effectiveness of role-based access control and geographic restrictions, calculated as percentage of unauthorized access attempts successfully blocked during access control testing.

Audit Log Completeness: Evaluates presence and accuracy of audit records for security-relevant events, measured as percentage of test actions that generate corresponding audit entries with complete contextual information.

Table 19. Technical Evaluation Metrics

Aspect	Metric	Measurement	Objective
Functional Accuracy	Feature success rate	% successful core functions	SO 1 - Correct functionality
Performance	Response time	Avg processing time	SO 1 - Timely processing
Reliability	System uptime	% operational time	SO 1,6 - Reliable operation
Security	Mitigation rate	% vulnerabilities addressed	SO 6 - Penetration testing
Access Control	Block rate	Unauthorized attempts blocked	SO 5 - RBAC and MFA

Post-Survey Evaluation (User-Based Assessment)

Post-survey evaluation is conducted after system testing and user interaction to gather user-perceived feedback on system usability, functionality, reliability, performance, and security. This evaluation complements technical testing by capturing human-centered insights that cannot be measured through automated testing alone. Post-survey evaluation provides valuable information about whether users find the system practical, trustworthy, and effective for their operational needs.

Survey Methodology

A structured questionnaire using a 5-point Likert scale is administered to system users following testing activities. Survey items are carefully aligned with defined KPIs and specific objectives to ensure that user feedback directly addresses evaluation criteria. The survey captures both quantitative ratings and optional qualitative feedback through open-ended questions. Participation is voluntary, anonymous, and conducted ethically with no personally identifiable information collected. Survey data is analyzed using descriptive statistics to calculate mean scores, identify trends, and assess user satisfaction across evaluation dimensions.

Survey Evaluation Dimensions

Functionality Perception: Users assess whether the system accurately performs document submission, approval, and verification functions as expected. Sample item: "The system correctly processes my document submissions and provides appropriate confirmation."

Usability Assessment: Users evaluate interface clarity, navigation ease, task completion efficiency, and learning curve. Sample item: "I can complete document submission tasks with minimal effort and without confusion."

Reliability Confidence: Users rate their confidence in system stability and data persistence. Sample item: "The system operates consistently without unexpected failures or data loss during my usage."

Performance Satisfaction: Users assess whether system response times meet their operational expectations. Sample item: "The system responds quickly enough that I do not experience frustrating delays during normal operations."

Security Trust: Users evaluate their trust in system security controls and data protection mechanisms. Sample item: "I trust that the system adequately protects sensitive document information from unauthorized access."

Table 20. *Post-Survey Evaluation Items Aligned with KPIs and Objectives*

Dimension	Survey Statement	Aligned KPI	Objective
Functionality	The system accurately processes documents.	Submission accuracy	SO 1
Usability	The interface is easy to navigate.	Task completion	SO 6
Reliability	The system operates without failures.	System uptime	SO 1,6
Performance	The system responds quickly.	Response time	SO 1
Security	The system protects documents.	User trust score	SO 4,6

Ethical Considerations: Survey participation is entirely voluntary. Respondents are informed that their feedback will be used for academic research purposes only. No personally identifiable information is collected or stored. Aggregate results will be reported in a manner that prevents identification of individual respondents. Survey data is handled responsibly and securely, with access limited to authorized research team members.

ISO/IEC 25010 Quality Characteristics

The ISO/IEC 25010 quality characteristics applied in this study extend FURPS by providing internationally standardized sub-characteristics that precisely define what is measured and how. This is particularly important for academic rigor, as the standard grounds evaluation in peer-reviewed engineering criteria. The table

below maps each relevant quality characteristic to its evaluation approach and aligned study objective:

Table 21. *ISO/IEC 25010 Quality Characteristics Applied to e-Patunay*

Quality Characteristic	Sub-characteristic / Focus	Measurement Approach	Objective
Functional Suitability	Correctness and completeness of cryptographic, document, and governance functions	% core functions passing system test cases	SO 1
Performance Efficiency	Time behaviour of encryption, signing, storage, and API response operations	Avg response time (upload, signing); JMeter load metrics	SO 6
Compatibility	Interoperability with Keycloak, EJBCA, and Azure Blob Storage integrations	% successful API handshakes and integration test passes	SO 1, 2, 4
Usability	Learnability and satisfaction of barangay officials and district officers	SUS score ≥ 70 ; User Accessibility Testing results	SO 6
Reliability	Fault tolerance, recoverability, and data durability under continuous operation	% uptime; mean time to recovery from simulated failures	SO 1, 6
Security	Confidentiality, integrity, non-repudiation, accountability, and authenticity	Encryption compliance; audit log completeness; pentest results	SO 2, 3, 6
Maintainability	Modularity and testability of system components for long-term governance use	Qualitative assessment: component isolation and testability	SO 1

The Compatibility characteristic is especially significant for e-Patunay given its reliance on three external integrations Keycloak, EJBCA, and Azure Blob Storage which must interoperate correctly for core workflows to function. Interoperability testing therefore constitutes a distinct evaluation criterion beyond what the FURPS KPI matrix alone captures.

NIST Cybersecurity Framework (CSF) Alignment

The NIST CSF alignment table below maps each of the five CSF functions to specific e-Patunay activities, evaluation criteria, and study objectives. This ensures that the system's security posture is evaluated not only at the point of attack resistance (STRIDE), but across the full security lifecycle from governance identification through operational recovery:

Table 22. *NIST CSF Alignment for e-Patunay*

NIST CSF Function	e-Patunay Activities	Evaluation Criteria	Objective
Identify (ID)	Asset inventory, risk assessment, and governance policy mapping for barangay documents	Risk register completeness; policy coverage assessment	SO 3, 6
Protect (PR)	RBAC and MFA access enforcement, AES-256 and TLS 1.3 encryption, EJBCA certificate issuance	% security controls implemented per design specification	SO 3
Detect (DE)	Hash-chain audit ledger tamper detection, OWASP ZAP log monitoring, anomaly visibility	% security-relevant events logged and detectable via audit trail	SO 2, 6
Respond (RS)	Incident response plan, alert escalation, and STRIDE-aligned remediation procedures	Qualitative assessment of documented response readiness	SO 6
Recover (RC)	Azure Blob archiving tiers for storage recovery, business continuity provisions (National Archives Act)	Mean time to recover; data restore success rate under test conditions	SO 1, 4

The Detect function is of particular relevance: the hash-chain audit ledger directly implements a tamper-detection mechanism aligned with NIST CSF continuous monitoring objectives. The Recover function addresses operational resilience dimensions, specifically whether e-Patunay can restore document access after failure events that pure penetration testing would not evaluate.

OWASP ASVS Application Security Criteria

The OWASP ASVS verification categories applied in this study are selected based on direct correspondence with the penetration testing tools listed in Specific Objective 6. Each ASVS category below maps to a specific tool and testing activity, ensuring that security evaluation is both application-layer specific and traceable to defined study objectives:

Table 23. *OWASP ASVS Application Security Criteria for e-Patunay*

ASVS Category	Verification Focus for e-Patunay	Testing Method	Objective
V1 – Architecture, Design & Threat Modeling	System design aligns with STRIDE threat model; secure design principles applied	Design review against STRIDE categories	SO 6
V2 – Authentication	Keycloak MFA enforcement; JWT token validation; session expiry controls	Auth bypass testing via Hydra; token replay tests	SO 5
V3 – Session Management	Session token security; token expiration; concurrent session handling	OWASP ZAP session tracking validation	SO 3, 5
V4 – Access Control	RBAC enforcement; geographic district restrictions; privilege escalation prevention	Burp Suite access control bypass testing	SO 4
V7 – Error Handling & Logging	Audit log completeness; error messages free of sensitive data exposure	Log verification; OWASP ZAP log integrity testing	SO 2, 5
V9 – Communications	TLS 1.3 enforcement on all API and data transmission channels	Nikto misconfiguration scan; SSL/TLS verification	SO 3
V13 – API & Web Service	RESTful API input validation; role-specific data retrieval enforcement	SQLMap injection testing; API authorization checks	SO 5,6

Alignment with Objectives and Security Models

The evaluation framework for e-Patunay is systematically aligned with the study's specific objectives, the CIA Triad cybersecurity model, and STRIDE threat

categories. This alignment ensures that system evaluation reflects both software quality and cybersecurity effectiveness, providing comprehensive assessment that addresses academic requirements, security principles, and practical operational needs.

Alignment with Specific Objectives:

Each evaluation criterion and KPI directly supports at least one specific objective stated in Chapter 1. For example, digital signature verification testing aligns with Objective 1 (implement EJBICA-based PKI); hash chain integrity testing aligns with Objective 2 (implement tamper-resistant audit ledger); access control testing aligns with Objective 5 (enforce RBAC through Keycloak); and encryption testing aligns with Objective 3 (implement TLS 1.3 and AES-256 encryption). This explicit mapping ensures that evaluation activities measure progress toward stated goals rather than arbitrary technical achievements.

Alignment with CIA Triad:

Evaluation criteria map to the three fundamental cybersecurity properties: Confidentiality is assessed through encryption testing, access control enforcement, and information disclosure penetration tests. Integrity is evaluated through hash verification, digital signature validation, tampering resistance testing, and audit log integrity checks. Availability is measured through system uptime metrics, load testing resilience, denial-of-service resistance, and disaster recovery capability. This comprehensive coverage ensures that all three pillars of information security are properly validated.

Table 24. CIA Triad Alignment

CIA Principle	Evaluation Criteria	KPI / Metric	Objective
Confidentiality	TLS 1.3 and AES-256 protection	Encryption compliance	SO 3
	MFA and RBAC enforcement	Unauthorized block rate	SO 4
Integrity	Hash-chaining and Merkle proofs	Audit ledger integrity	SO 2
	EJBCA PKI digital signatures	Signature success rate	SO 1
Availability	System uptime and continuity	Uptime percentage	SO 1,6

Alignment with STRIDE Threat Model:

Security testing activities directly correspond to STRIDE threat categories identified in Chapter 1. Spoofing threats are tested through authentication bypass attempts and identity impersonation scenarios. Tampering threats are evaluated through data modification attempts and integrity verification testing. Repudiation threats are assessed by validating non-repudiation mechanisms through digital signature testing and audit log completeness. Information Disclosure threats are tested through unauthorized data access attempts and confidentiality breach scenarios. Denial of Service threats are evaluated through load testing and availability testing. Elevation of Privilege threats are assessed through privilege escalation attempts and authorization bypass testing. This complete STRIDE coverage ensures that all identified threat categories are systematically evaluated.

Table 25. STRIDE Threat Alignment

STRIDE Threat	Evaluation Criteria	Testing Method	Objective
Spoofing	MFA prevents impersonation	Auth testing; MFA bypass	SO 3
Tampering	Hash-chain prevents modification	Integrity checks; alteration	SO 1,2
Repudiation	Audit ledger ensures accountability	Log verification; Merkle proof	SO 2
Info Disclosure	TLS 1.3 and AES-256 protect data	Encryption validation	SO 3
Denial of Service	Availability and resilience	Load testing; resource exhaustion	SO 1,6
Elevation of Privilege	RBAC prevents escalation	Privilege escalation attempts	SO 5

3.7.3 Statistical Treatment of Data

This subsection describes the statistical and analytical methods used to analyze collected data in order to answer the study objectives. Statistical treatment explains how data will be processed and summarized to support evaluation findings, without presenting actual computed values or interpretations. The e-Patunay study employs both quantitative and qualitative analysis methods appropriate to the types of data collected during technical testing and user evaluation.

3.7.3.1 Descriptive Statistical Methods

Descriptive statistics summarize and present data in meaningful ways without making inferences beyond the observed data. These methods are appropriate for BSIT-level studies and provide clear, objective analysis of system performance and user feedback.

3.7.3.2 Frequency and Percentage

Percentage will be used to summarize categorical data, such as test pass/fail results and vulnerability severity distributions. It will be utilized to calculate the success rate of various technical metrics, including functional accuracy, access control blocking rate, and the vulnerability mitigation rate.

Formula:

$$P = (f / N) \times 100$$

Where:

P = Percentage (Success Rate)

f = Frequency of successful operations (e.g., successful document validations, unauthorized attempts blocked, or vulnerabilities mitigated)

N = Total number of operations or attempts

3.7.3.3 Mean (Arithmetic Average)

Calculated for continuous metrics including response times, processing durations, and survey ratings. The mean provides a central tendency measure that summarizes typical performance or user perception across all observations.

3.7.3.4 Weighted Mean

The weighted mean will be applied to the Likert scale survey responses to compute overall usability scores, security trust ratings, and satisfaction measures. This calculates the central tendency to summarize the typical user perception across all respondents (Barangay Officials, District Officers, and BCRD Administrators).

Formula:

$$W M = (\Sigma (fw)) / N$$

Where:

$W M$ = Weighted Mean

f = Frequency of responses for a specific Likert scale option

w = Numerical weight/value assigned to the scale position (e.g., 5, 4, 3, 2, 1)

N = Total number of responses

3.7.3.5 Likert Scale Interpretation

A structured questionnaire utilizing a 5-point Likert scale will be administered to system users following the User Acceptance Testing (UAT) activities. Numerical values are assigned to scale positions (e.g., 5 = Strongly Agree, 1 = Strongly Disagree) to quantify user-perceived feedback on the system's functionality, usability, reliability, performance, and security

Scale Rating	Range of Weighted Mean	Verbal Interpretation	Descriptive Equivalent
5	4.21 – 5.00	Strongly Agree	Highly Acceptable
4	3.41 – 4.20	Agree	Acceptable
3	2.61 – 3.40	Neutral	Moderately Acceptable
2	1.81 – 2.60	Disagree	Unacceptable
1	1.00 – 1.80	Strongly Disagree	Highly Unacceptable

Range and Standard Deviation is used to describe variability in performance metrics such as response times. Range indicates the difference between minimum and maximum values, while standard deviation quantifies typical deviation from the mean, helping identify consistency or variability in system behavior.

Success Rate Calculation will be computed as the percentage of successful operations out of total attempts for metrics including functional accuracy, access control blocking rate, and vulnerability mitigation rate. Formula: $(\text{Successful Operations} / \text{Total Operations}) \times 100\%$.

3.7.3.6 Comparative Analysis

Comparative analysis techniques identify differences or similarities between data sets without making claims about statistical significance or causality. These methods help contextualize findings by comparing against baseline values or established benchmarks.

Before-After Comparison: If applicable, compares system metrics or vulnerability counts before and after implementing security controls, providing simple descriptive comparison of changes without claiming statistical significance.

Benchmark Comparison: Compares observed performance metrics against predefined targets or industry benchmarks (e.g., comparing actual response times against target thresholds, comparing vulnerability counts against baseline assessments).

Cross-Dimension Analysis: Compares evaluation results across FURPS dimensions to identify relative strengths and areas for improvement, presented descriptively without claiming causal relationships.

3.7.3.7 Qualitative Analysis Methods

Qualitative analysis methods process non-numerical data including open-ended survey responses, expert feedback, and observation notes to identify common themes, patterns, and insights that complement quantitative measurements.

Thematic Categorization: Open-ended survey responses are reviewed and grouped into thematic categories based on common topics, concerns, or suggestions. Frequencies of each theme are reported to identify the most common user observations.

Content Analysis: Written feedback from users or evaluators is analyzed to extract specific examples, identify recurring issues, and highlight positive experiences that provide context for quantitative findings.

3.7.3.8 Data Analysis Procedures

Technical test data including performance measurements, security test results, and functional accuracy metrics are collected in structured formats during testing activities. Survey responses are collected through online questionnaire tools and exported for analysis. All quantitative data is processed using spreadsheet software or statistical packages to compute descriptive statistics. Qualitative responses are reviewed manually to identify themes and patterns. Results from both quantitative and qualitative analysis are synthesized to provide comprehensive evaluation findings presented in Chapter 4.

3.8. Data Gathering Procedures

This section explains the organized steps used to collect information for building and evaluating the *e-Patunay* system. It looks at the project through five

main areas: where the information comes from, who the people involved are, what forms or checklists are used, how the information is actually collected, and what software or equipment is utilized to record it. By detailing these steps, it shows that the research is based on real and accurate information from the Barangay Community Relations Department.

3.8.1 Data Sources

Data for the study will be gathered from multiple sources to support system requirements analysis, security implementation, and comprehensive evaluation. These sources are categorized based on their purpose and application within the research methodology.

User and Stakeholder Inputs will be collected from BCRD personnel, district officers, and barangay officials to establish functional and security requirements. This includes understanding current document processing workflows, identifying pain points in manual verification processes, defining role-based access requirements for different user categories, and determining district-specific operational needs. Stakeholder inputs ensure that the system addresses real operational challenges rather than hypothetical scenarios, and that security controls align with actual user responsibilities and data sensitivity requirements.

System-Generated Data will be produced during controlled testing and pilot operation of the e-Patunay system. This includes audit log entries created during document submission and approval workflows, digital signatures generated by the EJBCA PKI infrastructure, hash chains and Merkle tree proofs produced by the cryptographic audit ledger, authentication and authorization logs from Keycloak, and performance metrics captured during system operation. System-generated data

provides objective evidence of how security controls function in practice and supports evaluation of cryptographic implementation correctness.

Cybersecurity Intelligence and Reference Materials consist of publicly available resources that inform security design and threat analysis. This includes vulnerability databases such as the National Vulnerability Database (NVD) and Common Vulnerabilities and Exposures (CVE) listings relevant to the technology stack employed, OWASP Top 10 documentation and testing methodologies, STRIDE threat modeling frameworks and best practices, Philippine legal and regulatory requirements including the Data Privacy Act (RA 10173), E-Governance Act (RA 12254), and related mandates, and NIST cybersecurity standards and guidelines applicable to PKI implementation and secure system development. These references ensure that security implementations align with established best practices and regulatory compliance requirements.

Evaluation and Testing Data will be collected through structured assessment activities conducted during system validation. This includes results from security testing using OWASP ZAP, Burp Suite, SQLMap, Nikto, and Hydra, User Acceptance Testing (UAT) feedback from BCRD stakeholders and pilot users, System Usability Scale (SUS) survey responses measuring perceived usability, performance testing metrics from Apache JMeter load tests measuring API response times and system behavior under stress, and security assessment findings documenting vulnerabilities identified and remediation actions taken. Evaluation data provides the empirical evidence necessary to validate that the system meets its stated objectives for functionality, security, usability, and performance.

3.8.2 Participants of the Study

BCRD Department Heads and Administrators serve as primary stakeholders providing strategic direction and high-level requirements for the system. These participants include the Acting City Government Assistant Department Head, division heads from Administrative, Operations & Monitoring, Planning & Programming, and Technical & Research Services divisions. Their participation involves semi-structured interviews to understand departmental goals, compliance requirements, security concerns, and operational constraints. BCRD administrators provide insights into how the system must align with city government policies, inter-agency coordination needs, and legal mandates including the Data Privacy Act (RA 10173) and E-Governance Act (RA 12254).

District Officers from the Operations & Monitoring Division represent the users who will directly process and approve document submissions. Participants from District I through District VI (with pilot focus on District II) contribute to requirements analysis by describing current manual workflows, identifying pain points in document verification processes, and defining role-based access requirements. District Officers participate in User Acceptance Testing (UAT) to validate that the review and approval workflow functions as intended and meets operational needs. Their feedback is critical for assessing whether the system successfully replaces manual verification processes while maintaining or improving efficiency.

Records Section Staff from the Administrative Division participate as users responsible for document archival and retrieval functions. These participants provide requirements related to long-term storage, document lifecycle management, and

retrieval processes. Records staff participate in UAT to validate that the Azure Blob Storage implementation with lifecycle management policies meets archival requirements per the National Archives Act (RA 9470) and supports efficient retrieval when documents are needed for audits or legal proceedings.

Barangay Officials represent the primary data submitters who will use the system to submit Personal Data Sheets, appointment papers, clearances, and other official documents. A representative sample of barangay officials from the districts included in the pilot deployment will participate in UAT to validate that the secure web forms, electronic attestation process, and document submission workflows are intuitive and practical. These participants provide critical feedback on usability, clarity of instructions, and any barriers to adoption that might impede transition from manual submission methods.

Barangay officials participating in UAT will complete representative tasks including user registration with Multi-Factor Authentication setup, document submission through secure web forms, electronic attestation to affirm data accuracy, and status checking to track submission progress. Their participation ensures that the system is designed for users with varying levels of technical proficiency and that security controls (MFA, electronic attestation) do not create undue burden preventing system adoption.

3.8.3 Data Gathering Instruments

System Monitoring and Logging Tools

Laravel Telescope - Built-in debugging and monitoring assistant for Laravel applications, used during development to inspect requests, exceptions, database

queries, and logs generated by the application. Telescope provides insights into system behavior during testing phases without requiring external monitoring infrastructure.

Azure Monitor - Cloud-based monitoring service tracking performance metrics, application logs, and resource utilization for Azure-hosted components (App Service, PostgreSQL, Blob Storage). Azure Monitor provides operational visibility into system health and performance during pilot deployment.

Postman / Insomnia - API testing tools used to validate RESTful endpoints, inspect request/response payloads, and generate synthetic API traffic for monitoring purposes. These tools support verification that APIs function correctly and return expected data structures.

3.8.3.1 Security Assessment Tools

OWASP ZAP (Zed Attack Proxy) - Open-source web application security scanner used for automated vulnerability assessment. ZAP performs active scanning to identify common vulnerabilities (SQL injection, XSS, CSRF) aligned with OWASP Top 10 standards.

Burp Suite Community Edition - Web application security testing toolkit used for manual security analysis. Burp Suite's proxy functionality enables interception and modification of HTTP requests to test authentication, authorization, and business logic vulnerabilities.

SQLMap - Automated SQL injection detection and exploitation tool used to validate that database queries are properly parameterized. SQLMap tests whether user inputs can be leveraged to manipulate database operations.

Nikto - Web server scanner used to identify server misconfigurations, outdated software versions, security headers, and exposed administrative interfaces. Nikto validates that the Azure App Service and web server configurations follow security best practices.

Hydra - Network authentication cracker used to test brute-force attack resistance. Hydra simulates credential-stuffing and password-guessing attacks against login endpoints to validate rate limiting and account lockout mechanisms.

3.8.3.2 Performance Testing Tools

Apache JMeter - Load testing tool used to measure system performance under stress conditions. JMeter simulates multiple concurrent users accessing the e-Patunay system, generating metrics on response times, throughput, and error rates. Performance testing validates that the system can handle expected operational loads without degradation.

Azure Application Insights - Performance monitoring service integrated with Azure App Service, providing detailed telemetry on request durations, dependency calls, and failure rates. Application Insights supports identification of performance bottlenecks during load testing.

3.8.3.3 User Evaluation Tools

System Usability Scale (SUS) Questionnaire - Standardized 10-item survey instrument measuring perceived system usability. The SUS provides a quantitative usability score (0-100 scale) enabling comparison against industry benchmarks and objective assessment of user experience. Administered via Google Forms for efficient distribution and response collection.

Google Forms - Survey platform used to administer the SUS questionnaire and

collect structured feedback from UAT participants. Google Forms enables automatic response aggregation, export to spreadsheet format for analysis, and visualization of quantitative results.

Observation Notes (Google Docs / Physical Notebooks) - Used during User Acceptance Testing sessions to document participant interactions with the system, recording navigation patterns, errors encountered, questions asked, and task completion sequences.

3.8.4 Data Collection Techniques

Requirement Elicitation Methods will be employed to gather user needs and system requirements from stakeholders. Semi-structured interviews will be conducted with BCRD administrators and district officers to understand current workflows, security concerns, and operational constraints. These interviews allow for in-depth exploration of specific challenges and enable researchers to probe beyond surface-level issues. Document analysis will examine existing BCRD policies, procedures, and record-keeping standards to identify compliance requirements and operational norms that the system must support. Observational studies of current manual processes will be performed where feasible and authorized, providing firsthand understanding of how documents are currently submitted, verified, and archived. These requirement elicitation methods ensure that system design is grounded in actual operational needs rather than assumptions.

System-Based Data Collection will occur through controlled execution of the e-Patunay system during development and pilot testing. Automated logging mechanisms embedded in the system will capture authentication events, document lifecycle state transitions, approval actions, and system errors without requiring

manual data entry. Database queries will extract system-generated data such as audit log entries, digital signature metadata, and user activity patterns for analysis. API monitoring tools will track request/response patterns during testing to identify performance bottlenecks or security anomalies. System-based collection provides objective, timestamped evidence of system behavior and security control effectiveness.

Security Assessment Activities will generate data through structured testing procedures designed to validate system resilience against identified threats. Security testing in a controlled environment will use industry-standard tools (OWASP ZAP, Burp Suite, SQLMap, Nikto, Hydra) to simulate attacks against the pilot system, with findings documented in detailed vulnerability assessment reports. Each penetration test will follow a defined scope and methodology, targeting specific STRIDE threat categories and system components. Configuration reviews will assess security settings in Azure services, Keycloak authentication, EJBCA PKI, and application code to verify alignment with security best practices and regulatory requirements. Code security analysis will examine critical security implementations such as cryptographic functions, input validation, and authorization logic to identify potential weaknesses before deployment.

User Evaluation and Feedback Collection will gather subjective assessments of system usability, functionality, and user experience. User Acceptance Testing (UAT) sessions will be conducted with representative users from each role category (barangay officials, district officers, BCRD administrators) to validate that workflows function as intended and meet operational needs. During UAT, participants will complete representative tasks while researchers observe and document issues,

questions, or difficulties encountered. The System Usability Scale (SUS) will be administered as a standardized questionnaire after UAT sessions, providing quantitative usability scores that enable objective comparison against established benchmarks. Optional open-ended feedback questions will allow participants to provide qualitative insights that structured assessments may not capture. User evaluation data ensures that security controls do not impede usability and that the system is practical for real-world operation.

3.8.5 Tools used for data gathering

Microsoft Teams / Zoom - Video conferencing platforms used for conducting remote interviews with BCRD stakeholders and district officers when physical meetings are not feasible. These tools enable virtual stakeholder consultations with recording capabilities (subject to informed consent) allowing accurate capture of requirements discussions.

Physical In-Person Interviews - Face-to-face meetings conducted at BCRD offices or designated locations for requirements elicitation. Physical interviews allow direct observation of current workflows, examination of existing manual processes, and building rapport with stakeholders. Interview sessions are audio-recorded (with participant consent) to ensure accurate documentation of requirements and feedback.

Audio Recording Devices / Smartphone Recording Applications - Used to capture audio during physical interviews, ensuring complete and accurate records of stakeholder statements, requirements, and concerns. Recordings serve as primary source material for transcription and analysis.

Google Docs - Cloud-based word processing tool used for transcribing interview recordings into text format. Google Docs enables collaborative transcription where multiple research team members can review and validate transcripts, add annotations, and organize requirements by category. The cloud-based nature allows secure sharing among team members while maintaining version control.

3.8.6 Ethical and Data Handling Considerations

All data collection activities will be conducted in accordance with ethical research principles and applicable data privacy regulations. Informed consent will be obtained from all participants in interviews and UAT (user acceptance testing) with participants informed of the study's purpose, data usage, and their right to withdraw at any time. Data minimization principles will be applied, collecting only information necessary for research objectives and avoiding unnecessary collection of personally identifiable information.

Synthetic and test data will be used for all system development, security testing, and evaluation activities. No real Personal Data Sheets, appointment records, or other sensitive government documents will be processed during research phases. Test data will be clearly marked and separate from any production data to prevent accidental disclosure or misuse.

Access controls and confidentiality will protect collected data throughout the research process. Interview transcripts, survey responses, and evaluation data will be stored securely with access limited to the research team. Any stakeholder feedback or vulnerability findings that could identify specific individuals or expose security weaknesses will be anonymized in research documentation and reports.

Compliance with the Data Privacy Act of 2012 (RA 10173) will be maintained throughout data collection and handling. No personal information will be collected without lawful basis and informed consent. All data will be processed fairly, transparently, and for legitimate research purposes only. Appropriate technical and organizational measures will protect collected data against unauthorized access, disclosure, or loss.

Responsible vulnerability disclosure will guide how security findings are handled. Any vulnerabilities identified during security testing will be documented confidentially and remediated before public disclosure or system deployment. Security assessment findings will be shared only with authorized BCRD stakeholders and academic advisers, not publicly disclosed in ways that could enable exploitation.

These ethical considerations ensure that data gathering activities respect participant rights, protect sensitive information, comply with legal requirements, and maintain research integrity throughout the study.

3.9 Ethical and Legal Considerations

This research adheres to strict ethical standards and data handling protocols in compliance with Philippine legal frameworks, including the Data Privacy Act of 2012 (Republic Act No. 10173), National Privacy Commission (NPC) Circulars, Republic Act No. 10175 (Cybercrime Prevention Act of 2012), and the E-Governance Act (Republic Act No. 12254). Formal written consent was obtained from the Barangay Community Relations Department (BCRD) of the Quezon City Government through a signed Memorandum of Agreement (MOA), authorizing the research team to access operational workflows and sensitive records, such as Personal Data Sheets (PDS) and

financial disclosures, for the exclusive purpose of developing and validating the cryptographic-assurance system.

The project will seek approval from an Institutional Review Board through the FEU Institute of Technology's Ethics Research Committee to ensure compliance with academic research ethics standards, with ongoing oversight provided by the thesis advisor, faculty mentor, and designated BCRD legal representatives who collectively verify adherence to ethical procedures throughout the research lifecycle. To demonstrate responsible data stewardship, synthetic data such as simulated barangay certifications and mock system logs will be generated and utilized during testing phases to avoid exposing real, sensitive public records.

In instances where real operational data from the BCRD is accessed for validation or cross-referencing purposes, such data will be retained only until project completion, at which point secure deletion procedures will be implemented in accordance with ISO/IEC 27037:2012 standards for digital evidence handling. Methodological integrity is ensured while protecting the privacy of government operations and personnel data by the deliberate use of synthetic data as the main training source, which is complemented carefully by anonymized real data for validation.

To keep BCRD information safe, we will use a multi-layered security approach that acts like a digital fortress. All sensitive work is done strictly at the BCRD or Quezon City National Headquarters on "air-gapped" computers that are physically disconnected from the internet, which completely removes the risk of online hacking or cloud leaks. Access is strictly limited to authorized team members who must pass double-identity checks to log in, and the system automatically records

every time someone views the data to ensure full accountability. To protect individual privacy, all files are scrambled with high-level encryption, and personal details like names or addresses are replaced with anonymous codes before any analysis begins. We also have a ready-to-use emergency plan that outlines exactly how to stop, fix, and report any security issues if they ever occur. By following these strict rules, the team ensures that the community's data is handled ethically, legally, and with the highest level of care.

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APPENDICES

APPENDIX A: Communication Letter



FEU INSTITUTE OF TECHNOLOGY

October 3, 2025

Hannah

BRCD Technical Staff

Barangay And Community Relations Department

2nd floor Community Center Building,

Quezon City Hall Compound,

Elipit Road, Barangay Central, Quezon City,

Metro Manila

Subject: Request for Collaboration in Cybersecurity Capstone Project

Dear Hannah,

Greetings of peace and progress!

We are third-year students currently enrolled in the **System Analysis and Design** under the Information Technology Department at FEU Tech. As part of our academic requirements, we are embarking on a year-long **Cybersecurity Capstone Project**, which will extend until our fourth year. This project involves the design, development, and implementation of a cybersecurity-oriented system that addresses real-world needs in organizations.

Your esteemed organization has been identified as a potential client and beneficiary of this project. The system will be developed **free of charge** by our research team, under the guidance of our faculty advisers, and will be **formally turned over to your organization upon completion**.

To provide you with a clearer view, below is the **project timeline**:

- **1st Trimester (Junior Year):** System Analysis, Title/Topic Defense
- **2nd Trimester:** Documentation of Chapters 1–3 and development of Prototype
- **3rd Trimester (Fourth Year):** Full System Development
- **4th Trimester:** Final Testing, Deployment, and Turn-over of the Project to your organization

In line with this, we respectfully request your support by:

1. Allowing us to conduct **interviews** with selected personnel to gather insights.
2. Providing access to **sample forms, workflows, and other relevant data** necessary for system analysis and prototype design.

We assure you that all information gathered will be treated with **strict confidentiality** and will be used **solely for academic and developmental purposes**.

Through this collaboration, we aim not only to develop an innovative system tailored to your organization's needs but also to contribute to making the **Philippines a more secure digital environment** by embedding cybersecurity principles in our solutions.

We sincerely look forward to a meaningful partnership with your organization and to co-creating a project that promotes both **innovation and security**. Should you have any questions or clarifications, please do not hesitate to contact us at quadsec.feutech@gmail.com.

Thank you very much for considering our request.

Respectfully yours,

Genro Baldemoro Project Manager

Junior Researchers | Cybersecurity Proponents

Approved by:


Abraham S. Tinga

Research Adviser | FEU Tech – Information Technology Department

astinga@feutech.edu.ph

Noted by:

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APPENDIX B: Client Profile



FEU Institute of Technology

INFORMATION TECHNOLOGY DEPARTMENT

GROUP PROFILE	
Course Code & Name:	IT0039 IT Project Management
Section:	TC31
Course Adviser:	Professor Abricam Tinga
Group/Team Name:	QuadSec
Group/Team Members: <i>(Include mobile number & e-mail address beside your name)</i>	1. Baldemoro Genro Gabriel D. - 09696109943, gdbaldemoro@fit.edu.ph 2. Berris Jonathan C. - 09761590768, jcberris@fit.edu.ph 3. Santos Jericho G. - 09098736522, jgsantos@fit.edu.ph 4. Sorrrera Lance Arthur J. - 09509986499, ljsorrera@fit.edu.ph
Project Adviser:	Dr. Juan Paulo Magcuyao
Approved Project Title:	e-Patunay: A Cryptographic-Assurance Governance System for the BCRD Enabling Storage Optimization and Inter-Agency Record Verification
Nature of the Project:	Record Management System

CLIENT PROFILE	
Client (Company Name):	Barangay Community Relations Department
Company brief description:	BCRD oversees the Barangay Council's programs, projects, and activities, providing technical and administrative assistance to the barangays. BCRD organizes training and seminars for barangay officials and personnel to develop their skills and competencies in creating appropriate plans for their communities in partnership with other local government agencies and civil society organizations. As part of its function to foster and continuously promote closer government-people relationships, BCRD organizes community conferences to explain city plans and programs, monitors and receives community complaints and problems, and provides development reports or updates on actions taken.
Nature of Business:	Barangay-level public service office responsible for community relations, coordination of local programs, handling citizen concerns, and supporting peace and order initiatives.
Contact Person:	Ms. Hannah
Position/Designation:	BCRD Technical and Research Services Division
Business Address:	2nd Floor, Community Center, Quezon City Hall Compound
Office Number:	(02) 8988-4242
Mobile Number:	N/A
Email Address:	BCRD@quezoncity.gov.ph

APPENDIX C: Transcripts of Interview

BCRD – QuadSec

Meeting Minutes

Meeting Summary – October 9 2025, 2:00 PM

QuadSec – Capstone Group

- Genro Gabriel D. Baldemoro
- Jonathan C. Berris
- Jericho G. Santos
- Lance Arthur J. Sorrrera

Speaker 2 – Client Reps

- Hannah Gabriella Gultiano – Technical and Research Services Division (TRSD Staff)
- Ricardo B. Corpuz – Department Head

Key Discussion Points:

1. Project Concept and Objective

- The client welcomed the team's proposal for **digitalization of barangay records**, especially focusing on **Barangay Officials' PDS (Personal Data Sheets)**.
- The **goal** is to transition from manual recordkeeping (physical files, printed PDS forms) to a **secure, centralized digital management system**.
- The system must **apply modern cybersecurity principles**, as required by the team's specialization (BSIT - Cybersecurity).
- The project will include both **record digitalization** and a **record management platform** accessible to authorized users.

2. Data Handling and Access Control

- Each barangay district currently maintains separate physical records with **no formal process for access control or consent**.
- Transfers of documents between districts are typically informal ("just giving the paper"), without tracking or authorization logs.
- Data sharing practices are **manual and untracked**, posing risks of data loss or unauthorized access.

3. Nature of Barangay Data

- The main datasets include:
 - **Personal Data Sheets (PDS)** of barangay officials (with full personal info, contact numbers, educational background, photo, signature, etc.)
 - **Barangay profiles** (public information such as subdivisions, emergency units, available services, etc.).
 - **Civil Society Organization (CSO)** records and accreditation data.
- **Sensitive personal data** is limited to officials and selected staff; residents' data are mostly public and non-personal.

4. Current Data Management Practices

- Records are stored both **physically (paper files, stored in the records room)** and **digitally (Google Sheets, Google Forms, and Docs)**.
- Google Sheets are used to collect and summarize barangay information submitted via forms.
- Data disposal: old documents (e.g., 2019–2022) are **archived or shredded**.

5. Proposed Technical Solutions

- **Centralized Barangay Management System (CBMS)** to handle PDS and record storage.
- **Integration of cybersecurity technologies**, including:
 - Access control and permissions per user or district.
 - Audit logging for every access or edit.
 - Possible use of **hashing** to verify data integrity.
 - **Encryption** of sensitive PDS information.
 - **Blockchain-inspired features** to prevent tampering or unauthorized record changes.
- Cloud-based storage for accessibility and scalability (with careful consideration of subscription or cost).

7. System Functionality Requirements (Client Suggestions)

- Track **status of barangay captains** (active, on leave, resigned, replaced).
- Manage **resignation and appointment records**, including attached letters and resolutions.
- Include **record disposal tracking** (archived, shredded, transferred).
- Generate **summaries and reports** for public information (e.g., barangay profiles).
- Limit exposure of **personal identifiers** (e.g., mobile numbers, addresses).

QuadSec:

Good afternoon. We're conducting client interviews to assess how we can best assist your organization. To start, could you tell us the **primary functions of the Barangay Community Relations Department (BCRD)?**

Hannah Gabriella Gultiano:

The BCRD mainly handles coordination with barangays. We deal with barangay councils and related community matters. Our responsibilities include community services, certifications, and assistance programs. For example:

- Burial benefits for barangay officials,
- BPSO (Barangay Public Safety Office) payroll processing,
- Handling **complaints involving barangay officials**,
- Coordination with **civil society organizations (CSOs)**.

So, our work is mostly focused on barangay and CSO concerns.

QuadSec:

What are your **day-to-day office activities?**

Hannah Gabriella Gultiano:

It depends on the task, but a common example is handling complaints. When a complaint is received, our head records officer routes it to the concerned department—often the legal division—based on the issue raised by the barangay or a constituent.

QuadSec:

What about **requests** from barangays?

Hannah Gabriella Gultiano:

Most requests involve **certifications**, particularly for barangay officials. We also issue **travel authorities** for government officials who need to travel abroad, which requires the Mayor's approval.

QuadSec:

Are these all handled using paper documents?

Hannah Gabriella Gultiano:

Yes, almost everything—complaints, certifications, payrolls—is on paper. We tried using email, but most barangays still prefer printed documents. They're very traditional, and there's a heavy reliance on physical receipts and paper files.

QuadSec:

So ideally, we'll be digitalizing your database—official records, complaints, and requests. Are you open to that?

Hannah Gabriella Gultiano:

Yes, definitely. The city has a centralized database, but we're not yet connected to it. There have been attempts to start digitalization, but nothing materialized.

QuadSec:

What's the current state of your records?

Hannah Gabriella Gultiano:

We still have documents dating back to the **1980s**. Some of them can already be disposed of, but we'd rather have them **digitized** for preservation.

QuadSec:

Who has access to these records?

Hannah Gabriella Gultiano:

Access is limited to our section's personnel. Each district has its own staff responsible for its records. For example, District 1 staff can't open District 2 files, but all records are stored in the same area.

QuadSec:

Have you ever experienced tampering or unauthorized access?

Hannah Gabriella Gultiano:

So far, no. Everything is still paper-based, so there's limited opportunity for that.

QuadSec:

How do you **track** requests or complaints?

Hannah Gabriella Gultiano:

We use a **logbook**—everything is recorded manually.

QuadSec:

So, based on what we've gathered, your main request is **digitalization of records**, with potential for system enhancements. As cybersecurity students, we'd also like to integrate **security and innovation**, not just create a basic database.

Hannah Gabriella Gultiano:

That's good. Our cybersecurity right now isn't strong. Each division has its own email account, but multiple people share them. Passwords must be changed every three months, and some users struggle to log in afterward.

QuadSec:

We could propose a **complaint management system** where each district has its own account. Complaints could be submitted, tracked, and monitored, with updates visible on a dashboard.

Hannah Gabriella Gultiano:

That would be very useful. Currently, when a complaint is received, we must endorse it within **five working days** to the proper department (for example, Fairview office). Fairview then has **15 working days** to respond. Considering there are **142 barangays**, this process takes time.

QuadSec:

Maybe we could **pilot test** the system in one district first—for example, **District 2**, since it only has five barangays.

Hannah Gabriella Gultiano:

Yes, that could work. The barangay counts per district are roughly:

- District 1: 37
 - District 2: 5
 - District 3: 37
 - District 4: 38
 - District 5: 15
 - District 6: 11
- That totals 142 barangays overall.

QuadSec:

We'll need to confirm our project scope with our professor to ensure it's appropriate—not too big or too small for a capstone project.

Hannah Gabriella Gultiano:

No problem. If you want to focus on the complaint tracking system, that's fine. But if possible, include other transactions as well—like certifications or requests.

QuadSec:

Understood. We'll explore secure, person-to-person transaction tracking too.

Hannah Gabriella Gultiano:

That's very technical. It might be complicated, but possible.

QuadSec:

Yes, our professor warned that a full complaint system involving all offices might be too complex, but we'll manage the scope carefully.

Hannah Gabriella Gultiano:

We receive complaints from various offices—the City Administrator, Mayor's Office, and even the **Presidential 888** program. They already have systems, but the data isn't unified or well-categorized. We'll check what information is available, like **age and sex demographics**, for city planning.

QuadSec:

Perfect. Regarding data, we'd also like to integrate security measures. For example, a system that **tracks who accessed records**, and if suspicious activity is detected, it can **temporarily block that user's access**. Would that kind of feature be practical?

Hannah Gabriella Gultiano:

Yes, that would help a lot. Currently, anyone can request data, so **restricted access per district** would be ideal.

QuadSec:

Got it. We'll also study your existing dashboard setup to integrate that data properly.

QuadSec:

One idea is to log every document-related activity—like a SOAR (Security Orchestration, Automation, and Response) mechanism. If someone with malicious intent accesses it, the system can automatically block them. We could also use hashing to detect any unauthorized changes in documents.

QuadSec:

For example, if someone wants to access files from another district—say District 2 requesting from District 3—how is that usually handled?

Hannah Gabriella Gultiano:

Currently, it's very informal. They just ask for the paper, and it's given. There's no consent or verification process.

QuadSec:

So, there's no approval or record of access?

Hannah Gabriella Gultiano:

None. No one has tried to formalize it before.

QuadSec:

That could be improved through a feature like blockchain—to verify access and integrity.

QuadSec:

If a complaint arises, who processes it?

Hannah Gabriella Gultiano:

We usually send an email or message to the appropriate authority. I've seen some dashboards using Microsoft platforms for barangay-level data, but they aren't yet implemented because of access issues.

QuadSec:

So, there needs to be access control in place.

Hannah Gabriella Gultiano:

Exactly. Even within the barangay, not everyone should have access. We should also prevent account sharing.

QuadSec:

Yes, that's important.

Hannah Gabriella Gultiano:

Let's brainstorm ideas first; we'll select which ones are feasible later.

QuadSec:

If we digitalize the physical records, access should still be restricted. A blockchain-like system would help identify who accessed or modified a document and ensure trustworthiness.

QuadSec:

Currently, do you need to file requests to access documents?

Hannah Gabriella Gultiano:

Not really. We just message the person-in-charge. There's no formal confirmation when documents are returned—so we easily lose track.

QuadSec:

So only specific people can enter the records room?

Hannah Gabriella Gultiano:

Yes, only authorized personnel. For data collection, we use Google Sheets and Google Forms. Access is limited to those with permission via email or link.

QuadSec:

What kind of data do you collect?

Hannah Gabriella Gultiano:

Mainly barangay official profiles. We also collect other data when needed—for example, if someone requests the number of barangay ambulances. We use Google Forms, summarize the responses, and forward them to the DRRM (Disaster Risk Reduction and Management) office.

QuadSec:

Do you store any personal data?

Hannah Gabriella Gultiano:

Yes, personal data sheets (PDS) for barangay officials. These include personal information like cellphone number, address, educational background, height, weight, blood type, picture, and signature.

QuadSec:

That's sensitive data—it's good to know for cybersecurity considerations.

Hannah Gabriella Gultiano:

Yes, it's necessary for our operations. We usually just provide a blank PDS in PDF format; they can fill it out manually or digitally.

QuadSec:

Can we take a picture of the records room for documentation?

Hannah Gabriella Gultiano:

Sure, as long as the door is open for transparency.

QuadSec:

Are older documents disposed of properly?

Hannah Gabriella Gultiano:

Yes. We shred them. Some payroll documents are rolled, but shredding is our usual process. Recently, we turned over 2019–2022 files to the General Services for archiving.

QuadSec:

What about your Google Docs data?

Hannah Gabriella Gultiano:

We're preparing to transfer new data to Sir Paul. Right now, we're collecting updated barangay

profiles—like lists of subdivisions. Most of these are public information and accessible through Google Sheets and Forms.

QuadSec:

If we created a centralized management system, could residents access certain data?

Hannah Gabriella Gultiano:

Yes, that's possible. The 2017 barangay profile was even posted on the BCRD website. However, it hasn't been updated since. Most of our data is public, except for personal details of officials and CSO (Civil Society Organization) leaders. For example, we don't publish their phone numbers or birthdays.

QuadSec:

That's good for cybersecurity and data protection.

Hannah Gabriella Gultiano:

Exactly. We're careful not to expose personal data.

Hannah Gabriella Gultiano:

When collecting PDS files from barangays, it's not just done once. Updates happen when changes occur, such as when an official is removed, replaced, or passes away. Each time that happens, the barangay submits a new PDS with updated information. So, it's not only once per term—it depends on personnel changes within the barangay.

Hannah Gabriella Gultiano:

Right, and if an official resigns, we don't delete their record. Their documents remain stored, marked as "resigned," with their resignation letter attached.

Hannah Gabriella Gultiano:

Exactly. The same applies if someone passes away—the record stays for reference, just labeled accordingly.

QuadSec:

How are these updates submitted?

Hannah Gabriella Gultiano:

Usually through formal letters. For example, when someone resigns or is replaced, the barangay sends a letter and resolution naming the replacement, along with a new PDS for the incoming official. Officers also submit their PDS this way—secretaries, treasurers, and others.

QuadSec:

So that means you exchange official correspondence regularly between the barangays and the district heads?

Hannah Gabriella Gultiano:

Yes, communication is constant. Each time there's a personnel change, a resolution and updated PDS are required. These documents form part of the barangay's official record.

QuadSec:

That falls under **personal information handling**, which connects directly to our course. For instance, when we request details like contact numbers or addresses of barangay captains, we

must manage that data securely. Access logs could help track who views or edits those records in our proposed system.

Hannah Gabriella Gultiano:

Agreed. Some officials go on official leave or travel for work, so their status can be marked as "On Leave" or "On Official Travel." There's also supporting documentation like travel authority letters or certifications.

QuadSec:

Right. They also have **leave and travel credits**, and there's a certification from the Civil Service Commission once they complete their full term.

Hannah Gabriella Gultiano:

We can provide you with a blank copy of the barangay PDS form. It's slightly different from the standard PDS used by regular government employees.

QuadSec:

That would be helpful. Thank you.

QuadSec:

Initially, our idea was the **digitalization of barangay records**, specifically for barangay officials. However, our professor advised that the project must include an **application of recent technology**, particularly with **cybersecurity integration**.

Earlier, we saw staff typing PDS (Personal Data Sheet) forms manually. That process inspired us to focus on developing a **Barangay PDS Management System** that could improve accuracy and data security.

Hannah Gabriella Gultiano:

All right. What are your current recommendations?

QuadSec:

Our **main recommendation** is still the **digitalization of records**—that's our foundation. But we plan to expand it with a **record management system** where data is segmented per district. For example, personnel in **District 2** cannot access records from **District 3** unless they request permission from that district's head.

We also plan to include a **status tracker for barangay captains** (e.g., active, on leave, replaced, or for replacement).

QuadSec:

We also want to include a small module for **recording complaints**, since that connects with the existing dashboard handled by ASOC.

Ricardo B. Corpuz:

When you say "digitalization," does that include scanning and uploading existing paper files?

QuadSec:

Yes, but ideally, the system will also allow **direct input of digital records**, so staff can encode data or test the program while transitioning from physical to digital records.

Ricardo B. Corpuz:

Understood. What specific type of data will this system handle?

QuadSec:

Mainly the **PDS (Personal Data Sheets)** of barangay officials.

Ricardo B. Corpuz:

So it's not HR data—it's the PDS of barangay council officials?

QuadSec:

Yes, sir. Exactly.

QuadSec:

We also plan to integrate a **record-access tracking feature**, somewhat similar to **blockchain**, but simpler. This will allow the system to log who accessed each record and detect any potential **malicious activity**.

Ricardo B. Corpuz:

How will the data be stored—on a **cloud server** or locally?

QuadSec:

We're still finalizing that, but most likely it will be **cloud-hosted**.

QuadSec:

Yes, we'll discuss it further with our professor, but a **cloud-based model** is preferred for scalability.

Ricardo B. Corpuz:

All right, but note that cloud systems usually require **subscriptions** or hosting fees.

QuadSec:

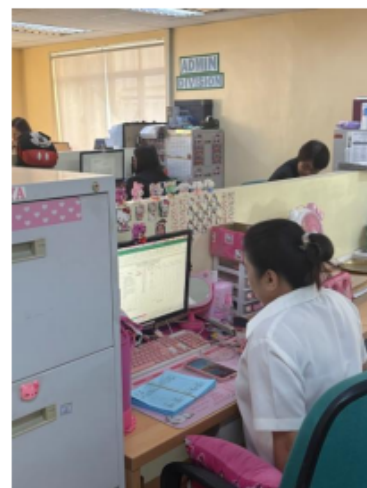
Yes, sir. For now, we'll create a **proof of concept** first. Any subscription or deployment decisions will depend on your approval after testing.

QuadSec:

For development and testing, our team will handle any temporary hosting subscriptions.

Ricardo B. Corpuz:

That's good. If your system performs well, we could include it in our **2027 budget** for possible implementation.



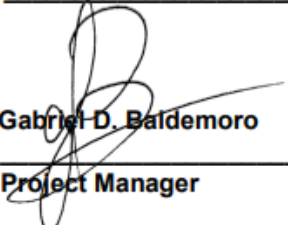
Meeting Adjourned

I hereby confirm that the information provided is correct and complete

Hannah Gabriella Gultiano

Technical and Research Services Division (TSRD Staff of BCRD)

Client


Genro Gabriel D. Baldemoro

Project Manager


Jonathan C. Berris

Document Specialist

BCRD – QuadSec

Meeting Minutes

Meeting Summary – November 21, 2025, 2:00 PM

QuadSec – Capstone Group

- Genro Gabriel D. Baldemoro
- Jonathan C. Berris C. Berris
- Jericho G. Santos
- Lance Arthur J. Sorrrera

Speaker 2 – Client Reps

- Hannah Gabriella Gultiano – Technical and Research Services Division (TRSD Staff)

Key Discussion Points:

1. Project Timeline and Launch

- **Development Phase:** The system development is estimated to span three semesters, totaling approximately 10 to 11 months.
- **Target Launch:** The team plans to launch the system in 2026 to coincide with the Barangay Elections scheduled for November of that year.

2. Document Management and Lifecycle

- **Personal Data Sheets (PDS):** These documents are primarily for storage with no additional transfers or complex processing involved.
- **SALN Handling:** The Statement of Assets, Liabilities, and Net Worth (SALN) is noted as "tricky" due to notarization requirements and data privacy concerns. These records are typically requested by and transferred to the Office of the Ombudsman.
- **Source of Records:** While PDS and SALN come directly from officials, other documents like the Oath of Office and Appointment papers often originate from the Office of the City Mayor.

3. Proposed System Services

- **Official Requests:** The system will allow appointed officials (such as Barangay Secretaries and Treasurers) to request certified copies of their appointments and Oaths of Office for official use.
- **Digitalized Services:** Proposed features include a dashboard for document management and the ability to request travel authority for personal travel.
- **Certification Process:** The system should accommodate the workflow where a records officer certifies copies of these documents.

Hannah Gabriella Gultiano :

We are going to have Barangay Elections next year, I think we could launch it next year on the barangay elections.

Jonathan C. Berris:

The system development and coding phase begins this semester. The project will span a total of three semesters, providing a development timeline of ten to eleven months.

Jonathan C. Berris:

Regarding the SALN (Statement of Assets, Liabilities, and Net Worth), that is the document that needs to be notarized, right? I was thinking of including the SALN, but I'm concerned that the record-keeping might become complicated due to the notarization requirement. While I think we could make it work, I'm not sure if you could utilize it immediately. Integrating the SALN is a bit complex. You also emailed a list of the records you keep from those records, which ones are typically sent by the barangay officials?

Hannah Gabriella Gultiano:

Everything is submitted, but copies of subpoenas come from the court. The PDS (Personal Data Sheet) and SALN are personal and come directly from the officials. Resolutions and various communications come from the barangay. As for the Oath of Office and Appointment papers, those come from other offices, I believe the Office of the City Mayor since that is where they take their oath before the mayor.

Jonathan C. Berris:

What else happens with the PDS? After you receive it from the barangay officials, are there any other processes involved throughout its lifecycle?

Hannah Gabriella Gultiano:

For the most part, it is simply stored. However, copies might be sent to the Office of the Ombudsman, but it is primarily for storage.

Jonathan C. Berris:

Are there times when someone requests a copy? Is it only the Ombudsman?

Hannah Gabriella Gultiano:

Yes, the Ombudsman, and they specifically request the SALN more often.

Jonathan C. Berris:

I believe the PDS is the first thing they send, right?

Hannah Gabriella Gultiano: Yes, and SALN

Jonathan C. Berris:

I am thinking they should be able to see a dashboard where they can manage and perhaps submit their other documents and monitor other processes.

Hannah Gabriella Gultiano:

Perhaps. Regarding submission requests, especially for barangay secretaries and treasurers since they are appointed officials they could request copies of their appointment papers, oath of office, and similar documents. We could include that feature if you'd like, as it is a service we provide. They often need to request those papers for official use.

Jonathan C. Berris:

So, specifically for appointments and such?

Hannah Gabriella Gultiano:

I will get back to you guys once the elections are over. The elections are scheduled for November 2026.

Jonathan C. Berris:

But regarding the PDS, it is strictly for storage with no other transfers involved?

Hannah Gabriella Gultiano:

The SALN is transferred to the Office of the Ombudsman, but the PDS is mostly just for storage.

Jonathan C. Berris:

Are there any documents I could show to prove that the PDS is for storage only and involves no other processes? Is there a formal statement confirming that the PDS is simply kept in storage?

Hannah Gabriella Gultiano:

None. It is literally just filed away.

Jonathan C. Berris:

I think this interview will serve as enough proof that it is strictly for storage. So, we will focus on the other services provided by the BCRD. Other than appointments, what else can we add to a barangay official's account once they have submitted their PDS?

Hannah Gabriella Gultiano:

I think we can add "requests for copies" (e.g., certifications) and personal documents like Travel Authority. Some travel is official and some is personal; official travel usually requires many attachments and is very paper-heavy, so perhaps we should just stick to personal travel for now.

Jonathan C. Berris:

Regarding the requests for copies of appointments, which appointments are these?

Hannah Gabriella Gultiano:

For the treasurer and barangay secretary; specifically, their appointment papers. Then there is the Oath of Office for the barangay officials and barangay *kagawads* (councilors).

Jonathan C. Berris:

What is the process for the appointment and oath of office? Is there a signing process?

Hannah Gabriella Gultiano:

There is a certification process, or rather, they are certified via a copy. Our records officer is the one who handles the certification.

Jonathan C. Berris:

So, for copies of appointments and the oath of office for barangay officers, they are the ones who submit those, correct?

Hannah Gabriella Gultiano:

Yes.

Jonathan C. Berris:

And for travel authority requests, is there anything else a barangay official does with the BCRD that we can include in the digitalization of the website?

Hannah Gabriella Gultiano:

Usually, those are the only services for them. I believe our services are listed in our Citizen's Charter, and those are really the only ones. There is a distinction between official and personal travel authority.

Jonathan C. Berris:

I will try to find a way to include the SALN, if the panelists will allow for a limited implementation.

Hannah Gabriella Gultiano:

Yes, that would be good since it involves data privacy since it concerns their personal finances.

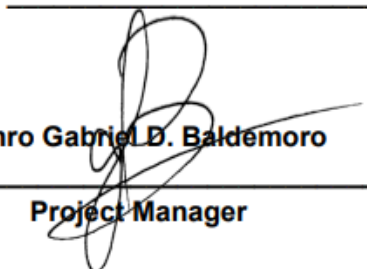
Meeting Adjourned

I hereby confirm that the information provided is correct and complete

Hannah Gabriella Gultiano

Technical and Research Services Division (TSRD Staff of BCRD)

Client


Genro Gabriel D. Baldemoro

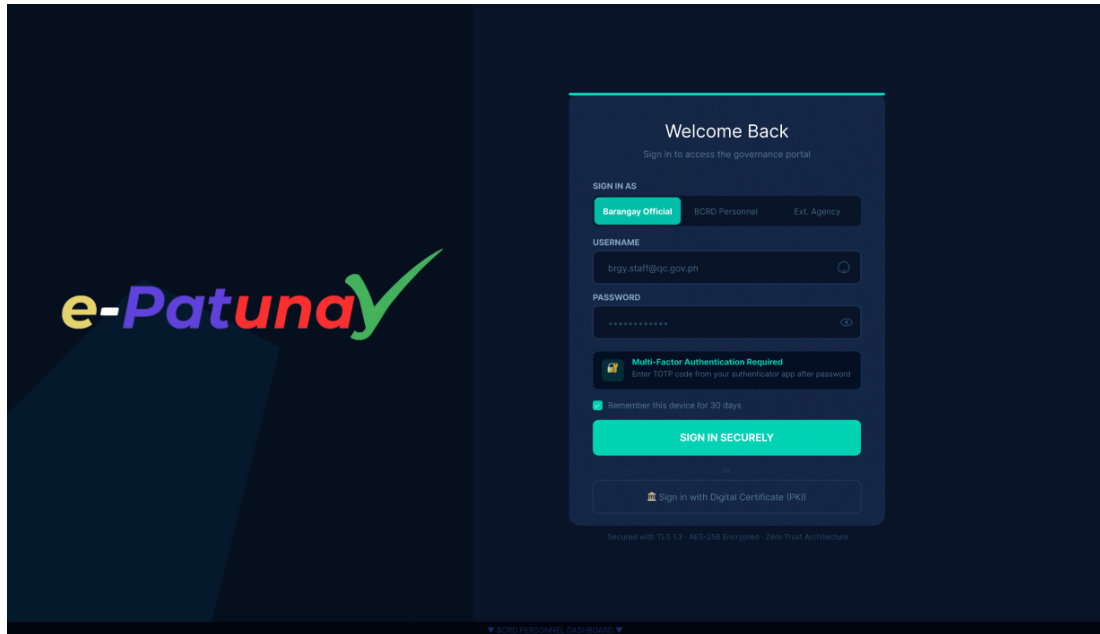
Project Manager


Jonathan C. Berris

Document Specialist

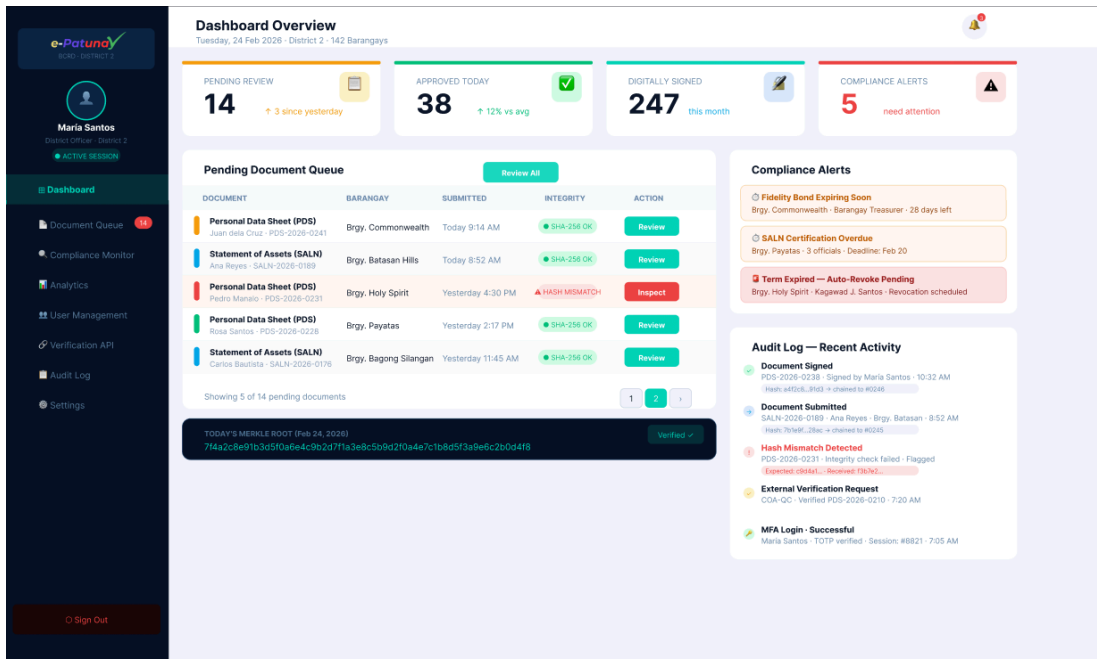
APPENDIX D: System Design

Login Page



The Login Page serves as the secure authentication gateway of the e-Patunay system. It requires users to enter registered credentials and complete Multi-Factor Authentication (MFA) before access is granted. This page enforces Zero Trust security principles by ensuring that no user is trusted by default, even if they are within the internal network. Credential transmission is encrypted, login attempts are monitored to prevent brute-force attacks, and session timeout mechanisms are implemented to reduce unauthorized access risks. Upon successful authentication, users are redirected to dashboards based on their assigned system roles.

BCRD Personnel Dashboard



Sidebar Persistent navigation menu showing the user's name, role, and active district.

The district label is a constant reminder of the officer's RBAC scope.

Four KPI Cards At-a-glance metrics: Pending Review (amber), Approved Today (green), Digitally Signed (blue), and Compliance Alerts (red). Colors signal urgency level.

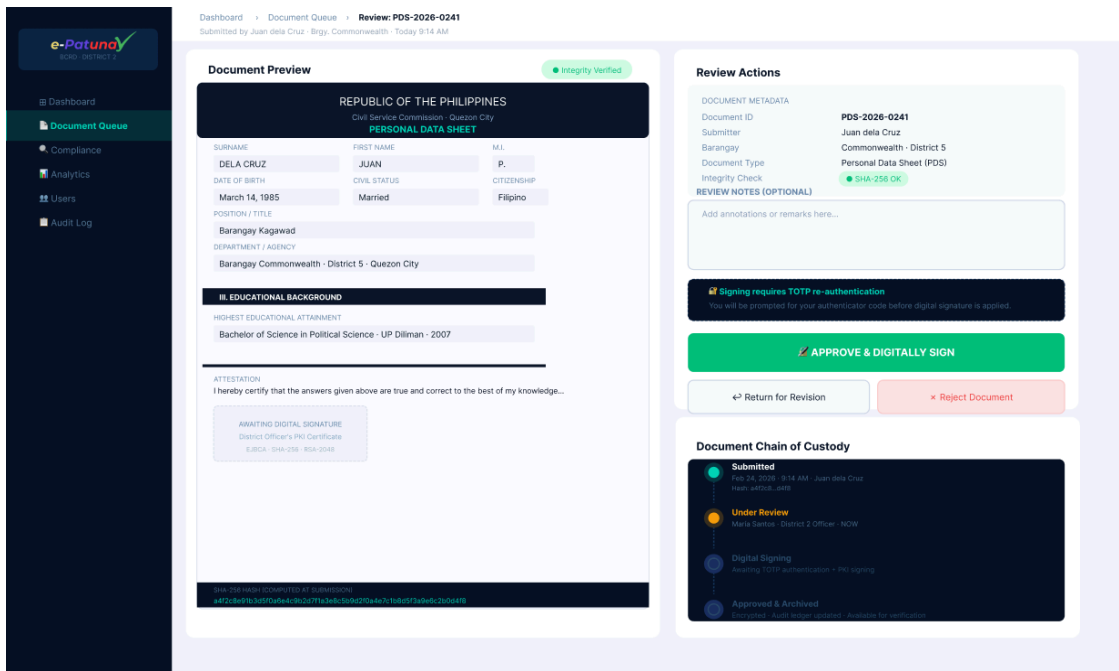
Document Queue Table Lists pending documents with submitter, barangay, timestamp, and a SHA-256 integrity check result. A red "HASH MISMATCH" badge flags tampered files before the officer even opens them.

Compliance Alerts Panel Shows certifications expiring within 30 days and officials whose terms have ended and are scheduled for automatic credential revocation.

Audit Log Feed A live feed of recent system events (submissions, approvals, errors) each stamped with a hash chain reference so officers can see the cryptographic record being built in real time.

Merkle Root Bar A dark bar at the bottom showing today's Merkle root has a single value that cryptographically summarizes all audit entries for the day.

Document Review & Signing



Document Preview Panel A read-only, full-fidelity rendering of the submitted government form (PDS or SALN). Officers can read but cannot edit the content. A green "Integrity Verified" badge confirms the file's hash matches what was submitted.

SHA-256 Hash Bar Displayed at the foot of the document the complete hash of the file, usable for independent verification.

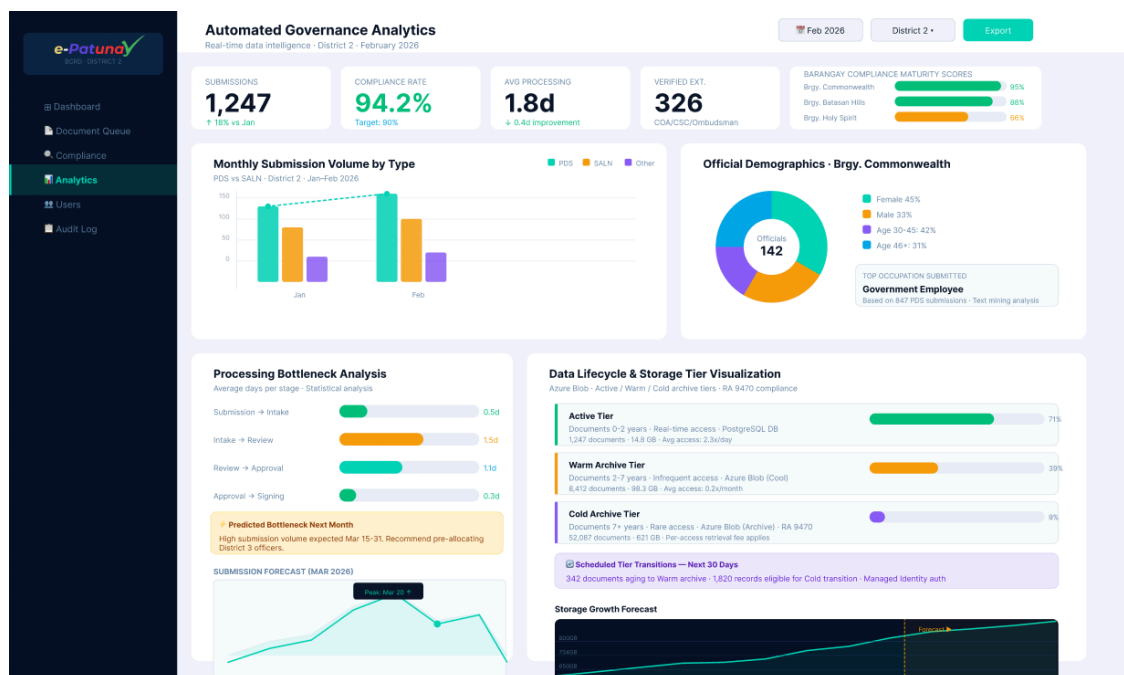
Review Notes Field A text box for the officer to record observations or rejection reasons. Required before returning or rejecting a document; stored permanently in the audit log.

MFA Re-authentication Notice A locked box placed directly above the signing button. Officers must re-enter their TOTP code before the signature is applied, even with an active session.

Approve & Sign / Return / Reject Buttons Three action buttons with distinct visual weight. Approve (teal, full-width) triggers PKI signing. Return (ghost) sends the document back for corrections. Reject (red) terminates the workflow permanently.

Chain of Custody Timeline A vertical step-by-step trace of the document's lifecycle who submitted it, who is reviewing it, and what happens next linking each step to its audit log entry.

Automated Governance Analytics



Five KPI Cards Top-line governance numbers: total submissions, compliance rate, average processing time, and count of external verifications. A mini compliance maturity chart ranks barangay performance.

Submission Volume Chart Grouped bar chart showing monthly document counts by type (PDS, SALN, Other), with a trend line for quick direction-reading.

Demographic Donut Chart Breaks down the age and gender profile of barangay officials based on PDS metadata, with a text-mining result for the most common occupation submitted.

Bottleneck Analysis Horizontal bars showing average days per workflow stage. Slow stages appear in amber. An advisory box predicts the next high-volume period and recommends pre-allocating staff.

Storage Tier Visualization Three rows (Active, Warm Archive, Cold Archive) showing document counts, storage volume, and usage bars for each tier, in compliance with the National Archives Act (RA 9470).

External Agency Verification Portal

The screenshot displays the 'e-Patuna' External Agency Verification Portal. At the top, it indicates 'EXTERNAL AGENCY VERIFICATION PORTAL - READ-ONLY ACCESS' and shows the authenticated agency as 'Commission on Audit (COA)'. The main interface is divided into three sections:

- Verify Document Authenticity:** A form for entering a document ID or uploading a signed PDF. The document ID 'PDS-2026-0238' is entered. A 'VERIFY DOCUMENT' button is present.
- DOCUMENT VERIFIED:** A panel showing the verification result. It states 'Cryptographic signature is VALID - PKI chain intact'. The document information includes ID 'PDS-2026-0238', Type 'Personal Data Sheet (PDS)', and Issuing Authority 'BCRD - Quezon City'. The cryptographic verification details show the signing officer 'Maria Santos (Active Officer)', timestamp 'Feb 24, 2026 - 10:32:14 AM PHT', certificate authority 'EJBCA - BCRD Subordinate CA - PNP/PI', certificate status 'VALID', document hash 'a1f1e8e1...d1f1e8e1', and audit ledger entry '#0246 - Merkle included ✓'.
- Verification Request History (This Session):** A table listing all verification requests during the current session.

DOCUMENT ID	TYPE	BARANGAY	VERIFIED AT	SIGNER STATUS	RESULT
PDS-2026-0238	PDS	Brgy. Commonwealth	10:34 AM	Active ✓	VALID
SALN-2026-0176	SALN	Brgy. Bagong Silangan	10:28 AM	Active ✓	VALID
PDS-2025-1120	PDS	Brgy. Holy Spirit	10:21 AM	Revoked ✗	CERT REVOKED
PDS-2026-0210	PDS	Brgy. Payatas	09:55 AM	Active ✓	VALID

At the bottom, a notice states: 'All verification requests are logged in the e-Patuna audit ledger and attributed to COA user session #7521. No personally identifiable information beyond verification result has been disclosed. Compliant with Data Privacy Act (RA 10173).'

Agency Badge Top-right card showing which external agency (e.g., COA) is currently authenticated. Read-only access is labeled prominently in the header.

Document ID Input / File Upload Two ways to verify: type the document ID directly, or drop the signed PDF file. The system checks the embedded PKI signature either way.

Verify Button Sends the query to the Verification API. No raw document content is ever returned, only the cryptographic result.

Verification Result Panel Shows whether the document is VALID or REVOKED with a clear green/red status banner. Below it: the signing officer's name and employment status, certificate authority, certificate status, document hash, and Merkle ledger entry.

Verification History Table A log of all documents checked during the current session, including result status. Revoked-certificate entries appear in red. A notice at the bottom confirms every request is itself logged in the system's audit trail.

APPENDIX E: Post-Test Questionnaire



FEU INSTITUTE OF TECHNOLOGY

Post-Test Questionnaire for BCRD

QuadSec

1. Baldemoro, Genro Gabriel D.
2. Berris, Jonathan C.
3. Santos, Jericho G.
4. Sorrera, Lance Arthur J.

E-Patunay: A Cryptographic-Assurance Governance System for the Barangay Community Relations Department Enabling Storage Optimization and Inter-Agency Record Verification

DATA PRIVACY POLICY STATEMENT

QuadSec, a group of researchers under FEU Institute of Technology, is committed to upholding the rights of individuals to data privacy in the conduct of our research activities.

This policy follows the principles of transparency, legitimate purpose, and proportionality in the collection, processing, and storage of personal data from research participants, employees, external collaborators, and other stakeholders. These principles guide our responsible handling of data throughout the research process.

- **Transparency.** Research participants must be fully informed about the nature, purpose, and scope of data collection and processing. This includes potential risks, data protection measures, the identity of the data controller, participants' rights, and how they can exercise those rights.

- **Legitimate Purpose.** All personal data gathered shall be used strictly for the declared research objectives and will not be used for purposes that violate ethical standards, laws, or public policy.

- **Proportionality.** Data collection shall be limited to what is necessary, relevant, and suitable for the research objectives. It shall not be excessive or beyond the intended scope of the study.

We adhere to all provisions of Republic Act No. 10173 (Data Privacy Act of 2012), its Implementing Rules and Regulations, relevant policies and issuances of the National Privacy Commission, and all other applicable laws and ethical guidelines. SD Games is committed to continuously improving our data security measures to protect the integrity, confidentiality, and availability of personal data.

☐ I acknowledge and consent to the use of my data as stated in the policy.

☐ I acknowledge but do not consent to the use of my data.



PERSONAL INFORMATION

Please enter the needed information below. Your inputs shall be in **UPPERCASE**.
(Maaring ilagay ang mga impormasyong kinakailangan sa ibaba. Ang lahat ng sagot ay nararapat na nakasulat sa **MALALAKING LETRA**.)

SURNAME [APELYEDO]:															
FIRST NAME [UNANG PANGALAN]:															
MIDDLE NAME [GITNANG PANGALAN]:															

AGE [EDAD]:	SEX [KASARIAN]:	CONTACT NUMBER:	E-MAIL ADDRESS:

Functionality

4 – Strongly Agree 3 – Agree 2 – Disagree 1 – Strongly Disagree

Question [Katanungan]	4	3	2	1
1. The system allows me to successfully upload and process documents. [Pinapayagan ako ng sistema na matagumpay na mag-upload, magpasa, at magproseso ng mga dokumento.]				
2. The Two-Step Login (MFA) works correctly every time I log in. [Gumagana nang maayos ang Two-Step Login (MFA) tuwing ako ay mag-login.]				
3. The digital signature is properly attached to the approved documents. [Ang digital signature ay maayos na nailalagay sa mga naaprubahang dokumento.]				
4. The document verification feature correctly shows if a document is valid or fake. [Tamang naipapakita ng verification feature kung ang isang dokumento ay totoo o peke.]				
5. The system automatically sends the correct alerts and notifications. [Awtomatikong nagpapadala ang sistema ng mga tamang alerto at abiso.]				



Usability

4 – Strongly Agree

3 – Agree

2 – Disagree

1 – Strongly Disagree

Question [Katanungan]	4	3	2	1
1. The system is easy to learn, even for beginners. [Madaling matutunan ang sistema, kahit para sa mga baguhan.]				
2. The design of the system is clean and easy to read. [Malinis at madaling basahin ang disenyo ng sistema.]				
3. I can easily find the buttons and menus I need to complete my tasks. [Madali kong nahahanap ang mga button at menu na kailangan ko para magawa ang aking mga gawain.]				
4. I can complete my tasks smoothly without getting confused. [Nagagawa ko ang aking mga gawain nang tuluy-tuloy at walang kalituhan.]				
5. The buttons and menus behave exactly as I expect them to. [Gumagana ang mga button at menu ayon sa eksaktong inaasahan ko]				

Reliability

4 – Strongly Agree

3 – Agree

2 – Disagree

1 – Strongly Disagree

Question [Katanungan]	4	3	2	1
1. The system does not crash or freeze while I am using it. [Hindi nagca-crash o nagha-hang ang sistema habang ginagamit ko ito.]				
2. I feel confident that the documents stored in the system are safe and protected. [Kampante ako na ligtas at protektado ang mga dokumentong nakatago sa sistema.]				
3. When I upload a document, I am sure it will still be there the next time I log in. [Kapag nag-upload ako ng dokumento, nakakasiguro akong nandiyan pa rin ito sa susunod na pag-login ko.]				
4. I have not experienced any unauthorized access to my account. [Wala akong naranasang ibang taong nakapasok sa aking account nang walang permiso.]				
5. The system clearly records every action made by the users, which makes the whole process feel secure. [Malinaw na inilista ng sistema ang bawat galaw ng mga user, kaya pakiramdam ko ay ligtas ang aming buong proseso.]				



Performance

4 – Strongly Agree

3 – Agree

2 – Disagree

1 – Strongly Disagree

Question [Katanungan]	4	3	2	1
1. The system loads quickly when I open it in my web browser. [Mabilis mag-load ang sistema kapag binubuksan ko ito sa internet browser.]				
2. Uploading and downloading files is fast. [Mabilis ang proseso ng pag-upload at pag-download ng mga file.]				
3. The system responds immediately when I click buttons or links. [Mabilis na tumutugon ang sistema kapag pumipindot ako ng mga button o link.]				
4. The verification process checks a document in just a few seconds. [Nabe-verify o nasusuri ng sistema ang isang dokumento sa loob lamang ng ilang segundo.]				
5. The system does not slow down even when many people are using it at the same time. [Hindi bumabagal ang sistema kahit maraming tao ang sabay-sabay na gumagamit nito.]				

APPENDIX F: Consent Statement Draft (MoA)

MEMORANDUM OF AGREEMENT

KNOW ALL MEN BY THESE PRESENTS:

This Memorandum of Agreement (hereinafter referred to as the “Agreement”) is made and entered into this [Date], by and between:

BARANGAY COMMUNITY RELATIONS DEPARTMENT (BCRD), with office address at the 2nd Floor Community Center Building, Quezon City Hall Compound, Elliptical Road, Barangay Central, Quezon City, represented herein by its Technical and Research Services Division, hereinafter referred to as the **"BCRD"**;

and

Quadsec, a student group from the Bachelor of Science in Information Technology with specialization in Cybersecurity at Far Eastern University Institute of Technology (FEU Tech), with principal address at P. Paredes Street, Sampaloc, Manila, represented by **Genro Gabriel D. Baldemoro, Project Manager**, hereinafter referred to as **"Quadsec"**;

BCRD and **Quadsec** are collectively referred to as the “Parties” and individually as a “Party”.

WITNESSETH:

WHEREAS, Quadsec is undertaking a capstone project titled “e-Patunay: A Cryptographic-Assurance Governance System for the Barangay Community Relations Department Enabling Storage Optimization and Inter-Agency Record Verification”;

WHEREAS, the BCRD recognizes the necessity of transitioning from manual, paper-based workflows to secure digital systems to comply with the E-Governance Act (RA 12254) and the Philippine e-Government Master Plan 2023-2028;

NOW, THEREFORE, for and in consideration of the foregoing premises and the mutual covenants herein set forth, the Parties hereby agree as follows:

ARTICLE 1
PURPOSE AND SCOPE OF THIS AGREEMENT

- 1.1. This Agreement establishes the terms for the development and deployment of e-Patunay, a web-based governance system designed to ensure document authenticity, data integrity, and non-repudiation for the BCRD.
- 1.2. This agreement covers the tasks required to design, construct, test, implement, and hand over the project in compliance with the roles and responsibilities specified herein and in line with relevant laws, regulations, and directives.

ARTICLE 2
OBLIGATIONS OF QUADSEC

- 2.1. Design, develop, and deploy the e-Patunay system as outlined in the Researchers' latest proposal dated [], subject to BCRD's review and acceptance milestones.
- 2.2. Treat all information shared by BCRD as confidential pursuant to a Non-Disclosure Agreement (NDA), using it solely for academic and developmental purposes under this Project.
- 2.3. Provide regular progress updates and demonstrations to designated BCRD representatives on a mutually agreed schedule and format.
- 2.4. Upon completion and acceptance of the Project, the final technological solution will be offered to the BCRD at minimal or no cost. This offering will include source code, documentation, deployment materials, and administrator guides, as applicable.

ARTICLE 3
OBLIGATIONS OF THE BARANGAY COMMUNITY RELATIONS
DEPARTMENT
(BCRD)

- 3.1. Grant Quadsec permission to conduct structured interviews with key personnel and provide access to sample forms and non-classified records (e.g., PDS, SALN) for system validation.

- 3.2. Provide access to sample forms, report templates, standard operating procedures, and other non-classified relevant data necessary to guide system design.
- 3.3. Designate an official point of contact to facilitate timely communication.
- 3.4. Provide feedback on design, functionality, and usability to ensure the Project aligns with operational needs and existing procedures.

ARTICLE 4 INTELLECTUAL PROPERTY

- 4.1. All intellectual property rights arising from the development of e-Patunay shall be jointly owned by the Researchers and BCRD, without prejudice to applicable academic and government policies.
- 4.2. The Researchers may use the Project for academic purposes, including presentation and publication, provided that any sensitive BCRD data is anonymized or removed in accordance with the NDA and applicable laws.
- 4.3. BCRD shall have the right to use, modify, and deploy the e-Patunay system for its internal operations without further compensation to the Researchers, subject to attribution consistent with academic norms where appropriate.

ARTICLE 5 CONFIDENTIALITY

- 5.1. Both Parties shall keep confidential all non-public information, data, and materials exchanged under this Agreement and shall use such information solely for the Project.
- 5.2. The obligations in this Article shall survive termination or expiration of this Agreement to the extent permitted by applicable law and the NDA.

ARTICLE 6 TERM

This Agreement shall commence on the date first written above and remain in force until completion and acceptance of the Project, unless earlier terminated in accordance with Article 7.

ARTICLE 7 TERMINATION

- 7.1. Either Party may terminate this Agreement, in whole or in part, for convenience by providing thirty (30) calendar days' written notice to the other Party.
- 7.2. Termination shall be without prejudice to either Party's accrued rights and obligations, including confidentiality and intellectual property provisions that expressly survive termination.
- 7.3. Termination is only available for the first and second week upon signing this Agreement. This provision would allow the Quadsec to find a new client and create a new project proposal for their capstone.

ARTICLE 8 ENTIRE AGREEMENT

This Agreement contains the entire agreement between the Parties with respect to the subject matter hereof and supersedes all prior agreements or understandings, whether oral or written, relating thereto.

ARTICLE 9 AMENDMENT

Any amendment or modification to this Agreement must be in writing and signed by authorized representatives of both Parties to be effective.

ARTICLE 10 SEVERABILITY

If any provision of this Agreement is held invalid or unenforceable, the remaining provisions shall remain in full force and effect, and such provision shall be enforced to the maximum extent permissible by law.

ARTICLE 11 NON-WAIVER OF RIGHTS

The failure of either Party to enforce any provision of this Agreement shall not be deemed a waiver of such provision or of the right to enforce it thereafter.

ARTICLE 12 PARAGRAPH HEADINGS

Headings are for convenience only and shall not affect the interpretation of this Agreement.

ARTICLE 13 DISPUTE RESOLUTION

- 13.1. The Parties shall strive to resolve any dispute arising from or relating to this Agreement through consultation and good-faith negotiations.
- 13.2. If unresolved, disputes involving national government agencies shall be resolved before the Office of the Solicitor General and/or the Department of Justice pursuant to applicable rules on alternative dispute resolution for disputes between national government agencies.

ARTICLE 14 GOVERNING LAW AND VENUE

This Agreement shall be governed by and construed in accordance with the laws of the Republic of the Philippines, and actions to enforce or interpret this Agreement shall be brought in the courts of the City of Manila to the exclusion of all other courts.

ARTICLE 15 EFFECTIVITY

This Agreement shall take effect upon execution by the Parties.

IN WITNESS WHEREOF, the Parties hereto have set their hands on the date and place first above written.

**BARANGAY COMMUNITY
RELATIONS
DEPARTMENT(BCRD)**

QUADSEC of FEU TECH

By:

By:

**TBA
TBA**

**GENRO GABRIEL D.
BALDEMORO**
Project Manager

SIGNED IN THE PRESENCE OF:

TBA
TBA

JONATHAN C. BERRIS
Quadsec, FEU Tech

APPROVED BY:

ABRICAM S. TINGA, MSIT
Project Advisor

DR. KIRK ALVIN AWAT
Cybersecurity Program Director of FEU
Tech

APPENDIX G: Curriculum Vitae

BALDEMORO, GENRO GABRIEL D.

INFORMATION TECHNOLOGY
CYBERSECURITY STUDENT



CONTACT

+63 9696109943

baldemorgenrogabriel@gmail.com

Block 6 lot 22 phase 2 Pinagsama
village

HARD SKILLS

- Proficiency English and Tagalog
- Blender 3D Modelling Experience
- Programming in C++, Java HTML5, SQL
- Davinci Resolve Video Editing

SOFT SKILLS

- Time management
- Flexibility
- Teamwork
- Communication
- Project Management

PROFILE

As an aspiring cybersecurity specialist, I am dedicated to delivering exceptional quality work with unwavering ethics and flexibility to meet project requirements. You can trust me to consistently deliver results on time and exceed your expectations.

EDUCATION

FEU Institute of Technology

2023-Present

- Junior Officer, Logistics Committee Association of Information Technology Students
- Logistics Head, UI/UXplosion Event Association of Information Technology Students
- Director, Logistics Committee Association of Information Technology Students

Far Eastern University High School

2021-2022

- Member, FEU HS Speech and Debate Society
- Academic Excellence Awardee (A.Y 2022-2023)
- Grade 12 Honors Student
- Grade 11 Honors Student

Future Generation Philippine International School

2009- 2020

- Loyalty Awardee
- 8th Grade Biology competition
- Member, FusiMatics Club
- Member, English Club

CERTIFICATIONS & AWARDS

- Pearson ITS Database
- Pearson ITS Java
- Pearson ITS Python
- Pearson CCST Networking
- CCNA Ethical Hacker
- CCNA Network Defense
- CCNA Introduction to Cybersecurity
- CCNA Junior Cybersecurity Analyst Career path
- CCNA: Switching, Routing, and Wireless Essentials

BERRIS, JONATHAN C.

INFORMATION TECHNOLOGY
CYBERSECURITY STUDENT



CONTACT

+63 9761490758

jcberris04@gmail.com

Block 3 Lot 118 Villa Vienna Phase 1-A

HARD SKILLS

- Proficiency English and Tagalog
- Basic Penetration Testing Experience
- Networking Experience
- Programming in C++, Java HTML5, PHP, SQL

SOFT SKILLS

- Problem Analysis
- Organization
- Team Collaboration

PROFILE

As an aspiring cybersecurity specialist, I am dedicated to delivering exceptional quality work with unwavering ethics and flexibility to meet project requirements. You can trust me to consistently deliver results on time and exceed your expectations.

EDUCATION

FEU Institute of Technology 2022-Present

- B.S. in Information Technology with Specialization in Cybersecurity

Cornerstone Christian School Academy 2018-2021

- Academic Excellence Awardee (S.Y 2020-2021)
- K-12 Graduate in Humanities and Social Sciences
- Grade 12 Honors Student
- Grade 11 Honors Student 2010-2017

**Operation Brotherhood
Montessori Inc. Fairview**

- Loyalty Awardee
- Elementary Graduate

CERTIFICATIONS & AWARDS

- Pearson ITS Database
- Pearson ITS Java
- Pearson ITS Python
- Pearson CCST Networking
- CCNA Ethical Hacker
- CCNA Network Defense
- CCNA Introduction to Cybersecurity
- CCNA Junior Cybersecurity Analyst Career path
- CCNA: Switching, Routing, and Wireless Essentials

SANTOS, JERICHO G.

INFORMATION TECHNOLOGY
CYBERSECURITY STUDENT



CONTACT

- 63+ 9098736522
- jericho.santos1620@gmail.com
- Sampaloc, Manila
- <https://www.linkedin.com/in/jericho-santos-777389358/>
- [JerichoSantos1620.github.io](https://github.com/JerichoSantos1620)

SOFT SKILLS

- Communication skills
- Problem-solving skills
- Organization and time management skills

TECHNICAL SKILLS

- Security Concepts - Network Segmentation, Vulnerability Assessment, Threat Analysis, Basic Penetration Testing.
- Operating Systems - Windows Server, Linux (Ubuntu/Kali Linux), macOS.
- Networking - TCP/IP, OSI Model, Wireshark, Basic Firewall Configuration.
- Programming/Scripting - Python (Basic), HTML, CSS, JAVA, C++ and SQL.

PROFILE

IT Cybersecurity Student at FEU Institute of Technology (FEU-TECH) focused on network security and data protection. I am highly organized, detail-oriented, and possess strong communication skills, which help me manage complex projects and analyze information diligently. Eager to begin a career and apply my technical knowledge in a challenging security environment.

EDUCATION

**Far Eastern University
Institute of Technology**
2023 - present
B.S. in Information Technology
with Specialization in
Cybersecurity

CERTIFICATIONS & AWARDS

- Cisco Certified Support Technician Networking (CCST Networking)
- Cisco Ethical Hacker Certificate
- Network Defense Certificate
- CCNA Certificate

SORRERA, LANCE ARTHUR J.

INFORMATION TECHNOLOGY
CYBERSECURITY STUDENT



CONTACT

- 63+ 9509986499
- sorrerance@gmail.com
- Pasig City, Manila
- www.linkedin.com/in/lance-arthur-sorrera-247bb4339

TECHNICAL SKILLS

- Security Concepts - Network Segmentation, Vulnerability Assessment, Threat Analysis, Basic Penetration Testing.
- Operating Systems - Windows Server, Linux (Ubuntu/Kali Linux), macOS.
- Networking - TCP/IP, OSI Model, Wireshark, Basic Firewall Configuration.
- Programming/Scripting - Python (Basic), HTML, CSS, JAVA, C++ and SQL.

PROFILE

Motivated and hardworking Cybersecurity student seeking an internship or entry-level role to apply academic knowledge in network security, threat analysis, and risk management. Committed to continuous learning and developing practical skills to help organizations defend against cyber threats.

EDUCATION

Far Eastern University Institute of Technology

2023 - present
B.S. in Information Technology with Specialization in Cybersecurity

Flora A Ylagan High School

2019 - 2022
Information and Communication Technology

San Jose Del Monte National High School

2017-2018

Heroesville Elementary School

2016

Pinyahan Elementary School

2010-2015

CERTIFICATIONS & AWARDS

- CCNA Ethical Hacker
- CCNA Network Defense
- CCNA Introduction to Cybersecurity
- CCNA Junior Cybersecurity Analyst Career path
- CCNA: Switching, Routing, and Wireless Essentials